LM2_LOOP_EQUATIONS

The equation of loops in types $(0,3)$ and $(1,1)$

A proof audit, exact finite computations, and corrected comparisons for §§36–47 of *La Longue Marche*

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Notation fixed at the outset. Locations in this table are scans of cote 140-3; the expanded dictionary in Part A gives folios, definitions, and the changes of notation between sections. Standard algebraic notation has its usual meaning; hats always mean full profinite completion.

Our symbol; manuscript symbol	Meaning fixed in this paper	Introduction / comparison
$F = F_2 = \Pi_{0,3}$; $\widehat{F}$	Tripod group, free on $x = \ell_1$, $y = \ell_0$; $z = \ell_\infty = (xy)^{-1}$.	85–94, 182
$B = \Pi'_{0,3} = \mathfrak{S}^+_{0,3}$	$C_2 * C_3 = \mathrm{PSL}_2(\mathbb{Z})$; $F \triangleleft B$ has index six.	18, 91–96
$B^\pm = \mathfrak{S}_{0,3}$	Extended modular group $\mathrm{GL}_2(\mathbb{Z})/\{\pm I\}$.	92–96
$H = \widehat{\mathrm{SL}_2(\mathbb{Z})}$	Full linear modular group; $H/\{\pm I\} = \widehat{B}$, not the congruence completion.	21–26, 176–178
$s = \sigma = \sigma_\infty$, $r = \rho$	Order-two and order-three projective generators; $\sigma(x,y) = (y,x)$, $\rho(x,y,z) = (z,x,y)$.	85–94
$J = \tau_\infty$; $c = \tau$	$J = \mathrm{diag}(1,-1)$; normalized complex conjugation $c = (-1,1)$.	87–93, 252–253
ε_i, ℓ_i	Three parabolic half-loops and their squares, $\ell_i = \varepsilon_i^2$; $e = sr$.	88, 92
$\widetilde{M}, \widetilde{M}^!$	Genuine compatible loop automorphisms; geometric kernels $\widehat{B}$, $\widehat{F}$.	182–184
$\widetilde{N}$, $C = \widetilde{N}^!$	Outer groups $S_3 \times C$ and $C = \widehat{\mathrm{GT}}_0$.	183–184
$\widetilde{\Gamma}$; older $\widetilde{N}$	The earlier quotient by $\widehat{B}$, equal to C; not the later $\widetilde{N}$.	170, 177–178
$\widetilde{N}_{1,1}$	Central linear pullback extension; its geometric group is H.	176–178
$K_q, \widetilde{M}^!(q)$	Chosen centralizer or cusp sections; $q = 1/2, 2, -1, 0$; order-three section only over $C(3)$.	206–227
$\mu = \lambda$, $\nu = (\mu-1)/2$	Unit multiplier and half-exponent; $f \in \widehat{F}'$ is the Drinfeld conjugator.	186–197
$g, h, f_G; \alpha, \beta; \xi, \eta$	Based conjugators and their ρ, σ cocycles; f_G is not the normalized f. The alternative ζ is not a new variable.	186–197
$\epsilon = \epsilon_2$, $\delta = \epsilon_3$	Global signs from the residues of μ modulo four and three.	154–168, 198–205
φ, C^φ; double marks	$(y,x,z) \mapsto (sr, s, r^{-1})$ and the additional compatible outer subgroup.	229–250
$\widehat{\mathrm{GT}}, \widehat{\mathrm{GT}}_0$; W^{lac}, d_n	Modern pentagon group, two-cycle group, graded loop envelope, and odd-generator free-Lie dimensions.	Comparisons B.m2, B.g1–B.g9
$G_\mathbb{Q}, \chi$; $\mathbb{\Gamma}_\mathbb{Q}$	Absolute Galois group and its cyclotomic character.	21–26, 155–170

Abstract

We distinguish the profinite group equations, their finite shadows, and the rational graded equations. The accompanying registers record the proof status of each numbered statement and the exact limits of the computations. Quotient solutions are never identified with liftable profinite solutions without a lifting argument. The attached unrefereed 2026 preprints are treated as sources whose used steps require audit, not as certified results. No priority is claimed.

Conventions and scope

We write $F_2 = \langle x, y\rangle$, $z = (xy)^{-1}$, and $[a, b] = aba^{-1}b^{-1}$. A hat on a discrete group means its full profinite completion; $\mathrm{SL}_2(\widehat{\mathbb{Z}}) = \prod_p \mathrm{SL}_2(\mathbb{Z}_p)$ means its congruence completion. These two operations are never interchanged. Homomorphisms of profinite groups are continuous. Arithmetic fundamental groups use a geometric or tangential base point; changing a path gives the indicated inner-conjugacy ambiguity. We write $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and χ for the cyclotomic character. The symbols in the manuscript are fixed in the dictionary in Part A; primes, exclamation marks, and tildes are mathematical data. The primary manuscript is cited as Grothendieck (1981).

Reading conventions. A reference "140-3:209, f. 403 verso" means the archive cote, scan, and handwritten folio; where folio attribution is not unambiguous, we retain the scan as the stable locator and explicitly say so. Corrected transcriptions are used as working text, with the selected original scans inspected whenever a conclusion depends on a disputed formula. Mathematical consistency provides editorial evidence, not a final palaeographic reading. Withdrawn assertions are marked as withdrawn. The notes, LM3 return, and reviews are leads rather than authority.

Status conventions. Proved includes a proof here; Known includes an exact source and the reduction required in this setting; Refuted includes an explicit counterexample and a corrected assertion; Open records the remaining precise assertion and the attack. Conditional use of an unrefereed preprint is named as such. A bibliography entry marked verified means that we opened the source; it does not mean that we independently verified its entire proof. The report distinguishes partly settled tasks from unfinished ones whenever the requested audit could not reach its stated standard.

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A. A dictionary which retains the primes and the extensions

We use ordinary left matrix multiplication. A prime on a group in the manuscript means a quotient by a specified central subgroup, or an orbifold quotient; it never means a commutator subgroup in this dictionary. We write $F = \langle x, y\rangle$, $z = (xy)^{-1}$, and $\widehat{F}$ for its full profinite completion. In the notation of §46 we choose

$$x = \ell_1, \qquad y = \ell_0, \qquad z = \ell_\infty, \qquad xyz = 1.$$

The switch of the first two labels is essential: the manuscript's product is $\ell_\infty \ell_1 \ell_0 = 1$. A symbol such as $\rho(a)$ below means conjugation by the indicated lift of ρ, not multiplication by that lift. Inner automorphisms are $\mathrm{Int}(a)(b) = aba^{-1}$. All automorphisms of profinite groups are continuous.

Our notation / his notation	Definition and present interpretation	Manuscript location (cote 140-3 unless indicated)
$F = \Pi_{0,3}$	Tripod fundamental group; discrete free group of rank two. Its completion is not the modular group.	§42, fols.332–336, scans85–94.
$B = \Pi'_{0,3} = \mathfrak{S}^+_{0,3} = \mathcal{G}^+_{0,3}$	Oriented modular orbifold group $C_2 * C_3 = \mathrm{PSL}_2(\mathbb{Z})$; F is its index-six level-two kernel.	§38, fol.297, scan18; §42, fols.334–336, scans91–94.
$B^\pm = \mathfrak{S}_{0,3}$	Extended modular group $\mathrm{GL}_2(\mathbb{Z})/\{\pm I\}$, equivalently the reflection group of signature $(2,3,\infty)$.	§42, fols.335–337, scans92–96.
$\sigma = \sigma_\infty, \rho$	Generators of B with $\sigma^2 = \rho^3 = 1$. On F, $\sigma(x,y) = (y,x)$ and $\rho(x,y,z) = (z,x,y)$.	§42, fol.334′, scan91, (33).
$J = \tau_\infty$; τ	τ is the outer complex-conjugation class. τ_∞ is its genuine lift commuting with σ_∞; $J = \mathrm{diag}(1,-1)$ projectively.	§42, fols.332′–335′, scans87–93.
$\varepsilon_0, \varepsilon_1, \varepsilon_\infty$	$\sigma\rho, \rho\sigma, \rho^{-1}\sigma\rho^{-1}$ in B; their squares are $\ell_0, \ell_1, \ell_\infty$.	§42, fol.333, scan88 (22), and scan92 (34).
$\widetilde{M} = \widetilde{M}_{0,3}$	Genuine loop automorphisms of $\widehat{F}$ whose outer class normalizes geometric S_3; equivalently the compatible automorphisms of $\widehat{B}$.	§46, fol.390, scan182 (1)–(3).
$\widetilde{N} = \widetilde{N}_{0,3}$	$\widetilde{M}/\mathrm{Inn}(\widehat{F})$; equals $S_3 \times C$, where C is the centralizer of S_3 in the label-preserving outer loop group.	§46, fols.390–391, scans182–184.
$C = \widetilde{N}^!_{0,3}$	The above centralizer; identified below with $\widehat{\mathrm{GT}}_0$, with the automorphism condition retained.	§46, scan183 (9bis)–(11).
$\widetilde{M}^! = \widetilde{M}^!_{0,3}$	Inverse image of C in $\widetilde{M}$; $1 \to \widehat{F} \to \widetilde{M}^! \to C \to 1$.	§46, scan183 (9bis).
$\widetilde{\Gamma}$; §45 $\widetilde{N}_{0,3}$	The quotient by $\widehat{B}$, hence C, before the renaming in §46. It must not be confused with §46's $S_3 \times C$.	§45, fol.384, scan170 (1); scans177–178 (20)–(21).

Our notation / his notation	Definition and present interpretation	Manuscript location (cote 140-3 unless indicated)
$\widetilde{N}_{1,1}$	Pullback of $\mathrm{GL}_2(\widehat{\mathbb{Z}}) \to \mathrm{GL}_2(\widehat{\mathbb{Z}})/\{\pm I\}$ along the claimed congruence homomorphism from $\widetilde{M}$; its geometric kernel is $\widehat{\mathrm{SL}_2(\mathbb{Z})}$.	§45, fols.387–388, scans176–178 (19)–(20).
$\widetilde{M}^![0]$	Normalizer of $\overline{\langle \ell_0 \rangle}$; kernel of its map to C is that procyclic group. Round brackets (0) denote a chosen section, after normalization; square brackets do not.	§46, fols.404–413, scans210–227.
$\widetilde{M}^!(1/2)$, (2), (-1)	Centralizers of the respective lifts $\sigma_\infty, \rho\sigma_\infty\rho^{-1}, \rho^{-1}\sigma_\infty\rho$; the latter two sections are conjugates of the first.	§46, fols.402–405′, scans206–213.
$\widetilde{M}^!(-\bar{j})$	Centralizer of ρ above the subgroup $\mu \equiv 1 \pmod 3$; a partial section, not a section over all C.	§46, fol.403′, scan209 (106)–(108).
$g, h, f_G; \alpha_0, \beta_0$	$u(\rho) = g\rho$, $u(\sigma) = h\sigma$, $u(\varepsilon_0) = f_G^{-1}\varepsilon_0^{\mu} f_G$; $g = \alpha_0\rho(\alpha_0)^{-1}$, $h = \beta_0\sigma(\beta_0)^{-1}$.	§46, fols.392–393, scans186–188.
$\alpha, \beta; \xi, \eta$	$\alpha = f_G\alpha_0$, $\beta = f_G\beta_0$; $\xi = \alpha\rho(\alpha)^{-1}$, $\eta = \beta\sigma(\beta)^{-1}$. Simultaneous left multiplication of α, β, f_G by ℓ_0^n is the gauge ambiguity.	§46, fols.393–394, scans189–190 (35)–(46).
ζ	Not a further independent unknown in the checked §46 system. In §44 the readings $\xi \mid \zeta$ refer to a cocycle letter, whose coordinate convention changes by $\underline{\xi} = \ell_0^{\nu}\xi\ell_1^{-\nu}$.	§44, scans151–153; §46, fols.397–397′, scans196–197.
$C^{\varphi} = \widetilde{N}_{0,3}^{!\approx}$	Classes in C compatible, up to a specified inner conjugacy in $\widehat{B}$, with $\varphi : \widehat{F} \twoheadrightarrow \widehat{B}$, $(\ell_0, \ell_1, \ell_\infty) \mapsto (\sigma\rho, \sigma, \rho^{-1})$.	§47, fols.415–420, scans229–240.
$\widetilde{M}^{!\approx}, \widetilde{N}^{\approx}, \widetilde{M}^{\approx}$	Inverse images of C^{φ}, $S_3 \times C^{\varphi}$, respectively, in the preceding exact sequences. The double mark imposes an additional compatibility.	§47, fols.420–421, scans239–242 (24)–(39).

A.1. Refuted; corrected Legendre identification proved [Refuted]. *The literal identifications* $M_{1,1}[2] \simeq U_{0,3}$ *and* $\Pi_{0,3} \simeq \mathfrak{S}_{0,3}^{+}$ *in the development prompt are false. Over* $S = \operatorname{Spec}\mathbb{Z}[1/2]$ *the correct first statement is*

$$\mathcal{M}_{1,1}[2] /\!\!/ \mu_2 \simeq U = \mathbb{P}^1_S \setminus \{0, 1, \infty\} = \mathcal{M}_{0,4}, \qquad \mathcal{M}_{1,1}[2] \simeq U \times B\mu_2.$$

Here $/\!\!/$ *denotes rigidification by the central automorphisms* $\{\pm 1\}$*; we use this symbol only in this display. The correct second statement is* $\Pi'_{0,3} \simeq \mathfrak{S}_{0,3}^{+}$*, with a prime. These distinctions are already explicit in the manuscript: cote 140-2, §37, fols.279–280, scans547–549, and cote 140-3, §38, fol.297, scan18 (24)–(25).*

Proof. An ordered full level-two structure labels the three nonzero two-torsion sections. The quotient $E/\{\pm 1\}$ is a projective line bundle with four ordered branch sections. Three of them

uniquely normalize its coordinate to $0, 1, \infty$; the fourth is a section λ of U. Conversely the double cover

$$E_\lambda : \quad v^2 = t(t-1)(t-\lambda)$$

with its point at infinity and its labeled points $(0,0), (1,0), (\lambda,0)$ is an elliptic curve over U: its discriminant is $16\lambda^2(1-\lambda)^2$, a unit. Every cover with these four ordered branch sections is an etale-local quadratic twist of this one. Its automorphisms over the ordered branch line are exactly the two deck transformations. Descent therefore identifies the groupoid over λ with the groupoid of μ_2-torsors, compatibly with base change. The exhibited family neutralizes the gerbe. Nontrivial inertia μ_2 proves that the unrigidified stack cannot be the scheme U. In characteristic two the four-point etale branching construction is unavailable, so extending this identification over $\mathbb{Z}$ would change the statement.

The tripod retracts onto a bouquet of two circles, hence its topological fundamental group is free of rank two and torsion-free (a nonempty cyclically reduced word has nonempty positive powers). The orbifold obtained by filling two of its punctures with stabilizers of orders two and three has presentation $\langle \ell_0', \ell_1', \ell_\infty' \mid \ell_\infty'\ell_1'\ell_0' = 1, (\ell_1')^2 = (\ell_\infty')^3 = 1\rangle$. Eliminating ℓ_0' gives $C_2 * C_3$, which contains nontrivial torsion. Thus it cannot be F. The exact quotient is $F/\langle\langle x^2, z^3\rangle\rangle$, after the change of labels above. □

A.2. the modular and reflection avatars [Proved]. *There are isomorphisms of discrete groups*

$$B = C_2 * C_3 \simeq \mathrm{PSL}_2(\mathbb{Z}), \qquad B^{\pm} \simeq \mathrm{GL}_2(\mathbb{Z})/\{\pm I\}, \qquad 1 \longrightarrow F \longrightarrow B \longrightarrow S_3 \longrightarrow 1.$$

The last map is reduction modulo two, and the indicated F is free on ℓ_0, ℓ_1. On complex orbifolds, B is the orbifold fundamental group of $[U/S_3]$, or of the rigidified moduli stack of elliptic curves; $B^{\pm}$ is its extension by complex conjugation. These identifications account for the reflection, cartographic, modular, and universal-surface avatars of §42, fols.336′–337, scans95–96. They induce isomorphisms of full profinite completions, and the index-six exact sequence remains exact after completion.

Proof. The matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

satisfy $S^2 = R^3 = -I$. The Euclidean algorithm shows that S and $SR = \left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$ generate $\mathrm{SL}_2(\mathbb{Z})$. The usual modular triangle in the upper half-plane has one ideal vertex and angles $\pi/2, \pi/3$; reflecting in its sides tiles the half-plane with disjoint interiors. Crossing a side records a reflection, and canceling a backtrack or a complete turn at either finite vertex gives precisely

$$r_0^2 = r_1^2 = r_\infty^2 = (r_0 r_\infty)^2 = (r_1 r_0)^3 = 1.$$

A reduced path through triangle interiors cannot close without these elementary cancellations: remove an outermost triangle in a finite disk bounded by the path, and induct on the number of triangles. This proves the presentation, rather than only a surjective matrix representation. Its even subgroup is $C_2 * C_3$. The reflection matrices can be taken as $r_0 = SJ$, $r_1 = R^{-1}r_0$, $r_\infty = J$, projectively.

The quotient $U \to U/S_3$ is the map

$$q(t) = \frac{27}{4}\frac{t^2(t-1)^2}{(t^2-t+1)^3}.$$

It has ramification orders two and three over $1, \infty$, and a missing cusp over 0. Local lifting of paths across a branch point replaces its loop by the corresponding order-two or order-three orbifold generator. This establishes the covering equivalence and the fundamental-group exact sequence. Its kernel is the fundamental group of U, hence is F_2. Directly, its peripheral generators are

$$L_0 = (SR)^2 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}, \quad L_1 = (RS)^2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad L_\infty = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}.$$

Their product $L_\infty L_1 L_0 = -I$ is the identity in B, not in $\mathrm{SL}_2(\mathbb{Z})$. Marking a fourth point on the three-punctured sphere identifies its moduli parameter with U; allowing permutations of the other three points gives the same quotient orbifold. An oriented marked torus has mapping class group $\mathrm{SL}_2(\mathbb{Z})$, as seen by straightening a meridian and longitude and then the complementary disk; quotienting the central elliptic involution gives B. Allowing orientation reversal gives $B^{\pm}$. Thus these interpretations preserve the named peripheral classes.

For a finite-index subgroup $F \subset B$, every finite-index normal subgroup of F contains the core in B of a finite-index subgroup, so the topology induced from B is its full profinite topology. Taking inverse limits gives the claimed exact sequence. No congruence-subgroup assertion is used. □

A.3. which group is $\widehat{\mathrm{GT}}_0$ [Known]. *For the §46 definitions,*

$$\widetilde{N}^! \simeq \widehat{\mathrm{GT}}_0, \quad \widetilde{N} \simeq S_3 \times \widehat{\mathrm{GT}}_0, \quad 1 \to \widehat{F} \to \widetilde{M}^! \to \widehat{\mathrm{GT}}_0 \to 1,$$

and $1 \to \widehat{B} \to \widetilde{M} \to \widehat{\mathrm{GT}}_0 \to 1$. *In particular, neither genuine-automorphism group $\widetilde{M}$ nor $\widetilde{M}^!$ is being identified with the pair group. The identification of the outer centralizer is Harbater and Schneps (2000), Main Theorem (a), or their §1.3, Theorem 2 (Harbater and Schneps, 2000); the reduction to their hypotheses follows below.*

Manuscript question: §45; fols384–388; scans170–178; §46; fols390–391; scans182–184.

Proof. Let O denote outer loop automorphisms, allowing permutation of the three peripheral classes and a common unit exponent. The geometric $S_3 \subset O$ maps isomorphically to the permutation group of these classes. If v normalizes this S_3, its conjugation on S_3 is inner: an automorphism of S_3 permutes its three order-two subgroups, and this embeds $\mathrm{Aut}(S_3)$ into their permutation group, already exhausted by conjugation. Since $Z(S_3) = 1$, there is a unique $s \in S_3$ such that $s^{-1}v$ centralizes S_3. Hence its normalizer is $S_3 \times C_O(S_3)$. An element centralizing S_3 has a peripheral permutation in $Z(S_3)$, so preserves labels. Under $\pi_1(\mathcal{M}_{0,4}) = F_2$, this is exactly the quasi-special symmetric group $\mathrm{Out}_4^{\#}$ of Harbater–Schneps: the S_4 action on the cross-ratio line factors through $S_4/V_4 = S_3$, and all their inertia classes are the three peripheral classes. Their theorem therefore identifies $C_O(S_3)$ with $\widehat{\mathrm{GT}}_0$. Statement B.m2 supplies the explicit coordinate conversion and checks both directions of the equations. Taking inverse images in genuine automorphisms gives the first extension; taking the inverse image of the geometric S_3 gives $\widehat{B}$, by the preceding based orbifold construction and the trivial center of $\widehat{F}$. Quotienting gives the second extension. This also explains the change of notation between §45 and §46. □

A.4. finite checks and the P060 alternatives [Proved]. *The displayed matrix relations have been checked exhaustively at levels* $2, 3, 4, 6, 12$ *(and additionally* $5, 7, 8, 24$*) by* `code/dictionary_matrices.py`. *The distinction $L_\infty L_1 L_0 = -I$ persists at every level except two. At level two S and R generate S_3, while the image of F generated by L_0, L_1 is trivial. Thus calling $\mathrm{SL}_2(\mathbb{Z}/2)$ a quotient of F via these particular matrices would lose the actual map.*

For P060 scan18, the first and second returns give terminal factors $\tau'_\infty \tau'_0$ and $\tau'_0 \tau'_\infty$ in the expression for ℓ_1. Both yield the same element, since $(\tau'_0 \tau'_\infty)^2 = 1$ and both factors are involutions.

Mathematics does not choose between those two readings. In the same scan, $(\ell_1')^2 = 1$ *is correct, whereas* $(\ell_\infty')^2 = 1$ *is false: the latter generator has order three. On scan19 the avatar is a genuine automorphism group, not the outer group (which in the discrete tripod case is finite). Scan20's "rank one" is incompatible with* $\Im\tau \neq 0$*; the lattice has rank two. These conclusions are editorial evidence; they do not replace the scan readings.*

Manuscript question: §38–39; fols297–298; scans18–20; §42; fol336; scan94.

Proof. The polynomial matrix multiplications used in Statement A.2 establish the identities over $\mathbb{Z}$, so reduction proves their validity at every finite level. The program independently forms the finite generated groups and tests each identity, including the central signs, by direct multiplication. For the two scan18 words, swapping the last two factors is allowed by their proved commutation, giving equality before any completion. A nonreal τ makes $1, \tau$ linearly independent over $\mathbb{R}$, so the integer span is a discrete rank-two subgroup of $\mathbb{C}$; it cannot have rank one. □

A.5. All ten avatars, with the fixed fourth puncture retained [Known]. *The ten items on §42, fols.336′–337, scans95–96, give the following realizations of* $B^{\pm}$*; taking orientation-preserving morphisms gives* B *in each row. A conformal groupoid in this table permits antiholomorphic morphisms. Ordinary holomorphic orbifold fundamental groups give only the oriented subgroup.*

Item	*Precise realization of* $B^{\pm}$
1	$\mathrm{Aut}_{\mathrm{lac}}(F)$*, genuine discrete loop automorphisms.*
2	*Peripheral outer automorphisms of the four-punctured sphere fixing the fourth peripheral class and permuting the first three.*
3	*Components of the homeomorphism group of that sphere fixing the fourth puncture.*
4	*Fundamental group of the quotient groupoid* $[U/(S_3 \times \langle \textit{conjugation} \rangle)]$.
5	*Extended orbifold group of the sphere with one cusp and orbifold orders two and three.*
6	*The three-reflection cartographic group with the presentation in A.2.*
7	*Fundamental group of the universal conformal tripod, whose total groupoid is the quotient in item 4.*
8	$\mathcal{T}_{1,1}/\{\pm 1\}$*, the torus mapping class group after quotienting its center.*
9	*Fundamental group of the rigidified conformal elliptic moduli groupoid* $[\mathbb{H}/(\mathrm{GL}_2(\mathbb{Z})/\{\pm I\})]$.
10	$\mathrm{GL}_2(\mathbb{Z})/\{\pm I\}$.

The topological input for items 2–3 is the punctured Dehn–Nielsen–Baer theorem and the Birman exact sequence: Farb–Margalit, version 5.0, Theorems 8.8 and 4.6, which we opened and checked in the hyperbolic cases used here. We give the reduction so that a full four-point mapping class group is not substituted for its fourth-point stabilizer.

Proof. For item 1, a discrete loop automorphism has a peripheral permutation and a common orientation sign. After removing representatives of these in the reflection group, it preserves the three peripheral conjugacy classes with positive sign. Conjugate to make it fix x; write its image of y as gyg^{-1}. It preserves the conjugacy class of xy. Choose a shortest representative of the double coset $\langle x \rangle g \langle y \rangle$. If nonempty, it begins with a y-letter and ends with an x-letter. Then $xgyg^{-1}$ is cyclically reduced of length $2 + 2|g| > 2$, contradicting conjugacy to xy. Thus $g = x^m y^n$ before minimization, and the automorphism is inner. This proves that the outer group is precisely $S_3 \times C_2$. Conjugation from $B^{\pm}$ has that quotient and geometric kernel F. Its kernel is trivial, since a matrix commuting projectively with both L_0, L_1 is scalar. Hence it is the required genuine automorphism group.

For items 2–3, Dehn–Nielsen–Baer applies to the hyperbolic four-punctured sphere and respects the peripheral permutation, so restricts to the subgroup fixing the fourth puncture. The Birman sequence for adding that point to the hyperbolic tripod has kernel $\pi_1(U) = F$. Point pushing acts on the tripod group by the corresponding inner automorphism. The resulting homomorphism to item 1 is the identity on F and on the outer quotient $S_3 \times C_2$; exactness makes it an isomorphism. This verifies every hypothesis and the precise fixed-label restriction of the imported theorems (Farb and Margalit, 2011, Theorems 4.6 and 8.8).

Items 4–7 are the quotient orbifold, its covering presentation, and its chamber presentation proved in A.2. The universal family of a tripod has no remaining complex moduli, so its total groupoid is the tripod with its conformal symmetry group acting; this identifies item 7 with item 4. For items 8–9, the marked elliptic Teichmüller space is the contractible upper half-plane, and change of an integral homology basis acts by the matrices and the conjugation rule of scan92 (37). The orbifold universal cover therefore has deck group $\mathrm{GL}_2(\mathbb{Z})$; rigidification removes exactly the central elliptic involution. This gives item 10 and its oriented subgroup, as asserted. None of these comparisons asserts a congruence-completion isomorphism. □

B. The equations, their graded forms, and finite tests

The exact §46 equations and the conversion to Drinfeld coordinates

B.m1. the loop equation, with its generation condition [Proved]. *Let $\widehat{B} = \widehat{C_2 * C_3}$, with generators $s = \sigma_\infty$, $r = \rho$, and let $\widehat{F}$ be its level-two kernel. Put $e = sr$, $y = e^2 = \ell_0$, $x = (rs)^2 = \ell_1$. For $\mu \in \widehat{\mathbb{Z}}^\times$ put $\nu = (\mu - 1)/2 \in \widehat{\mathbb{Z}}$. A triple $h, g, f_G \in \widehat{F}$ specifies a loop automorphism inducing the identity on S_3 by*

$$u(s) = hs, \quad u(r) = gr, \quad u(e) = f_G^{-1} e^\mu f_G$$

if and only if

$$h\sigma(h) = 1, \qquad g\rho(g)\rho^2(g) = 1,$$
$$\sigma(h)g = \sigma(f_G)^{-1} x^\nu \rho(f_G),$$

and hs, gr topologically generate $\widehat{B}$. Here f_G is defined modulo left multiplication by y^n, $n \in \widehat{\mathbb{Z}}$. The generation condition is part of the theorem; it is not a consequence claimed from the displayed identities.

Equivalently, after composing with $\mathrm{Int}(f_G)$*, one obtains $u_0(s) = \eta s$, $u_0(r) = \xi r$ with*

$$\eta\sigma(\eta) = 1, \qquad \xi\rho(\xi)\rho^2(\xi) = 1, \qquad \xi = \eta x^\nu. \tag{1}$$

The gauge transformation is

$$(\xi, \eta, f_G) \longmapsto (y^n \xi x^{-n}, y^n \eta x^{-n}, y^n f_G).$$

These are §46, fols.392–394 and 407–408, scans186–190 and 215–217, equations (25), (29), (44), (46), (133), (134), (144).

Proof. The universal property of a free profinite product makes $s \mapsto hs$, $r \mapsto gr$ an endomorphism precisely when $(hs)^2 = (gr)^3 = 1$, which expands into the two cocycles. It induces the identity modulo $\widehat{F}$. Its image is the closed subgroup generated by its two indicated images. A surjective endomorphism of a finitely generated profinite group is injective: for each n there are only finitely

many open normal subgroups of index at most n, and the inverse-image map permutes this finite set; an element in the kernel lies in all of them. This proves the role of the generation condition.

Expanding $u(sr) = f_G^{-1}(sr)^\mu f_G$ gives $f_G h\sigma(g\rho(f_G)^{-1}) = y^\nu$. Apply σ and use $\sigma(y) = x$ to obtain the asserted equation. Set $\eta = f_G h\sigma(f_G)^{-1}$ and $\xi = f_G g\rho(f_G)^{-1}$. The two cocycles telescope under this change, and the loop equation becomes $\xi = \eta x^\nu$. Conversely these substitutions recover g, h and all the original equations. Finally $\rho(y) = \sigma(y) = x$ proves the gauge formula. Centralizers of the free generators are their procyclic closures, so the freedom in the conjugator for y^μ is exactly the stated y^n. □

B.m2. An explicit equivalence with (I) and (II) [Proved]. *Modulo the gauge above, solutions of* (1) *satisfying the generation condition are in bijection with the pairs* (λ, f) *defining* $\widehat{\mathrm{GT}}_0$*, with* $\lambda = \mu$ *and*

$$f \in \widehat{F}', \quad f(y,x)f(x,y) = 1, \quad f(z,x)z^\nu f(y,z)y^\nu f(x,y)x^\nu = 1.$$

The pair acts by $x \mapsto x^\mu$, $y \mapsto f^{-1}y^\mu f$ *and is required to be an automorphism. If the abelianization of* ξ *is* $a[x] + b[y]$*, the exact change of variables is*

$$a + b = \nu, \qquad f = y^{-b}\xi x^{-a}. \tag{2}$$

An inverse is $\xi = y^\nu f$, $\eta = y^\nu f x^{-\nu}$.

Manuscript question: §46; fols393–394,407–408; scans189–190,215–217.

Proof. Abelianize $\eta = \xi x^{-\nu}$ and the σ-cocycle, where σ interchanges x, y. Its sum is $(a + b - \nu)([x] + [y]) = 0$, so $a + b = \nu$. Thus (2) belongs to the closed commutator subgroup. Under gauge by y^n, the exponents become $(a - n, b + n)$ and the expression for f stays unchanged. Choose $n = a$, giving $\xi = y^\nu f$, as claimed.

Substitution into the first equation gives

$$\eta\sigma(\eta) = y^\nu f(x,y)f(y,x)y^{-\nu},$$

so it is equivalent to (I). Since $\rho(x,y,z) = (z,x,y)$, substitution into the second gives

$$y^\nu f(x,y)x^\nu f(z,x)z^\nu f(y,z) = 1.$$

Moving its first two factors cyclically to the end gives exactly (II). A cyclic permutation of a product is conjugate to that product; thus this step is an equivalence, not an implication after abelianization.

The automorphism u_0 fixes y to y^μ. Since $x = \rho(y)$, it sends x to $\xi x^\mu \xi^{-1}$. Consequently $\mathrm{Int}(\xi^{-1})u_0$ restricts to $x \mapsto x^\mu$, $y \mapsto \xi^{-1}y^\mu\xi = f^{-1}y^\mu f$. Conversely a pair satisfying (I),(II) makes (1) define an endomorphism of $\widehat{B}$. Its restriction to $\widehat{F}$ is conjugate to the stated automorphism and its quotient on S_3 is the identity; it is therefore surjective and hence an automorphism. This proves both directions, including the generation condition. The change of variables in arbitrary gauge is implemented on genuine automorphisms by $\mathrm{Int}(x^a\xi^{-1}f_G)u$. □

This proof is independent of any proposed graded identification. It agrees with Harbater–Schneps (2000), §1.2 Proposition 2 and §1.3 Propositions 3–4, Theorem 2, which we read. Their normalized lift supplies the same two-cycle and three-cycle relations. There is no five-point configuration in the proof.

B.m3. the additional double-marked condition has separate content [Proved]. *The §47 condition is compatibility with the specific continuous surjection*

$$\varphi : \widehat{F} \twoheadrightarrow \widehat{B}, \qquad (y,x,z) \longmapsto (sr, s, r^{-1}).$$

For a representative u_0 fixing e to e^μ, strict commutation $\varphi \circ (u_0|_{\widehat{F}}) = u_0 \circ \varphi$ is equivalent to

$$\xi = (\varphi(\xi)\sigma(\varphi(\xi))^{-1})x^\nu. \tag{3}$$

In particular, deriving (I),(II) from §46 does not prove that every element of $\widehat{\mathrm{GT}}_0$ satisfies §47. This is the precise extra condition defining C^φ in the dictionary. The passage is §47, fols.416–420, scans231–240, especially (14),(15),(20),(30).

Proof. The elements x, y freely generate $\widehat{F}$, so equality of the homomorphisms need only be checked there. For y, the two images are e^μ. For x, the two images are $\varphi(\xi)s^\mu\varphi(\xi)^{-1}$ and ηs. Since s has order two and μ is odd, equality is exactly $\eta = \varphi(\xi)\sigma(\varphi(\xi))^{-1}$. Use $\xi = \eta x^\nu$ to get (3). The subgroup of strictly commuting representatives is closed under composition and inversion because this is an equality of typed homomorphisms. Outer commutation allows a further conjugating element in $\widehat{B}$; it is a different assertion until a normalization has been made. □

B.m4. Proved in the stated representatives; global classification open [Proved]. *The three fundamental cases of §46 are: $u(\sigma) = \sigma$, equivalent to $h = 1$ and, when a coboundary conjugator is chosen, $\beta = f_G$; $u(\rho) = \rho$, equivalent to $g = 1$ and $\alpha = f_G$; and normalization of $\langle \ell_0 \rangle$, equivalent to taking $f_G = 1$ after the allowed change of representative. If $K_{1/2}$ is the centralizer section, then*

$$K_2 = \rho K_{1/2}\rho^{-1}, \qquad K_{-1} = \rho^{-1}K_{1/2}\rho$$

with the same formulas for the homomorphisms after identifying their common outer quotient. The order-three centralizer can project only to $\mu \equiv 1 \pmod 3$.

Manuscript question: §46; fols402–405 verso; scans206–213.

Proof. The equivalences follow by substituting the definitions of g, h, α_0, β_0. The cyclic action on the three fixed points is the same action that permutes the three puncture labels, so conjugating the subgroup and its section gives the two displayed sections; no choice of a new path is made. The abelianization of $\widehat{B}$ is $C_2 \times C_3 = C_6$. A loop automorphism sends the image of e to its μth power. Its C_3 coordinate is the coordinate of r. If $u(r) = r$, that coordinate is unchanged, forcing $\mu \equiv 1 \pmod 3$. Conversely, the required adjustment to a centralizing representative is a nonabelian H^1 problem, treated with its ambiguity in Part D; a matrix cocycle alone does not prove that adjustment. □

B.m5. A conjugator in the smaller group does not define an outer condition [Refuted]. *The literal condition on §47, fol.415, scan229 (4), that the conjugator Λ for $\varphi u = \mathrm{Int}(\Lambda)u\varphi$ belong to $\widehat{F}$, is not invariant under changing a representative by $\mathrm{Inn}(\widehat{F})$. It cannot, with that literal target, define the claimed inverse image of an outer subgroup containing every geometric inner automorphism. The correct ordinary outer-compatibility condition allows $\Lambda \in \widehat{B}$. The stronger condition $\Lambda \in \widehat{F}$ is a separate relative condition. The margin on scan230 already notices that $\varphi(f)$ need not lie in $\widehat{F}$.*

Proof. Take $u = \mathrm{Int}(x)$, where $x = \ell_1 \in \widehat{F}$ and $\varphi(x) = s$. Then $\varphi u = \mathrm{Int}(s)\varphi$ and $u\varphi = \mathrm{Int}(x)\varphi$. Surjectivity of φ and $Z(\widehat{B}) = 1$ imply that the only possible conjugator is $\Lambda = sx^{-1}$. Its reduction in $B/F = S_3$ is the nontrivial transposition s, so it is not in $\widehat{F}$. The identity representative of the same outer class has conjugator $1 \in \widehat{F}$. This is a complete counterexample to representative invariance; the program checks the obstructing reduction modulo two. Allowing $\Lambda \in \widehat{B}$ removes precisely this obstruction, since inner changes on either side merely change a target inner

conjugator. Statement B.m3 uses strict commutation for an explicitly normalized representative, and remains valid without either ambiguity. □

B.m6. The alternate §47 formulas and their finite labels [Proved]. *Assume the system of B.m1, the generation condition, and strict φ-compatibility for u_0. Put $a_3 = (1-\varepsilon_3(\mu))/2$ and $\widetilde{\alpha} = \alpha J^{a_3} \in \widehat{F}$. There are unique $i \in \mathbb{Z}/2$ and $j' \in \mathbb{Z}/3$ such that*

$$\varphi(\xi) = \beta s^i, \qquad \varphi(\rho^{-1}(\xi)) = r^{j'} s^{a_3} \widetilde{\alpha}^{-1}.$$

Conversely either displayed compatibility formula, together with the loop system and its generation condition, implies the other. Reduction in $B/F = S_3$ gives exactly (51) of scan247 and uniquely determines i, a_3, j' from these two reductions. Moreover $i \equiv \nu \pmod 2$, so $i = (1-\varepsilon_2(\mu))/2$; no analogous determination of j' from μ is asserted here.

When $\mu \equiv 1 \pmod 3$, the full compatible loop system can equivalently be written as the conjunction

$$\xi = w\rho(w)^{-1}, \qquad w = \varphi(\rho^{-1}(\xi^{-1})),$$
$$\xi = \varphi(\xi)\sigma(\varphi(\xi))^{-1} x^{\nu},$$

again retaining the generation condition. These are (57),(58), §47, fols.423–425, scans245–250. The reading "l'ensemble" on scan250 is supported by this equivalence. We do not replace it by "l'une" without specifying additional standing hypotheses.

Proof. Statement B.m3 says that checking strict compatibility on the free generator x is equivalent to $\varphi(\xi)s\varphi(\xi)^{-1} = \beta s \beta^{-1}$. The centralizer of s in $\widehat{B}$ is $\langle s \rangle$, so this is equivalent to the first displayed formula, with unique i.

One may instead check on the free basis (y, z). The y equation holds by normalization. The formula $z = r^{-1} y r$ gives $u_0(z) = \mathrm{Int}(\rho^{-1}(\xi^{-1}))(z^{\mu})$. Write $v = \varphi(\rho^{-1}(\xi))$. Compatibility on z, using $\varphi(z) = r^{-1}$, is $v\alpha r \alpha^{-1} v^{-1} = r^{\mu} = r^{(-1)^{a_3}}$. The element $\kappa = sJ$ inverts r. Hence $\kappa^{a_3} v \alpha$ centralizes r, so equals r^j for a unique $j \in \mathbb{Z}/3$, by the free-product centralizer theorem used in C.d1. Solving and setting $j' = (-1)^{a_3} j$ yields $v = r^{j'} s^{a_3} (\alpha J^{a_3})^{-1}$, as claimed. This proves the equivalence by checking the two free bases, with all comparison maps specified. Reduction gives the stated finite formulas because $\widetilde{\alpha}, \beta$ belong to $\widehat{F}$. To determine i, the sign character of S_3 sends both $\varphi(x)$ and $\varphi(y)$ to -1. Thus $(-1)^i = \mathrm{sgn}(\varphi(\xi)) = (-1)^{a+b} = (-1)^{\nu}$, by B.m2. In the unnormalized formula $\varphi(\xi) = e^r \beta s^i$, the same argument gives $i + r \equiv \nu \pmod 2$; setting $r = 0$ by gauge gives the stated value.

If $a_3 = 0$, the second formula says $w = \alpha r^{-j'}$; taking its ρ-coboundary gives the first of (57),(58). The second is B.m3. Conversely the first is a coboundary and therefore satisfies the ρ-cocycle by telescoping. The second makes $\eta = \varphi(\xi)\sigma(\varphi(\xi))^{-1}$ a σ-coboundary and supplies the loop relation. B.m1 with the generation condition reconstructs the automorphism, and B.m3 proves strict compatibility. Thus the conjunction is exactly sufficient as well as necessary. □

B.m7. Complex conjugation is self-centralizing already in $\widehat{\mathrm{GT}}_0$ [Known]. *Let $c = (-1, 1)$ in the normalized pair group. Then $C_{\widehat{\mathrm{GT}}_0}(c) = \{1, c\}$. The profinite fixed-subgroup input is $\widehat{F}^c = 1$ for $c(x) = x^{-1}, c(y) = y^{-1}$, Lochak–Schneps (1997), Proposition 4(i); their Proposition 4(ii) proves self-centralization in $\widehat{\mathrm{GT}}$ (Lochak and Schneps, 1997). We opened both statements and their proof in §2, printed p.581; the latter uses relation (II) and no pentagon. We derive the larger-group assertion using the genuine loop equations, without the pentagon. This proves the proposition of §47, fols.426–427, scans252–253, and corrects its final $\nu = \pm 1$ to $\nu \in \{0, -1\}$.*

Proof. One can identify the fixed-subgroup input with the finite-factor centralizer theorem: the semidirect product $\widehat{F} \rtimes \langle c \rangle$ has the free profinite product presentation $C_2 \amalg C_2 \amalg C_2$, on c, xc, yc. The centralizer of the first factor's nontrivial element is that factor; its intersection with $\widehat{F}$ is trivial. This is the same standard profinite centralizer input isolated in D.3; Proposition 4(i) of the opened primary article gives the needed consequence directly.

For a normalized pair (μ, f) commuting with c, compare their genuine normalized representatives. On y this gives conjugation of $y^{-\mu}$ by $c(f)$ and by f. Their quotient centralizes y, so is a power of y; its abelianization is zero because $f, c(f) \in \widehat{F}'$. Thus $c(f) = f$, and the fixed-subgroup result forces $f = 1$. The inverse construction in B.m2 first gives $\xi = y^{\nu}$, $\eta = y^{\nu}x^{-\nu}$. Gauge by $y^{-\nu}$ changes these to $\xi = x^{\nu}$, $\eta = 1$, without changing the pair.

For this actual profinite solution, C.d1 supplies the matrix lifts $\eta = I$ and $\xi = \varepsilon L_1^{\nu}$ with one global sign ε. Its trace constraint is $1 = \operatorname{tr}(\xi R) = \varepsilon(2\nu + 1) = \varepsilon\mu$. Hence $\mu = \varepsilon$ is globally 1 or -1, not an independently chosen sign at each prime. These two pairs are precisely $1, c$, and both commute with c. This completes the reduction and the claimed correction of the manuscript's last line. □

The graded loop equations: three spaces that must be distinguished

The manuscript anchors are §41, folio 315 (cote 140-3, scan 54), §42, folio 331 (scan 85), and its explicit-solution question at folio 349 (scan 120). The computations below answer the added graded questions while keeping their profinite lifting problem explicit.

We fix a prime ℓ, put $k = \mathbb{Q}_\ell$, and use positive word weight, so that the corresponding motivic weight is $-2n$. Write $F^{(\ell)} = \langle X, Y \rangle_{\text{pro-}\ell}$, $Z = (XY)^{-1}$, and $\mathfrak{L} = \operatorname{Lie}_k\langle x, y \rangle$, with $x = \log X$, $y = \log Y$ in unipotent completion. In the associated graded Lie algebra, and only there, $z = -x - y$. In the filtered Lie algebra $\log Z = -\operatorname{BCH}(x, y)$. This distinction changes the answer to the dimension question.

B.g1. The precise leading-term map and its limitation [Proved]. *Restrict normalized loop automorphisms to $\lambda = 1$, writing $\alpha_f(X) = X$, $\alpha_f(Y) = f^{-1}Yf$ with f in the closed commutator subgroup. Give this group the filtration in which f has lower-central weight at least n. For $n \geq 2$ its degree-n leading term is*

$$\psi = (\log f)_n \in \mathfrak{L}_n, \qquad D_\psi(x) = 0, \quad D_\psi(y) = [y, \psi].$$

This gives an injective map on the associated graded of the normalized pro-ℓ group, after tensoring with k, into outer special derivations. If the outer automorphism commutes with the geometric $\mathfrak{S}_3$, its image is $\mathfrak{S}_3$-invariant. Its bracket is the Ihara bracket

$$\{\psi, \phi\} = D_\psi(\phi) - D_\phi(\psi) + [\psi, \phi].$$

We do not identify the image of this map with the entire space of invariant special derivations.

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. The graded Lie algebra of a free pro-ℓ group for the closed lower central series is the free Lie algebra over $\mathbb{Z}_\ell$; its rationalization is $\mathfrak{L}$. One may obtain this description from the integral Lyndon basis and the Magnus embedding. Its compatibility with the rational unipotent completion is supplied by Hain and Matsumoto (2003, Theorem A.6 and Corollary A.10). In logarithmic coordinates, $\alpha_f(y) = \exp(-\operatorname{ad}\log f)y = y + [y, \psi] + \text{terms of degree greater than } n + 1$. A Lie element of degree greater than one commuting with y is zero: its associative expansion would be a polynomial in y, whereas the ordinary free Lie algebra on one generator has no such degree. Thus the normalized leading term is recovered from the automorphism. Changing a general lift

of an outer automorphism adds an inner derivation. The normalized lift has no ambiguity in degrees $n > 1$, since a possible inner change must centralize x and hence vanish in that degree.

Preservation of the conjugacy class of Z implies that the leading variation of z is $[c, z]$ for some $c \in \mathfrak{L}_n$. The geometric permutations of X, Y, Z act on the associated graded by permuting x, y, z; all higher terms in $\log Z$ disappear in this leading comparison. Commutation modulo inner automorphisms therefore implies invariance of the outer derivation class. Finally, applying $[D_\psi, D_\phi]$ to y and using Jacobi gives $[D_\psi, D_\phi](y) = [y, \{\psi, \phi\}]$. These arguments establish the inclusion and the bracket; they contain no assertion that an arbitrary graded derivation has a compatible integral lift at every subsequent degree. □

For clarity we define the exact formal system we compute. Let $\mathcal{H}_n \subset \mathfrak{L}_n$ be the solution space of

$$\psi(x, y) + \psi(y, x) = 0, \tag{4}$$

$$\psi(x, y) + \psi(y, z) + \psi(z, x) = 0, \qquad z = -x - y. \tag{5}$$

The normalization $f \in (F^{(\ell)})'$ removes $\mathcal{H}_1$. Let W_n^{lac} denote the verified graded loop-and-symmetry system: the $\mathfrak{S}_3$-invariant outer derivations satisfying $D(x) = [a, x]$, $D(y) = [b, y]$, $D(z) = [c, z]$. This is the complete infinitesimal envelope of loop preservation and commutation with $\mathfrak{S}_3$. Congruence conditions that do not descend to this quotient are not silently inserted into its definition. In particular, W_n^{lac} is not notation for an unproved surjectivity statement about the associated graded of the full profinite solution group.

B.g2. The missing special identity [Proved]. *For $n > 1$, normalized representatives identify W_n^{lac} with the polynomials satisfying (4), (5), and*

$$J(\psi) := [x, \psi(x, y)] - [x + y, \psi(-x - y, y)] = 0. \tag{6}$$

Indeed antisymmetry and (6) already imply (5) in this normalized setting.

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. Subtract $\operatorname{ad} a$ from a special derivation to make its value on x zero. The remaining value on y has the form $[y, \psi]$, uniquely in degree $n > 1$. Let $s(x) = y, s(y) = x$ and $r(x) = y, r(y) = z, r(z) = x$. The difference $sD_\psi s^{-1} - D_\psi$ is inner precisely when $\psi(y, x) = -\psi(x, y)$: its value on y forces the inner conjugator to be ψ, and its value on x gives this equation. Similarly

$$(rD_\psi r^{-1})(x) = -[z, \psi(y, z)], \qquad (rD_\psi r^{-1})(y) = 0.$$

The difference is $\operatorname{ad}\psi$ if and only if $[x, \psi(x, y)] = [z, \psi(y, z)]$, which by antisymmetry is (6). Adding its three cyclic transforms gives $\operatorname{ad}(\psi + r\psi + r^2\psi) = 0$. The free Lie algebra has zero centre, so this is (5). Conversely this same computation gives invariance under s and r. Moreover $[y, \psi] = -[z, \psi + r\psi]$, so $D_\psi(z) = -[y, \psi]$ is tangential at z. Thus the derivation is special as well. □

B.g3. Exact dimensions through weight twelve [Proved]. *The independently recomputed dimensions are as follows. The column $\mathcal{H}_n$ is literally the solution space of the two displayed linear equations, before removing its weight-one line.*

n	$\dim \mathfrak{L}_n$	$\dim \mathcal{H}_n$	$\dim W_n^{\mathrm{lac}}$	d_n
1	2	1	0	0
2	1	0	0	0
3	2	1	1	1
4	3	1	0	0
5	6	2	1	1
6	9	3	0	0
7	18	6	2	1
8	30	10	1	1
9	56	19	4	1
10	99	33	2	1
11	186	62	9	2
12	335	112	7	2

The review's invariant counts in weights one through eight are correct for special outer derivations. Their identification with the solution space of (4)–(5) alone is false. The first excess over d_n is weight four for the normalized $\mathcal{H}_n$, and weight seven for W_n^{lac}.

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. Here is a dimension calculation independent of Gaussian elimination. Let V be the standard two-dimensional representation of $\mathfrak{S}_3$ with coordinates x, y, z, $x+y+z=0$. For $g \in \mathfrak{S}_3$ put

$$q_n(g) = \frac{1}{n} \sum_{d \mid n} \mu(d) \big(\operatorname{tr}(g^d \mid V)\big)^{n/d}.$$

The Witt character formula gives $q_n(g) = \operatorname{tr}(g \mid \operatorname{Lie}_n V)$. It follows by taking the logarithm of the tensor-algebra character identity, or equivalently the PBW product, and applying Möbius inversion. The traces of $1, s, r$ on V are $2, 0, -1$. In a standard representation the s-odd subspace is one-dimensional and $1 + r + r^2$ vanishes; in the sign and trivial representations the cyclic sum does not vanish. Hence

$$\dim \mathcal{H}_n = \frac{q_n(1) - q_n(r)}{3}.$$

For $n > 1$ the conjugator triples form $\mathfrak{L}_n \otimes k\{x, y, z\}$, and the map $(a, b, c) \mapsto [a, x] + [b, y] + [c, z]$ onto $\mathfrak{L}_{n+1}$ is surjective because $[\mathfrak{L}_n, x] + [\mathfrak{L}_n, y] = \mathfrak{L}_{n+1}$. There is no conjugator ambiguity in degree greater than one. Dividing its kernel by the diagonal inner copy of $\mathfrak{L}_n$ gives, as a virtual $\mathfrak{S}_3$-representation,

$$\operatorname{OutSDer}_n = V \otimes \mathfrak{L}_n - \mathfrak{L}_{n+1}.$$

Taking invariants, exact in characteristic zero, gives

$$\dim W_n^{\mathrm{lac}} = \frac{q_n(1) - q_n(r)}{3} - \frac{q_{n+1}(1) + 3q_{n+1}(s) + 2q_{n+1}(r)}{6}.$$

Degree one is handled separately: all special derivations there are inner. Finally PBW for one generator in every odd weight at least three gives $\prod_{n \geq 1} (1 - t^n)^{-d_n} = (1 - t^2)/(1 - t^2 - t^3)$. Substitution yields the table.

The program `graded_dimensions.py` separately constructs standard Lyndon brackets in the free associative algebra. It forms the exact symmetry projection $(1 - s)(2 - r - r^2)$, extracts an independent rational basis, and computes the rational nullspace of (6) on it. A nonzero minor modulo 1000003 proves independence; the character formula supplies the matching characteristic-zero upper bound, so there is no unproved modular-rank inference. The final kernel calculation uses exact rational arithmetic, and agrees in every weight with the formula above. The code, full output, CSV and integer witness coefficients are included. □

Write b_w for the standard Lyndon bracket of a word w in $x < y$; thus $b_x = x, b_y = y$, and $b_w = [b_u, b_v]$ when v is the longest proper Lyndon suffix and $w = uv$.

B.g4. A fourth-weight obstruction to the naive linearization [Refuted]. *The polynomial*

$$q_4 = b_{xxxy} - 4b_{xxyy} + b_{xyyy}$$

satisfies the two-cycle and three-cycle linear equations but is not a leading term of any solution of the genuine group equations (I),(II) over a characteristic-zero field. The failure already occurs in weight five.

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. Direct substitution verifies both linear equations. Its special defect is

$$J(q_4) = 2b_{xxxxy} - 3b_{xxxyy} + b_{xxyxy} - 3b_{xxyyy} - 2b_{xyxyy} + 2b_{xyyyy} \neq 0.$$

Thus B.g1–B.g2 already obstruct an actual loop-and-symmetry lift. For a direct check in Drinfeld's group equations, write $f = \exp(q_4 + q_5 + \cdots)$ and $\log Z = z_0 + z_2 + \cdots$, where $z_0 = -x - y$ and $z_2 = -[x, y]/2$. Through weight five the equation $f(Z, X)f(Y, Z)f(X, Y) = 1$ is linear in $\log f$ because products of terms of weight four have weight at least eight. Its weight-five inhomogeneous term is

$$R_5 = (\partial_1 q_4)(z_0, x)z_2 + (\partial_2 q_4)(y, z_0)z_2.$$

Consequently $q_5 \in \mathfrak{L}_5$ must satisfy $q_5 + sq_5 = 0$ and $(1 + r + r^2)q_5 = -R_5$. The supplied exact program `graded_first_lifts.py` constructs this six-unknown system: its coefficient matrix has rank four, whereas its augmented matrix has rank five. The output includes a left-kernel vector annihilating the matrix and pairing to one with the right-hand side, a complete rational inconsistency certificate. Thus no higher coefficients can repair this leading term. □

B.g5. An explicit seventh-weight loop solution failing the pentagon [Proved]. *The polynomial*

$$\begin{aligned} q_7 = {} & 3b_{xxxyxyy} + 3b_{xxxyyxy} + b_{xxyxyxy} - 3b_{xxyxyyy} \\ & - 3b_{xxyyxyy} - 3b_{xxyyyxy} - b_{xyxyxyy} \end{aligned}$$

belongs to W_7^{lac} but does not satisfy the infinitesimal pentagon. This is an explicit rational graded counterexample; it is not asserted to be an element of $\widehat{\mathrm{GT}}_0$ or of its pro-ℓ image. Moreover $\dim W_7 = 1$ *follows without the 2026 all-weight upper bound.*

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. The polynomial is an integer linear combination of two independently computed basis vectors of W_7^{lac}. The program expands it in associative words and verifies all three displayed linear identities exactly. For a certified nonzero pentagon defect use the following representation of $\mathfrak{t}_4$ on $\mathrm{Lie}\langle a, b, c\rangle$. For $i, j \leq 3$, t_{ij} is the derivation sending $a_i \mapsto [a_i, a_j]$, $a_j \mapsto [a_j, a_i]$ and the third generator to zero, where $(a_1, a_2, a_3) = (a, b, c)$; t_{i4} acts by $\mathrm{ad}\, a_i$. All defining relations hold: for example the derivation t_{ij} fixes $a_i + a_j$, so it commutes with $\mathrm{ad}(a_i + a_j)$; the remaining relations follow by Jacobi or disjoint support. The program also checks all eighteen such relations directly.

Let $P(\psi)$ be the left side minus the right side of the pentagon (1.4) in the prompt. Under this representation, the coefficient of the associative word a^5bcb in $P(q_7)(a)$ is 16. It is therefore nonzero in characteristic zero, and hence the original pentagon defect in $\mathfrak{t}_4$ is nonzero. No

faithfulness of the representation is needed for this implication. `graded_pentagon_witness.py` prints this coefficient and writes the entire image; its exact output is part of the archive. Since $W_7 \subset W_7^{\mathrm{lac}}$ by B.g8, a nonzero pentagon map on the latter two-dimensional space leaves a kernel of dimension at most one. The classical free odd-weight subalgebra of B.g7 supplies dimension at least one. Thus $\dim W_7 = 1$. □

Comparison with the Grothendieck–Teichmüller Lie algebra

B.g6. The typed rational comparison chain [Known]. *Let* $\mathrm{GT}_\ell^{\mathrm{eq}}$ *mean the group of normalized invertible pairs satisfying the pro-*ℓ *forms of (I),(II),(III). This notation is distinct from the image of* $\widehat{\mathrm{GT}}$ *in that group. There are continuous homomorphisms*

$$G_{\mathbb{Q}} \hookrightarrow \widehat{\mathrm{GT}} \longrightarrow \mathrm{GT}_\ell^{\mathrm{eq}} \longrightarrow \mathrm{GT}(\mathbb{Q}_\ell).$$

The last arrow is injective for normalized free-group pairs; no surjectivity of either completion arrow is asserted. On the characteristic-zero algebraic group with $\lambda = 1$*, a choice of associator gives*

$$\mathrm{GRT}_1(k) \simeq \mathrm{GT}_1(k), \qquad \operatorname{gr} \mathfrak{gt}_1(k) = \mathfrak{grt}_1(k).$$

The first isomorphism depends on the associator, while the induced map on the graded Lie algebra is the identity in the standard coordinates. The space W *in the prompt is* $\mathfrak{grt}_1$*.*

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. The arithmetic inclusion and the normalized pair convention are those of Harbater and Schneps (2000, Proposition 1, Proposition 2 and Main Theorem): Belyi supplies injectivity of the outer tripod action, and the normalized representative is unique. Projection of the free profinite group to its characteristic pro-ℓ quotient sends each defining relation to the corresponding pro-ℓ relation and sends an automorphism and its inverse to inverse maps. This defines the middle arrow; it does not show that every solution in the new category comes from a profinite one.

The continuous unipotent comparison is Hain and Matsumoto (2003, Theorem A.6 and Corollary A.10). Its hypotheses hold for the finitely generated discrete free group F_2. The Magnus expansion embeds its free pro-ℓ completion in $1 + \mathbb{Z}_\ell\langle\langle u, v\rangle\rangle$. The expansion can be recovered from all its finite-degree truncations, and scalar extension to $\mathbb{Q}_\ell$ loses no coefficient. Consequently the free pro-ℓ group embeds in its unipotent completion. Equality of two normalized pairs after completion therefore gives equality of their scalar parameters and of f; this proves the stated injectivity.

Drinfeld defines the pro-unipotent group in §4 of Drinfeld (1990). His Propositions 5.1 and 5.5 give the left $\mathrm{GT}(k)$-torsor and right $\mathrm{GRT}(k)$-torsor of associators. Proposition 5.6 is the exact theorem identifying these groups after an associator has been chosen; it also explicitly states $\operatorname{gr} \mathfrak{gt}_1 = \mathfrak{grt}_1$ and that the induced graded map is the identity. The Lie equations are (5.17)–(5.20), and Proposition 5.7 removes the redundant special identity. Thus these are comparisons of proalgebraic groups, their Lie algebras and their weight filtrations, not comparisons of arbitrary finite congruence quotients. □

B.g7. The arithmetic lower bound and its hypotheses [Known]. *Let* $I_\ell^n G_{\mathbb{Q}}$ *be the kernel of the outer action on* $F^{(\ell)}/L^{n+1}F^{(\ell)}$*. The positive graded arithmetic Lie image, tensored with* $\mathbb{Q}_\ell$*, contains the free odd-weight Lie algebra and therefore has dimension at least* d_n *in weight* n*. In the standard mixed-Tate comparison this image is itself freely generated by one element in each odd weight at least three. Hence any correctly derived graded condition on the Galois image must leave at least* d_n *independent solutions. Our table satisfies this test in every computed weight.*

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. The exact results needed from Hain and Matsumoto (2003), *Weighted completion of Galois groups and Galois actions on the fundamental group of* $\mathbb{P}^1 - \{0, 1, \infty\}$, are Theorem 7.3, Theorem 8.4, Lemma 8.5 and Remark 8.6. The action factors through the Galois group unramified outside ℓ, as used explicitly in their §8.4. For this curve $H^1_{\text{et}}(X, \mathbb{Q}_\ell(1))$ is a trivial Galois module because the three punctures are rational. Thus Theorem 8.4 applies: its graded outer Lie image is precisely the rationalization of $\mathrm{gr}^{>0}_{I_\ell} G_{\mathbb{Q}}$. Theorem 7.3 supplies generators in odd weights, initially including weight one; Lemma 8.5 kills weight one in the outer action, and Remark 8.6 identifies the nonzero higher odd abelian classes by Soulé's characters. Hain–Matsumoto alone proves generation, not freeness, a limitation stated explicitly in their introduction.

Brown's *Mixed Tate motives over* $\mathbb{Z}$, Theorem 1.1 and its stated consequence called Conjecture 1 in the introduction, prove faithfulness of the mixed Tate action (Brown, 2012). The unipotent motivic Lie algebra is free on one generator in each odd weight at least three, as recalled in that paper's §1.1. Its action respects the braid associativity and symmetry operations, so its image satisfies the pentagon and the two cycle equations. This gives the free odd-weight subalgebra of W. For the ℓ-adic arithmetic comparison the same generators are identified by their nonzero Soulé abelian classes: the crystalline mixed Tate completion in Hain–Matsumoto, Theorem 9.2, Proposition 9.6 and Theorem 9.8, has the same free graded source with weight one removed. The realization comparison on abelian generators is the regulator $K_{2n-1}(\mathbb{Z}) \otimes \mathbb{Q}_\ell \to H^1_f(G_{\mathbb{Q}}, \mathbb{Q}_\ell(n))$. Its isomorphism is the final assertion of Theorem 9.2, with the crystalline modification stated immediately after Proposition 9.6; the proof explicitly includes $\ell = 2$ via Rognes–Weibel, and odd primes via Soulé. Thus it is an isomorphism on these one-dimensional odd abelian generators. A graded map between free Lie algebras on these generators that is invertible on their abelianizations is surjective degree by degree; their dimensions agree by PBW, so it is an isomorphism. Brown's faithfulness then rules out relations in the arithmetic image. This is also the Deligne–Ihara consequence stated at the beginning of Brown's paper. Applying an actual graded condition to this embedded free subalgebra proves the inequality. This argument uses positive word weight, not Hain–Matsumoto's negative doubled motivic weight without reindexing. □

B.g8. The special identity follows from the pentagon [Proved]. *Over any field of characteristic zero, the two-cycle, three-cycle and infinitesimal pentagon imply* (6). *Thus this part of the loop system is of type (a) in the requested classification; it does not depend on the 2026 upper bound.*

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. We give the reduction rather than only invoke Drinfeld's Proposition 5.7. In degree greater than two forget strand 2 in $\mathfrak{t}_4$ and write $a = t_{12}, b = t_{23}, c = t_{24}, u = t_{13}, v = t_{34}$. The kernel is the free Lie algebra on a, b, c; the base acts by the usual braid derivations. This follows directly from the presentation: t_{ij} acts on the free fibre generators s_i, s_j by $[s_i, s_j], [s_j, s_i]$ and kills the others; these derivations satisfy all the base braid relations. Put $S = a + b + c$ and

$$F_a = \psi(u + b, v) - \psi(u, v), \quad F_b = \psi(u + a, v + c) - \psi(u, v), \quad F_c = \psi(u, v + b) - \psi(u, v).$$

All three belong to this free kernel. The braid relations give

$$[S, F_b] = [a, \Delta_a] + [c, \Delta_c], \qquad \Delta_a = F_b - F_a, \quad \Delta_c = F_b - F_c.$$

The pentagon and its image under exchange of strands $1, 4$ give

$$\begin{aligned} \Delta_a &= \psi(a, b + c) - \psi(a, b) - \psi(b, -b - c), \\ \Delta_c &= -\psi(c, a + b) + \psi(c, b) + \psi(b, -b - a). \end{aligned}$$

Both vanish at $S = 0$. In that quotient the actions of u, v are $\operatorname{ad} c, \operatorname{ad} a$; hence $[b, F_b + \psi(c, a)] = 0$. The degree-$n > 1$ centralizer of b is zero, so $F_b|_{S=0} = -\psi(c, a)$.

Every degree-n Lie polynomial linear in a free direction T and otherwise involving a is a scalar multiple of $\operatorname{ad}_a^{n-1} T$. Write $\partial_2\psi(a, -a)T = k \operatorname{ad}_a^{n-1} T$. Differentiating $\psi(a, -a) = 0$ gives the same expression for $\partial_1\psi(a, -a)T$. Now differentiate the displayed free-kernel identity with $c = S - a - b$, at $S = 0$ in direction T. Substitution of the two expressions for Δ gives

$$-[T, \psi(c, a)] = k(\operatorname{ad}_a^n - \operatorname{ad}_c^n)T + [a, \partial_2\psi(b, a)T] - [c, \partial_2\psi(b, c)T].$$

Set $G(a, b) = [a, \psi(b, a)] + [c, \psi(b, c)]$, $c = -a - b$. The three-cycle and antisymmetry turn its derivative into

$$d_a G(T) = -k(\operatorname{ad}_a^n - \operatorname{ad}_c^n)T.$$

Equality of mixed derivatives in directions U, V, specialized to $a = a_0, b = -a_0, c = 0, U = a_0, V = T$, gives $(n + 1)k \operatorname{ad}_{a_0}^n T = 0$. Indeed the two derivatives of ad_a^n are respectively $n \operatorname{ad}_{a_0}^n T$ and $-\operatorname{ad}_{a_0}^n T$, while the derivatives at $c = 0$ vanish. Thus $k = 0$. All a-derivatives of G vanish; characteristic zero and evaluation at $a = 0$ give $G = 0$. Using antisymmetry, this is $J(\psi) = 0$. Weights one and two vanish for the full system: the degree-one pentagon forces both coefficients to vanish, and the degree-two three-cycle is $3[x, y]$ times its scalar coefficient. This completes the proof, including the low weights. The free-fibre calculation is also the argument of OpenAI, Lemma 2.4; its logic has been reproduced here independently of their all-weight estimate. □

B.g9. The conditional consequence of the 2026 upper bound [Proved]. *Assume OpenAI (2026, Corollary 5.4), namely* $\dim W_n \le d_n$ *in every weight. Then the free arithmetic graded subalgebra in B.g7 equals* $W \otimes \mathbb{Q}_\ell$ *in every weight. Consequently every homogeneous linear relation on that Lie algebra which holds on the arithmetic graded image holds on all of* $W \otimes \mathbb{Q}_\ell$*. When its premise is supplied by the attached manuscript, the equality is labelled* claimed (OpenAI, 2026, unrefereed)*. It does not identify* $G_{\mathbb{Q}}$ *with* $\widehat{\mathrm{GT}}$*.*

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. The injected free subalgebra has dimension d_n in degree n by PBW. The opposite inequality forces equality of each finite-dimensional weight piece. Taking their products gives equality after weight completion because each coefficient of a bracket involves finitely many weight pieces. A linear relation is a linear map out of such a piece; a map vanishing on a spanning subspace vanishes on the whole space. For a claimed nonlinear condition one would additionally need a specified algebraic equation and the appropriate Zariski-density argument; the existence of a tangent space by itself is insufficient. A filtered or graded Lie algebra does not recover the finite, disconnected, torsion or integral data lost under completion, so none of these arguments imply the profinite assertion. □

Audit of the imported upper bound. We opened the attached 41-page manuscript and followed the chain from §2 to Corollary 5.4, rather than using its abstract as a theorem. We also opened the attached manuscript of Willwacher (2026), its introductory statements and the beginning of its alternative construction. The following distinctions record the extent of our check.

1. *Integral reduction (OpenAI, Lemmas 2.1–2.4 and Proposition 2.5).* The centralizer, free-fibre and derivative argument for the special identity is reproduced in B.g8. A rational solution space intersects the finite free $\mathbb{Z}_{(2)}$ Lie lattice in a saturated submodule: if $2v$ belongs, then v belongs to the rational solution space. Thus $L \cap 2K = 2L$, and reduction embeds with dimension unchanged. Integral Lyndon bases and the iterated free-fibre PBW basis make the

two equation targets torsion free. These are exactly the hypotheses required to reduce the rational equations, and we checked them. No claim that the reduction equals *every* solution in characteristic two is used.

2. *The alphabet $A = x^2, C = [x, y], B = y$ (Lemma 3.1).* With $x > y$ the leading words xx, xy, y form a prefix code, proving associative injectivity. Lazard elimination uses $e_j = \mathrm{ad}_x^j y$; in characteristic two $e_{2i} = \mathrm{ad}_A^i B$ and $e_{2i+1} = \mathrm{ad}_A^i C$. We checked the inference that every ordinary Lie element of weight greater than one has this expression. It does not identify x^2 itself as an ordinary Lie element in x, y.
3. *The transfer of the pentagon (Lemma 3.3 and Proposition 3.2).* We independently evaluated all 63 source commutator cases needed for the incident and nonincident index patterns in the free associative fibre over $\mathbb{F}_2$; all agree with the displayed formulas. The ordering argument uses the decreasing pair (number of base letters, number of base–fibre inversions), so its termination does not require preservation of symbolic length. We checked why source corrections increase the u-count, target corrections decrease it, and the common quadratic parts agree. The independent source normal forms come from the prefix code XX, XU_i, U_i. These are the logical points on which the claimed transfer rests. We have not supplied a second all-weight construction of the complete target rewriting calculation: the transferred Proposition 3.2 remains an imported assertion in our conditional use of Corollary 5.4, *claimed, steps not checked by us* beyond the checks just specified.
4. *Deletion and injectivity (§4).* The integer edge valuations use the distinct rational hyperplanes $z_i - z_j = 0$ and $z_i + z_j = 0$ before reduction. They must not be merged in characteristic two. We checked the logarithmic triangle/residue argument for the incidence relations, including repeated letters. Independently, `graded_preprint_audit.py` finds the thirteen codimension-two flats of the B_3 arrangement and verifies 13,104 exact integral deletion incidence relations on all words of lengths zero through five. We followed the minimal-A argument: an e before a nonempty $p, 0$ block cannot be deleted before that block, and its final p or 0 has coefficient zero modulo two. Reversal excludes the remaining terminal-e words. In Proposition 4.4 the two-marked-position test forces all A, B coefficients to agree, and the three-marked-position test kills their common value when there are at least two A's. We checked separately the one-A exceptions $r = 1, 2$. These observations check the stated argument conditional on the transferred relation; bounded deletion tests alone are not an all-weight proof.
5. *The image bound (Lemmas 5.1–5.2, Proposition 5.3).* The e_j expansion makes the number of even elimination letters equal to the B-count. The Ore algebra with $DB = C, DC = 0, X^2 = 0$ is valid because $D^2 = 0$ in characteristic two. We checked the lower-count argument proving $f(C, B + s(C)) = f(C, B)$. The subsequent Vandermonde matrix is inverted over $\mathbb{F}_2(t_0, \ldots, t_{r-1})$, where its determinant is a nonzero polynomial; it is not evaluated on the two-element field, and there is no division by the length r. Divisibility by every $s_{i-1} - s_i$ excludes $g_0 = B$ from the associative expansion in $g_j = \mathrm{ad}_C^j B$; the retraction $g_0 \mapsto 0$ preserves ordinary Lie membership. We checked this logical passage and the finite filtration sum in Corollary 5.4. It remains conditional here on the preceding imported transfer.
6. *Material not imported as checked.* OpenAI's holonomy construction in §§6–8, especially Proposition 7.1, is *claimed, steps not checked by us*; Brown supplies the independent lower bound used here. Willwacher's Theorems 1.1 and 1.3 are *claimed, steps not checked by us*: we have not checked his complete alternative proof. The attached Lean comparator ends with `theorem main : MainStatement := by sorry`; it is a precise statement, not an attached proof certificate. The separate scope note does not change this fact.

Torsion, lifting and what a congruence condition can retain

B.g10. Torsion in the normalized pro-ℓ kernel [Proved]. *The subgroup $\lambda = 1$ of normalized loop-and-symmetry automorphisms of the free pro-ℓ group has no nonidentity element*

of finite order. The same is true in characteristic-zero unipotent completion. For an element of finite order in the full pro-ℓ solution group, its order equals the order of its scalar image in $\mathbb{Z}_\ell^\times$. Thus its possible order divides $\ell-1$ for odd ℓ, and divides two for $\ell=2$. Every divisor of $\ell-1$ is realized in the pro-ℓ arithmetic image when ℓ is odd. Finite subgroups of a fixed order in the full pro-ℓ solution group are then conjugate by its $\lambda=1$ kernel. For $\ell=2$ only the trivial group and order two occur; we make no conjugacy assertion at two. The pair $(-1,1)$ has order two. The odd-prime existence statement does not give explicit normalized coefficients f.

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. The normalized representative is unique by the free-group centralizer argument used in B.g1, so a finite-order outer element has a finite-order normalized representative. Suppose $\alpha\neq 1$ acts trivially on abelianization. The Magnus filtration is separated; take its first nonzero derivation term D in positive degree. Composition adds this term, so the first term of α^m is mD. The module of homogeneous derivations embeds in a finite direct sum of free $\mathbb{Z}_\ell$ Lie pieces and is torsion free. Hence $mD\neq 0$ for every nonzero integer m, contradiction. The same proof works over $\mathbb{Q}_\ell$, or follows from exponential and logarithm for a prounipotent group in characteristic zero. If an element g has finite order and its scalar has order e, then g^e is a finite-order element in the torsion-free kernel, so $g^e=1$. The reverse divisibility is immediate. The torsion units of $\mathbb{Z}_\ell^\times$ are $\mu_{\ell-1}$ for odd ℓ and $\{\pm 1\}$ for two. Finally $X\mapsto X^{-1}, Y\mapsto Y^{-1}$ is an involution and represents $(-1,1)$ in the normalized convention. For completeness, the normalized $\lambda=1$ kernel is a pro-ℓ group. The kernel of the action of $\mathrm{Aut}(F^{(\ell)})$ on the Frattini quotient is pro-ℓ: on each successive finite lower exponent-ℓ central quotient, automorphisms trivial on the preceding quotient differ by homomorphisms to an elementary abelian ℓ-group. Taking inverse limits proves the assertion, and the normalized kernel is a closed subgroup of it. The cyclotomic character of $G_{\mathbb{Q}}$ is onto $\mathbb{Z}_\ell^\times$. For odd ℓ, the inverse image of $\mu_{\ell-1}$ in the arithmetic image is therefore an extension of this finite cyclic group by a pro-ℓ group. It has a complement, unique up to conjugation by that kernel. Here is the requisite form of Schur–Zassenhaus: for a finite ℓ-group kernel, induct on its order using a characteristic elementary abelian subgroup of its centre; on that subgroup the obstruction and ambiguity are $H^2(C_m,V)$ and $H^1(C_m,V)$, both zero since multiplication by m is invertible and averaging contracts the positive-degree cochain complex. Induction gives existence and conjugacy. Passage to the inverse limit follows by compactness of the finite nonempty sets of complements and conjugators. The same argument in the full solution group conjugates any two finite subgroups with the same scalar image. A complement for $\mu_{\ell-1}$ contains its subgroups of every divisor order. At two the displayed complex-conjugation pair supplies the nontrivial allowed order; no coprime argument applies there. Finite nilpotent or finite congruence quotients may acquire torsion because their graded pieces are finite; the torsion-freeness argument does not apply to those quotients. Conversely, these separate pro-ℓ results do not classify torsion in the full profinite group unless a further joint injectivity theorem is supplied. □

B.g11. The remaining infinite lifting problem, with a verified finite jet [Open]. *For the explicit q_7 of B.g5, we construct*

$$F=q_7+F_9+F_{11},\qquad f=\exp F,$$

satisfying the genuine group equations (I),(II) with $\lambda=1$ through weight twelve. The complete rational Lyndon coefficients are supplied: F_9 has 46 nonzero coefficients with common denominator dividing 720, and F_{11} has 160 with common denominator dividing 8640; the even corrections through weight twelve are zero. Reduction gives a solution in the finite nilpotent Lazard group

$$\exp\bigl(\mathrm{Lie}_{\mathbb{F}_{13}}\langle x,y\rangle/\mathfrak{L}_{>12}\bigr),\qquad |G|=13^{747}.$$

Its leading seventh-weight infinitesimal pentagon defect remains nonzero. We have not settled whether q_7 admits compatible coefficients in an actual free pro-ℓ group satisfying (I),(II) in every degree, or whether any such element lifts to $\widehat{\mathrm{GT}}_0$. The first uncomputed rational correction is weight thirteen; for $\ell = 13$ this is also the first class outside the class-less-than-prime Lazard argument used for the finite example.

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof of the finite assertion and record of the obstruction. Put $Z = (XY)^{-1}$ and substitute the actual $\log Z = -\mathrm{BCH}(x, y)$, not $-x - y$. Since F starts in degree seven, every commutator between two values of F has degree at least fourteen. Thus through degree twelve the logarithm of (II) is exactly

$$F(\log Z, x) + F(y, \log Z) + F(x, y).$$

Equation (I) is likewise $F(x, y) + F(y, x) = 0$ in this range. The program `graded_rational_jets.py` computes the required BCH terms, imposes the additional simplifying ansatz that even corrections vanish, and solves all the remaining exact rational constraints while retaining the free parameters from earlier weights. There are 242 unknown Lyndon coefficients, and the full system has rank 229. It then substitutes the resulting F_9, F_{11} back into both identities and checks every associative coefficient through weight twelve. Both defects are zero.

The retained failed greedy computation is relevant to the claimed "successive approximation" argument in the prompt: choosing all free ninth-weight coefficients to be zero after the preceding equations produces an inconsistent tenth-weight system. This is not an obstruction to every lift; preserving those parameters and solving the coupled constraints gives the successful jet just described. Thus unrestricted choices at one stage do not prove an integral infinite lifting theorem.

For the finite example, class 12 is smaller than 13, so the BCH and exponential denominators in this class are units in $\mathbb{F}_{13}$. The displayed rational corrections also have unit denominators at thirteen. The truncated free Lie algebra therefore defines its finite BCH group, and its dimension is $\sum_{n=1}^{12} \dim \mathfrak{L}_n = 747$. The independent program `graded_lazard13.py` repeats the substitutions directly modulo thirteen and verifies both defects are zero. The normalized substitution has identity linear part; solving successively by degree gives its inverse, so it is an automorphism of this finite nilpotent group. The leading pentagon coefficient computed in B.g5 reduces to $16 \equiv 3 \pmod{13}$. This establishes the stated finite equation solution and its infinitesimal pentagon failure without enumerating 13^{747} elements.

An infinite pro-ℓ lift requires compatible integral choices at all further nilpotent and ℓ-power levels. Rational exponential alone does not control denominators in unbounded degree. Moreover a pro-ℓ solution need not lie in the image of the profinite solution group. Those are the precise unproved steps; neither is supplied by our finite computation or by the additive formal exponential of B.g14. □

B.g12. Failure of literal descent of finite congruence maps [Proved]. *A homomorphism $\widehat{F}_2 \to H$ to a finite group factors through the maximal pro-ℓ quotient only if its image is an ℓ-group; this condition is also sufficient. No nontrivial finite quotient of $\widehat{F}_2$ extends as a homomorphism from all the rational points of its characteristic-zero unipotent completion. Consequently a congruence condition imposed through a specified finite matrix quotient has no automatic informative graded form in the system above.*

Manuscript question: §41; folios315–316; cote140-3 scans54–57.

Proof. Every continuous finite image of a pro-ℓ group is an ℓ-group, which proves necessity. Sufficiency is the universal property of the maximal pro-ℓ quotient. A characteristic-zero

unipotent group, and its inverse limit, has a unique mth root for every positive integer m, given by $\exp((\log g)/m)$. If a homomorphism to a finite group of exponent m were nontrivial, write $g = h^m$; its image would nevertheless be one. Thus every such homomorphism is trivial. A finite congruence predicate can still imply some rational Lie relation, but that requires an explicit comparison argument. It cannot be inferred by substituting $z = -x - y$ in the original finite quotient. □

We found no map identifying the degenerate pentagon (3.5), or a 2-adic version of it, with one of Grothendieck's level-two or level-four congruence conditions.

B.g13. The characteristic-two arrangement is not naive reduction of its hyperplanes [Proved]. *The Lie algebra used in the degenerate pentagon of OpenAI, Proposition 3.2, retains nine degree-one generators when $d = 3$. The arrangement obtained by identifying the equations $z_i - z_j = 0$ and $z_i + z_j = 0$ after reduction to $\mathbb{F}_2$ has only six hyperplanes. Their holonomy Lie algebras are therefore not identified by the reduction used in that proposition.*

Manuscript question: §38; fol.297; cote140-3 scan18; added Recent results task4.

Proof. The defining incidence relations are brackets, so impose no relations in degree one. Over $\mathbb{Q}$ the three coordinate hyperplanes and six signed diagonal hyperplanes are distinct. Reduction of the integral incidence *presentation* preserves these nine generator symbols. Reducing the hyperplane *equations* first identifies each signed pair, leaving three coordinate and three diagonal hyperplanes. The degree-one dimensions, nine and six, are already different. Our deletion program accordingly constructs all thirteen rational codimension-two flats before any scalar reduction. This checks a concrete obstruction to the proposed naive comparison; it neither proves nor presupposes a link to a modular congruence quotient of level two or four. □

B.g14. An infinite element in the additive graded formal loop group [Proved]. *Let $\mathcal{A}^{\mathrm{add}}$ be the group of continuous outer k-Lie automorphisms of $\widehat{\mathfrak{L}}$, equal to the identity on abelianization, preserving the three adjoint-conjugacy classes of $x, y, z = -x - y$, and commuting with the linear $\mathfrak{S}_3$ action. The exponential of D_{q_7} defines a nonidentity element of $\mathcal{A}^{\mathrm{add}}$ outside $\mathrm{GRT}_1(k)$. This supplies an explicit infinite element in this additive graded formal group. We do not identify $\mathcal{A}^{\mathrm{add}}$ with $\mathrm{GT}_0(k)$ or with any pro-ℓ group.*

Manuscript question: §42; folios 331, 349; cote 140-3 scans 85, 120.

Proof. The derivation raises degree by seven, so $\exp D_{q_7} = \sum_{j\geq 0} D_{q_7}^j/j!$ converges for the weight filtration, with only finitely many terms affecting a fixed weight. It is an automorphism, with inverse $\exp(-D_{q_7})$. Tangentiality integrates directly. If $D(u) = [u, c]$, set $U(t) = \exp(tD)u$ and $C(t) = \exp(tD)c$. Then $U'(t) = [U(t), C(t)]$. The equation $g'(t) = g(t)C(t)$, $g(0) = 1$, has a unique solution weight by weight in the completed enveloping algebra, using integration of polynomials in t over the characteristic-zero field. The solution is group-like because $C(t)$ is primitive, and $U(t) = g(t)^{-1}ug(t)$. Apply this to x, y, z. In the quotient by inner derivations, B.g2 says that D_{q_7} is $\mathfrak{S}_3$-invariant. Exponential commutes with conjugation and with this quotient, so its outer class centralizes $\mathfrak{S}_3$. Finally the degree-seven logarithmic term of an element of $\mathrm{GRT}_1(k)$ must satisfy the infinitesimal pentagon. B.g5 proves that q_7 does not; higher terms cannot cancel that lowest defect. Thus the constructed element is outside $\mathrm{GRT}_1(k)$, as asserted. □

B.g15. A genuine finite non-arithmetic outer solution [Proved]. *Let N_2 be the characteristic finite quotient of $\widehat{F}_2$ given by the free class-twelve BCH group over $\mathbb{F}_{13}$ in B.g11. The normalized automorphism defined there solves (I),(II) but is not the projection of an element*

of $\widehat{\mathrm{GT}}$. *Its outer class is not in the image of* $G_{\mathbb{Q}}$ *in* $\mathrm{Out}(N_2)$. *This gives a finite non-arithmetic equation solution, of order-quotient* $|N_2| = 13^{747}$*; it does not prove that the solution lifts to* $\widehat{\mathrm{GT}}_0$. *It is an additional nilpotent quotient, not one of the small matrix or permutation quotients prescribed in B(e).*

Manuscript question: §42; folio349; cote140-3 scan120.

Proof. We first supply an actual finite pure-braid target for the pentagon; replacing it by the exponential of a braid Lie algebra without a comparison would not suffice. Forgetting the fourth strand gives $1 \to F_3 \to P_4 \to P_3 \to 1$, with fibre generators $a_i = A_{i4}$, $1 \le i \le 3$. Conjugation gives an action of P_4 on its normal free fibre. Let

$$N_3 = \exp\bigl(\mathrm{Lie}_{\mathbb{F}_{13}}\langle a, b, c\rangle/\mathfrak{L}_{>8}\bigr).$$

The fibre action descends to N_3. The full twist is central in P_4, hence acts trivially on the fibre; thus the action factors through $P_4/Z(P_4) = \pi_1(M_{0,5})$. This identifies a finite quotient of the actual pentagon target. The split free-fibre sequence follows directly from forgetting a point in the ordered configuration space: locally the punctures move by disjoint-disc isotopies, its fibre is a three-punctured plane, and induction gives asphericity and the stated fundamental sequence. A far-away fourth point supplies a section.

Here are explicit signs for its action. In the inverse Artin convention the base half-twists act by

$$\beta_i(a_i) = a_{i+1}, \qquad \beta_i(a_{i+1}) = a_{i+1}^{-1} a_i a_{i+1},$$

fixing the other generator. The base pure generators are β_1^2, $\beta_2\beta_1^2\beta_2^{-1}$, β_2^2. Their first Lie terms are precisely the derivations t_{12}, t_{13}, t_{23} used in B.g5. The fibre generators act by inner conjugation and have first terms $\mathrm{ad}\, a_i = t_{i4}$. The program `graded_braid_transfer.py` verifies the Artin relation as reduced free words, all six first Lie terms, and the identity that the base full twist acts by conjugation with $(abc)^{-1}$; fibre conjugation by abc cancels it. Thus the signs and central quotient in this transfer are checked without an associator or a braid-formality assertion.

The image lies in the group of automorphisms inducing the identity on $H_1(N_3, \mathbb{F}_{13})$. If IA_i denotes those raising Lie degree by at least i, then $[\mathrm{IA}_i, \mathrm{IA}_j] \subset \mathrm{IA}_{i+j}$ and $\mathrm{IA}_8 = 1$. This finite target has class at most seven. As a linear operator $\phi = 1 + T$ on the truncated Lie algebra, $T^8 = 0$, whence $\phi^{13} = 1 + T^{13} = 1$ in characteristic thirteen. Truncated exponential and logarithm identify these degree-raising automorphisms with degree-raising derivations. To check the latter statement directly, $\exp(t \log \phi) = \phi^t$ for all $t \in \mathbb{F}_{13}$ preserves brackets. In every relevant homogeneous component the bracket-preservation polynomial has degree less than thirteen in t, so it vanishes identically; differentiating at zero shows that $\log \phi$ is a derivation. All denominators used have index at most eight and are units. Thus the free BCH group N_2 admits evaluation homomorphisms into this target for arbitrary pairs of its elements: apply the universal property of the free truncated Lie algebra to their logarithms. The same property shows that the kernel defining N_2 is fully invariant in $\widehat{F}_2$, hence characteristic.

Evaluate the genuine group pentagon, equation (4.10) of Drinfeld (1990) in this finite pure-braid target. Every occurrence of f starts in degree seven. Since the target has class seven, its value depends only on q_7 and on the degree-one parts of the braid arguments. Products of braid arguments give the sums of chords in (1.4), so the lowest group pentagon defect is exactly the infinitesimal defect evaluated in B.g5. On a its coefficient at the word a^5bcb is $16 \equiv 3 \pmod{13}$. This word has length eight and survives in N_3; the defect is nonidentity. The corrections F_9, F_{11} vanish in this target and cannot cancel it. Therefore our finite pair is not the projection of a pair satisfying the group pentagon.

Finally we check that taking outer classes cannot remove the obstruction. In N_2,

$$C_{N_2}(X) = \langle X\rangle Z(N_2), \quad C_{N_2}(Y) = \langle Y\rangle Z(N_2), \quad Z(N_2) = \exp \mathfrak{L}_{12}.$$

Indeed the associative centralizer of a free letter consists of its power series, and its intersection with the ordinary free Lie algebra below the top degree is just the line spanned by that letter. The invertible factor in $\exp(\operatorname{ad} x)-1 = \operatorname{ad} x\,(1+\cdots)$ identifies the group and Lie centralizer conditions. Hence the displayed formulas hold in this class-less-than-prime quotient. If two normalized pairs $f, g \in [N_2, N_2]$ give inner-conjugate automorphisms fixing X, their inner conjugator is $X^a z$ with z central. Comparing their values on Y gives a difference by an element of $\langle Y\rangle Z(N_2)$; abelianization forces both powers of X, Y to be zero. Thus f and g differ only by $Z(N_2)$. That difference is invisible in the class-seven pentagon target, so both have the same nonzero defect. Any arithmetic outer class equal to ours has scalar 1 modulo thirteen by its action on abelianization, and has such a normalized representative. The Galois image lies in $\widehat{\mathrm{GT}}$, so its pentagon defect must vanish. This contradiction proves the outer-class assertion as well. □

What a finite quotient can, and cannot, test

B.f1. An arbitrary two-generator finite quotient does not carry the full system [Refuted]. *A surjection $F_2 \to G$ alone does not define the substitutions $f(x,y) \mapsto f(y,x)$ and $f(x,y) \mapsto f(y,z)$ on elements of G. Nor does it define a pentagon test. A correct finite test either retains the word and all its evaluations, or uses a common invariant quotient for all required maps. In particular the Legendre images at even level n are not the full groups $\mathrm{SL}_2(\mathbb{Z}/n)$ requested literally in B(e).*

Manuscript question: §§38–39; fols.297–300; scans18–24.

Proof. In S_3 choose $x = (12)$ and $y = (123)$. The relation $x^2 = 1$ holds, but its image under swapping the two generators is $y^2 \neq 1$. Thus the swapping map does not descend. This counterexample is checked in `finite_congruence.py`. The pentagon has five evaluations in a quotient of the geometric group of $M_{0,5}$; one quotient of F_2 does not specify that target or its five structure maps. Finally both Legendre matrices $L_0 = \left(\begin{smallmatrix} 1 & -2 \\ 0 & 1 \end{smallmatrix}\right)$, $L_1 = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$ are identity modulo 2, whereas $\mathrm{SL}_2(\mathbb{F}_2)$ has order six. The projective image is the kernel of reduction to S_3 at even level, as follows from the index-six modular covering and reduction; its orders are computed below. These are failures of the proposed formulation, not counterexamples to a correctly formulated Grothendieck–Teichmüller relation. □

For a finite two-generated G, put

$$N_G = \bigcap_{q:F_2 \twoheadrightarrow G} \ker q, \qquad \overline{G} = F_2/N_G.$$

The intersection is finite. It is invariant under all automorphisms of F_2, including the swapping and cyclic substitutions. More strongly, any generating pair of $\overline{G}$ is the image of its distinguished pair under an automorphism: every epimorphism to G is accounted for in the intersection. We use the finite shadow $\mathcal{D}(G)$ consisting of outer automorphisms of $\overline{G}$ that send x to a conjugate of x^k, with k prime to its order, and commute in the outer group with $\theta : (x,y) \mapsto (y,x)$ and $\delta : (x,y) \mapsto ((xy)^{-1}, y)$. This is Guillot's $GT(G)$, whose inverse limit is $\widehat{\mathrm{GT}}_0$, not his notation for the pentagon group (Guillot, 2015, §§2.1–2.4). Its subgroup with a representative $k = 1$ is denoted $\mathcal{D}_1(G)$. At a finite level the multiplier need not be unique, so this subgroup is not automatically the image of the global subgroup $\lambda = 1$.

B.f2. Complete matrix residue enumeration at the requested levels [Proved]. *Let $Q_n = \mathrm{SL}_2(\mathbb{Z}/n)/\langle -I\rangle$ and $H_n = \langle L_0, L_1\rangle \subseteq Q_n$. Write $R = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ and $S = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, so that $\rho = \operatorname{Ad} R$, $\sigma = \operatorname{Ad} S$ on H_n. The following is an exhaustive enumeration of*

$$\xi\rho(\xi)\rho^2(\xi) = 1, \quad \eta\sigma(\eta) = 1, \quad \xi = \eta L_1^{\nu}, \qquad 2\nu + 1 \in (\mathbb{Z}/n)^{\times}, \tag{B.f2.1}$$

with ν taken modulo the order of L_1. The columns headed cycles enumerate (I),(II) with $f \in H_n'$ in the convention of B.m2.

Manuscript question: §46; fols.393–394,407–408; scans189–190,215–217.

n	$\lvert H_n\rvert$	$\lvert H_n'\rvert$	pairs	gauge orbits	descend	cycles	generating cycles
2	1	1	1	1	1	1	1
3	12	4	8	4	6	8	6
4	4	1	4	2	4	2	2
5	60	60	20	4	20	20	20
6	12	4	8	4	6	8	6
7	168	168	42	6	42	42	42
8	32	2	16	4	16	4	4
12	96	8	24	4	24	12	12
24	768	16	96	8	96	24	24

Proof. The algorithm constructs Q_n by enumerating determinant-one matrices, identifying a matrix with its negative, and closing L_0, L_1 under multiplication. It constructs H_n' from all conjugates of $[L_1, L_0]$; in a two-generator group their normal closure is the derived group. For each allowed ν and each $\xi \in H_n$, η is forced by (B.f2.1), after which both products are evaluated exactly. Gauge orbits use $(\xi, \eta) \mapsto (L_0^a \xi L_1^{-a}, L_0^a \eta L_1^{-a})$; the program checks that every such orbit stays in the solution set.

"Descend" is a separate test: the proposed images $S \mapsto \eta S$, $R \mapsto \xi R$ are propagated along the Cayley graph of Q_n and checked on every edge. Consistency and bijectivity are both required. Thus it is a test of an automorphism of this particular finite quotient, not a presumed consequence of two order relations. The cycle columns enumerate $f \in H_n'$ with the swapping and cyclic maps actually defined on H_n, then test whether $L_1^{2\nu+1}$ and $f^{-1} L_0^{2\nu+1} f$ generate H_n. All loops are finite and exhaustive; all solutions, not just the counts, are retained in `finite_matrix_cycles.json`. Level 1 is the single trivial solution. No conclusion about lifting any remaining residue to $\widehat{F}_2$ is built into the algorithm. □

For example, the first failed quotient descent at level 3 has $\nu = 0$ and $\xi = \eta = S$. It sends S to 1, and hence is not an automorphism. This is why Grothendieck's generation condition cannot be discarded. The outer images among the descending pairs have orders $1, 2, 1, 2, 2, 2, 4, 4, 8$ in the displayed level order. The projective level-four outer image is trivial because $Q_4 \simeq S_4$ has no outer automorphisms; it must not be substituted for the nontrivial outer image on $\mathrm{SL}_2(\mathbb{Z}/4)$.

B.f3. The characteristic S_3 calculation [Proved]. *The characteristic envelope $\overline{S_3}$ has order* 108 *and exponent* 6*, with derived group of order* 27*. There are exactly twelve pairs* $(\lambda \bmod 12, f)$ *satisfying (I),(II), with $f \in \overline{S_3}'$ and an induced automorphism. Their outer image is*

$$\mathcal{D}(S_3) = \mathcal{D}_1(S_3) \simeq C_2.$$

Complex conjugation is trivial in this particular outer quotient.

Manuscript question: §42; fols.331 verso–337; cote 140-3 scans85–96.

Proof. There are three generating-pair orbits modulo $\mathrm{Aut}(S_3)$. They give an embedding $\overline{S_3} \subseteq S_3^3$. It is the subgroup in which the product of the three signs is $+1$: its image on signs has order four, and the squares of the distinguished generators and their product generate the three independent C_3 factors. Thus its order is $4 \cdot 3^3 = 108$.

`finite_shadows.py` enumerates all four unit residues modulo 12 and all 27 commutator elements. It computes the actual substitution automorphisms and checks both cycle products,

generation, and the induced homomorphism on every Cayley edge. An independent enumeration uses the definition of $\mathcal{D}(S_3)$: fix $y \mapsto y^k$, range over every conjugate of x^k, and check commutation with both θ, δ modulo every possible inner conjugator. The two enumerations yield the same two outer classes. The twelve pairs and their orders are printed in the JSON certificate. The involution $x \mapsto x^{-1}, y \mapsto y^{-1}$ has the same outer key as the identity. This last computation prevents an erroneous identification of every finite order-two Galois image with complex conjugation. □

B.f4. The arithmetic image of the S_3 envelope is all of C_2 [Proved]. *The outer Galois image in B.f3 is C_2, with fixed field $\mathbb{Q}(\sqrt{3})$. Consequently the images of $G_{\mathbb{Q}}$, $\widehat{\mathrm{GT}}$ and $\widehat{\mathrm{GT}}_0$ in $\mathcal{D}(S_3)$ all coincide; there is no nonarithmetic outer shadow at this level.*

Manuscript question: §44; fols.365 verso–368 verso; cote 140-3 scans152–158.

Proof. Use the rational S_3-Galois cover

$$b(t) = \frac{27t^2(t-1)^2}{4(t^2-t+1)^3}, \qquad a(t) = \frac{t^3-3t+1}{t(t-1)}, \quad w = a - \frac{3}{2}.$$

The rational transformations $t \mapsto 1-t, 1/t$ generate its deck group; the degree is six. The three distinct orders-of-inertia patterns are obtained by taking $b = T, 1/T, 1/(1-T)$, so these are exactly the three epimorphism kernels used in B.f3. Direct rational identities give

$$a(1/(1-t)) = a(t), \quad a(1-t) = 3 - a(t), \quad w^2 = \frac{27}{4}\frac{1-b}{b}.$$

Thus the quadratic subfields of the three covers have square classes $3T(1-T)$, $3(T-1)$, and $-3T$. Their product is the square class of 3. Their compositum therefore contains $\sqrt{3}$ as a constant. Its geometric Galois group has order 108 by B.f3, whereas its arithmetic group is a subgroup of S_3^3 of order at most 216. The constant quadratic extension forces its order to be at least 216; both bounds are equal. No other constant is possible.

The centralizer of the geometric subgroup in S_3^3 is trivial, since its projection to every center-free factor is surjective. An element outside the geometric subgroup cannot act by an inner automorphism of it: multiplying it by the inverse of a putative inner conjugator would centralize that subgroup. Hence the constant-field quotient acts faithfully in the outer group. This proves the claim. All rational identities are independently checked in `finite_s3_fields.py`. The arithmetic character is $\chi_{-4}\chi_{-3}$, in agreement with C.a2. □

B.f5. Exact simple-group finite shadows, recomputed [Known]. *For the characteristic envelopes attached to A_5 and $\mathrm{PSL}_2(\mathbb{F}_7)$,*

$$\mathcal{D}_1(A_5) = 1, \quad |\mathcal{D}(A_5)| = 2, \qquad \mathcal{D}_1(\mathrm{PSL}_2(\mathbb{F}_7)) \simeq C_2^3 \times D_8^2, \quad |\mathcal{D}(\mathrm{PSL}_2(\mathbb{F}_7))| = 1024,$$

where D_8 has order eight. The first of these full shadows is entirely arithmetic, with cyclotomic quadratic field $\mathbb{Q}(\sqrt{5})$. We do not determine the order of the arithmetic image in the second. The full 1024-element shadow has exactly one element of order one, 575 of order two, and 448 of order four; its exponent is four.

Manuscript question: §42; fol.349 verso; scan120.

Proof. We use and verify the mechanism of Guillot (2015, Theorem 4.9 and Proposition 4.12). For a nonabelian simple G, the distinct epimorphism kernels give $\overline{G} = G^r$. This follows by induction from the subdirect-product lemma: a proper subdirect subgroup of a product with one new simple factor identifies that factor with an old quotient, contradicting distinctness of

the kernels. Every automorphism of G^r permutes its simple normal factors and acts on each by $\mathrm{Aut}(G)$. Consequently its outer group acts faithfully on generating pairs modulo simultaneous inner conjugacy.

On this set, take the centralizer of the permutations induced by $\mathrm{Out}(G)$, swapping the pair, and $(x,y)\mapsto((xy)^{-1},y)$. Intersect it with the permutations preserving the ordered pair of conjugacy classes. The result is exactly $\mathcal{D}_1(G)$: the class-preserving condition translates coordinatewise into conjugacy of the distinguished elements in G^r. The program constructs the set, these permutations, and every color-preserving equivariant bijection between their orbits. If a block contains r isomorphic colored orbits with a equivariant self-maps, its factor has order $a^r r!$. It also generates every element of the resulting group and checks the centralizer and color conditions anew.

For A_5 the set has 38 elements (from 2280 generating pairs), and the centralizer intersection is trivial. For $\mathrm{PSL}_2(\mathbb{F}_7)$ it has 114 elements (from 19152 generating pairs); the blocks give three C_2 factors and two $C_2 \wr S_2 = D_8$ factors. The orders of its elements are 1 once, 2 in 287 cases, and 4 in 224 cases. These are independent recomputations of the corresponding entries of Guillot (2015, Theorem 5.2).

The full shadow is obtained by allowing a common power k on all conjugacy classes. Every unit power pattern is realized by Galois, because the cyclotomic character is surjective and inertia is raised to that power; this is also Guillot (2015, Lemma 2.3). Two such patterns give the same coset precisely when every class is preserved by their ratio. The power-class kernel has index two in both examples, as verified for every group element. For A_5 it is the quadratic character modulo five. Since the residual class-preserving group is trivial, the arithmetic image is the entire order-two group. For $\mathrm{PSL}_2(\mathbb{F}_7)$ this argument only gives the quotient of order two, not all 512 elements in its kernel.

For the full finite-order calculation, adjoin to the recorded subgroup $\mathcal{D}_1$ the permutation

$$c : [(x,y)] \longmapsto [(x^{-1},y^{-1})]$$

of the 114 inner generating-pair classes. The power class of -1 is nontrivial modulo seven, so $c \notin \mathcal{D}_1$. The established index-two result identifies the generated group with the full shadow. The program `finite_shadow_orders.py` verifies that c normalizes all nine recorded generators, enumerates the full group, checks all 1024^2 products for closure, and computes every element order. The coset $c\mathcal{D}_1$ contains 288 involutions and 224 elements of order four. The other three named full shadows are C_2, so their only orders are one and two. These are finite-shadow orders; the full arithmetic subgroup at $\mathrm{PSL}_2(\mathbb{F}_7)$ remains undetermined. □

A complete additional finite case: the characteristic $\mathfrak{S}_4$ envelope

B.s4.1. The characteristic $\mathfrak{S}_4$ shadow is arithmetic of order two [Proved]. *Let N_G be the intersection of the kernels of all continuous epimorphisms $\widehat{F}_2 \twoheadrightarrow G$, and put $\overline{G} = \widehat{F}_2/N_G$. For $G = \mathfrak{S}_4$, let $\mathrm{GT}(G)$ and $\mathrm{GT}_1(G)$ have Guillot's finite-shadow definitions (power-conjugacy and commutation with the two standard outer symmetries, not an asserted image of the full profinite pentagon group). Then*

$$|\overline{\mathfrak{S}_4}| = 442368 = 108\cdot 2^{12}, \qquad \mathrm{GT}(\mathfrak{S}_4) = \mathrm{GT}_1(\mathfrak{S}_4) = \mathrm{im}(G_{\mathbb{Q}} \to \mathrm{Out}(\overline{\mathfrak{S}_4})) \simeq C_2. \quad \text{(B.s4.1.1)}$$

There is therefore no nonarithmetic finite-shadow element in this quotient. This supplements the requested finite range of B(e); it asserts no surjectivity theorem for arbitrary finite shadows.

Manuscript question: §42; fol.349 verso; cote140-3 scan120; added finite-quotient task B(e).

Proof. The envelope and its kernel. All automorphisms of $\mathfrak{S}_4$ are inner. Its 216 ordered generating pairs form nine simultaneous-conjugacy classes, represented by the pairs printed in `arithmetic_s4.json`. Write $x, y \in \mathfrak{S}_4^9$ for their two coordinate tuples and $H = \langle x, y \rangle$. By the intersection-of-kernels definition, $H = \overline{\mathfrak{S}_4}$. Project each factor through $\mathfrak{S}_4 \twoheadrightarrow \mathfrak{S}_3$ by its action on the three partitions into two pairs. Its kernel is the Klein four group V. The projected group Q has order 108: each of the three generating-pair classes of $\mathfrak{S}_3$ occurs three times among the nine projections, and after conjugating coordinates it is the threefold repeated embedding of

$$Q_0 = \{(a, b, c) \in \mathfrak{S}_3^3 : \operatorname{sgn}(a) \operatorname{sgn}(b) \operatorname{sgn}(c) = 1\}. \tag{B.s4.1.2}$$

The program verifies this by a 108-element traversal, independent of a large permutation-group package. A Schreier transversal in H over Q gives 216 generators for the kernel $V_0 = H \cap V^9$. Identify V^9 with $\mathbb{F}_2^{18}$. Their span has rank 12, with the full bit-vector basis printed in the output; Schreier's lemma proves that it is exactly the kernel. Hence $|H| = 108 \cdot 2^{12}$.

The permutation action has trivial image. Let $\mathcal{P}_c$ be the nine generating-pair classes. The two symmetries are

$$[a, b] \longmapsto [b, a], \qquad [a, b] \longmapsto [(ab)^{-1}, b].$$

Their permutations on the program's zero-based indices are respectively

$$(2, 5, 0, 6, 7, 1, 3, 4, 8), \qquad (7, 4, 5, 8, 1, 2, 6, 0, 3).$$

The blocks obtained by fixing the separate conjugacy classes of the two entries are

$$\{0\}, \{1\}, \{2\}, \{3, 4\}, \{5\}, \{6, 7\}, \{8\}.$$

Of the four block-preserving permutations, only the identity commutes with both displayed permutations, as direct substitution verifies. Thus $\mathcal{S}(\mathfrak{S}_4) = 1$ in the terminology of Guillot (2015, Proposition 4.3, Lemmas 4.5–4.7, §4.3). Every unit power of a permutation has the same cycle type. Consequently the action of the entire finite $\mathrm{GT}(\mathfrak{S}_4)$, not just its multiplier-one subgroup, preserves these blocks. It has trivial image on $\mathcal{P}_c$.

Bounding the kernel of the action. Put $N = N_{\mathfrak{S}_4^9}(H)$ and $W = V^9$. An automorphism of H acting trivially on $\mathcal{P}_c$ is, in every coordinate, conjugation by some element of $\mathfrak{S}_4$. Conversely such a tuple defines an automorphism of H exactly when it belongs to N. The centralizer of H in $\mathfrak{S}_4^9$ is trivial, because its coordinate projections are surjective and $Z(\mathfrak{S}_4) = 1$. Modulo inner automorphisms of H, the kernel is therefore exactly N/H. In particular $\mathrm{GT}(\mathfrak{S}_4) \subseteq N/H$.

We compute this normalizer quotient without enumerating its ambient group. The normalizer of the repeated group Q in $\mathfrak{S}_3^9$ lies in the corresponding repeated copy of $\mathfrak{S}_3^3$: within any repeated block, two conjugators must induce the same conjugation on the surjective $\mathfrak{S}_3$ factor, and so coincide, since its centre is trivial. By (B.s4.1.2) the resulting quotient of normalizer by Q has order two. Next

$$(W/V_0)^Q = 0. \tag{B.s4.1.3}$$

There is a conceptual check as well as the rank certificate: the normal odd-order subgroup $A = C_3^3$ of Q_0 projects onto C_3 in every coordinate, which acts without fixed vectors on each copy of $V \simeq \mathbb{F}_2^2$. Hence $W^A = 0$. Taking invariants under A is exact in characteristic two, by averaging over its odd order, and so $(W/V_0)^A = 0$ and a fortiori (B.s4.1.3). The program independently computes the two-generator fixed-vector equations: their rank is 6 on W, their solution space has dimension 12, and quotienting by the 12-dimensional V_0 leaves dimension zero. If $w \in N \cap W$, then $[w, H] \subseteq V_0$, so (B.s4.1.3) implies $w \in V_0$. Thus $N \cap W = V_0$ and

$$N/H \hookrightarrow N_{\mathfrak{S}_3^9}(Q)/Q \simeq C_2.$$

An explicit nontrivial finite-shadow class. The code solves the normalizer equations in the 64-element vector-space quotient W/V_0, and prints a tuple $n \in N \setminus H$. In the nine coordinates and using permutation arrays on $\{0,1,2,3\}$, it is

$$n = (1, (01), 1, 1, 1, (13), 1, 1, (13)). \tag{B.s4.1.4}$$

Here 1 denotes the identity permutation; the cycle notation is in the zero-based labeling just specified. The output additionally prints conjugators *in* H showing that $n^{-1}xn$ is conjugate to x and $n^{-1}yn$ to y. These tests use the Schreier transversal and exact linear solving in V_0, and verify the resulting conjugators by direct multiplication. The kernel N/H, being the kernel of the natural action of $\mathrm{Out}(H)$ on $\mathcal{P}_c$, is normal in that outer automorphism group. Since it has order two, all outer automorphisms centralize it, in particular the two standard symmetries. Therefore the class of n is in $\mathrm{GT}_1(\mathfrak{S}_4)$, proving both finite shadow orders in (B.s4.1.1). We emphasize the extra conjugacy check: rational conjugacy classes of $\mathfrak{S}_4$ alone would not prove conjugacy by an element of the proper subgroup $H \subset \mathfrak{S}_4^9$.

The arithmetic lower bound. The projection to Q identifies it with $\overline{\mathfrak{S}_3}$, so $N_{\mathfrak{S}_4} \subseteq N_{\mathfrak{S}_3}$ and the arithmetic outer action descends. We verify the nontriviality of this descended action by the explicit fields, rather than assuming every finite shadow is arithmetic. The rational quotient map

$$\beta(t) = \frac{27t^2(t-1)^2}{4(t^2-t+1)^3}$$

is invariant under $t \mapsto 1-t$ and $t \mapsto 1/t$, has degree six, and is a regular $\mathfrak{S}_3$ cover. Its branch orders at $0, 1, \infty$ are $(2,2,3)$. The covers obtained by the equations $\beta(t) = T$, $\beta(t) = 1/T$ and $\beta(t) = 1/(1-T)$ therefore represent all three generating-pair classes. If

$$w = \frac{t^3 - 3t + 1}{t(t-1)} - \frac{3}{2},$$

then $w^2 = \frac{27}{4}(1-\beta)/\beta$; this rational-function identity and the invariances are verified in `finite_s3_fields.py`. The three quadratic subfields have square classes

$$3T(1-T), \quad 3(T-1), \quad -3T.$$

Their product is $27T^2(T-1)^2$, so their compositum contains $\mathbb{Q}(\sqrt{3})$. Its geometric group has the already computed order 108, while its arithmetic group is a subgroup of $\mathfrak{S}_3^3$, of order at most 216. The degree of its constant field is consequently at most two and at least two; it is exactly $\mathbb{Q}(\sqrt{3})$, and the arithmetic group is all of $\mathfrak{S}_3^3$. An element outside the geometric subgroup acts noninner on it: otherwise the element divided by an appropriate geometric conjugator would centralize that subgroup, whose centralizer in $\mathfrak{S}_3^3$ is trivial by the surjective coordinate projections. Thus the arithmetic outer image on $\overline{\mathfrak{S}_3}$ has order two. The image on H has order at least two, and is contained in the finite shadow of order two just proved. This finishes (B.s4.1.1). □

B.s4.2. What the simple-group shadow counts do and do not prove [Proved]. *The finite calculations for* A_5 *and* $\mathrm{PSL}_2(\mathbb{F}_7)$*, when interpreted by Guillot (2015, Theorem 4.9), give*

G	$\lvert\mathrm{GT}_1(G)\rvert$	$\lvert\mathrm{GT}(G)\rvert$	$\lvert \mathrm{im}\, G_{\mathbb{Q}}\rvert$
A_5	1	2	2
$\mathrm{PSL}_2(\mathbb{F}_7)$	512	1024	*not determined here*

The lower bound on the last unknown size is two. Neither the colored centralizer computation nor cyclotomic surjectivity proves its equality to 1024.

Manuscript question: §42; fol.349 verso; cote140-3 scan120; added finite-quotient task B(e).

Proof. Both groups are finite, nonabelian and simple, exactly the hypothesis of Guillot's Theorem 4.9. The standard outer involution is conjugation by an odd permutation for A_5, and by a matrix of nonsquare determinant for $\mathrm{PSL}_2(\mathbb{F}_7)$. These generate their respective outer automorphism groups of order two; this is also verified by enumerating all automorphisms in `arithmetic_simple_audit.py`. The permutations used in `finite_shadows.py` are exactly the swap, $(g,h) \mapsto ((gh)^{-1}, h)$, and those outer involutions appearing in the theorem. The color is the pair of separate conjugacy classes, not an incorrectly assumed action of $G \times G$ on generating pairs. Thus the computed colored centralizer orders 1 and 512 have precisely the stated multiplier-one meaning.

For completeness we justify the passage to the full finite group. Guillot's Lemma 4.10 gives $\overline{G} = G^r$. Therefore x^k is conjugate to x in $\overline{G}$ exactly when each coordinate has the same conjugacy class under the power k. In these two groups every nontrivial conjugacy class occurs in a generating pair; this too is checked in the audit. The kernel of the power action on all conjugacy classes is

$$\begin{array}{r|l} A_5,\ e=30 & 1, 11, 19, 29 \pmod{30} \\ \mathrm{PSL}_2(\mathbb{F}_7),\ e=84 & 1, 11, 23, 25, 29, 37, 43, 53, 65, 67, 71, 79 \pmod{84}. \end{array}$$

Its index among all units is two in each case. The quotient of the full finite shadow by its multiplier-one subgroup embeds in this power-action group, by its definition, and the cyclotomic character of Galois realizes every residue unit. Hence it is all of that group and doubles the orders. For A_5 this same arithmetic lower bound fills the entire group of order two. For $\mathrm{PSL}_2(\mathbb{F}_7)$ it says nothing about how much of the kernel of order 512 is realized by Galois. The smallest unsettled question within this row remains the exact image on this specified characteristic quotient, not the polynomial-time computation of its finite-shadow upper bound. □

B.f6. A noncongruence finite quotient detects information lost by matrices [Proved]. *The continuous map $\widehat{\mathrm{SL}_2(\mathbb{Z})} \to \mathrm{SL}_2(\widehat{\mathbb{Z}})$ is not injective. The corresponding loss is already present in the index-six free subgroup after passing to projective groups. Thus the matrix equations are necessary conditions and are not an equivalent encoding of the full profinite equations merely because their matrix calculation is correct.*

Manuscript question: §§39,42; fols.299–301,349 verso; scans22–26,120.

Proof. Let $\mathbb{F}_8 = \mathbb{F}_2[a]/(a^3 + a + 1)$. The matrices

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & a \\ a^{-1} & 1 \end{pmatrix}$$

have orders two and three and generate $\mathrm{SL}_2(\mathbb{F}_8)$ of order 504. The program `finite_noncongruence.py` verifies generation and simplicity: for every nonidentity conjugacy class its normal closure is the whole group. This is a complete finite simplicity certificate, not a probabilistic test. Hence $C_2 * C_3$ surjects onto this group, and the map extends to its profinite completion.

No congruence quotient has this simple quotient. By the Chinese remainder theorem and the successive reduction kernels, the nonabelian simple composition factors of $\mathrm{SL}_2(\mathbb{Z}/n)$ are $\mathrm{PSL}_2(\mathbb{F}_p)$ for primes $p \geq 5$; the kernels between prime powers are p-groups and the cases $p = 2, 3$ are solvable. The order $p(p^2-1)/2$ is 168 at $p = 7$ and 660 at $p = 11$, strictly increasing, and never 504. A continuous map from the congruence completion to a finite group factors through some finite level. Thus the quotient of order 504 cannot factor through that completion. If the completion map were injective it would be an isomorphism of compact groups, which would force just such a factorization.

Its projective kernel lies in the kernel of reduction modulo two, which is the free subgroup F. If the comparison restricted to F were injective, that projective kernel would vanish. This proves the free-group version as well. What is lost is the closed congruence kernel and the action on all its finite quotients, not merely a sign. □

B.f7. The exact remaining finite-image and pentagon problem [Open]. *For $G = S_3, S_4, A_5$, the outer characteristic shadows of $G_{\mathbb{Q}}$, $\widehat{\mathrm{GT}}$, and $\widehat{\mathrm{GT}}_0$ are all the computed order-two group. For $G = \mathrm{PSL}_2(\mathbb{F}_7)$, the unresolved arithmetic image is a subgroup of the explicitly given group of order 1024, surjecting onto its power-class quotient C_2. Determining this subgroup requires the Galois action on the 57 regular covers and on their labeled structures, not the conjugacy-class calculation alone. In addition, the raw matrix residues in B.f2 do not come with the compatible five-puncture quotient required to test (III); the uncomputed datum is that quotient and its five evaluation maps, already at level $n = 3$, the first nontrivial Legendre matrix image after the trivial level-two image.*

Manuscript question: §42; fol.349 verso; scan120.

The attack is B.f2–B.f5 and B.s4.1: complete residue lists and finite shadow groups are available, and arithmetic fields close three of the four named cases. No finite shadow outside the arithmetic image has been established in this range. We have neither equated a finite shadow with the image of the inverse limit nor reported a pentagon count for an undefined finite target. The first unresolved member of the four named characteristic examples is the group of order 168; a universal characteristic-envelope pentagon calculation for all the separately listed $\mathrm{SL}_2(\mathbb{Z}/n)$ is not supplied by these matrix tests.

B.f8. The BISITE endomorphism is invisible to every nilpotent completion [Proved]. *The endomorphism $v : F_2 \to F_2$ defined by $v(x) = x$, $v(y) = y[x, y]$ induces an automorphism of every lower-central nilpotent quotient, every free pro-ℓ completion, and every characteristic-zero pro-unipotent completion. It is not an automorphism of the full profinite completion. It preserves the first two peripheral conjugacy classes, but not the third; thus it is not a counterexample inside $\widehat{\mathrm{GT}}_0$ or the group of all loop automorphisms.*

Manuscript question: §42; fol.349 verso; cote140-3 scan120; added BISITE B.11 follow-up.

Proof. It is the identity on abelianization. The graded Lie ring of a free group is generated by degree one, so the induced maps on every lower-central graded quotient are the identity. Induction on the central extensions proves the nilpotent quotient assertion. For the pro-ℓ completion the images generate the Frattini quotient, so they topologically generate the free pro-ℓ group. The resulting surjection is an automorphism: a finitely generated profinite group is Hopfian, since it has only finitely many open normal subgroups of any fixed index, and a surjective endomorphism permutes these subgroups. Alternatively the Magnus realization and the preceding nilpotent argument prove the same result. In the pro-unipotent Lie algebra the map is a filtered identity on the associated graded, and is inverted recursively degree by degree.

On three letters put $x = (12)$, $y = (13)$. Exact multiplication gives $y[x, y] = (12)$. The old pair generates S_3 and is transitive, while the new pair generates C_2 and fixes the third letter. Precomposing a transitive finite action with a surjective profinite endomorphism would preserve its image and its orbits. Hence the profinite map is not surjective. The full calculation is in `finite_congruence.json`. Finally $y[x, y] = (yx)y(yx)^{-1}$ is conjugate to y, but $v(xy) = xyxyx^{-1}y^{-1}$ is cyclically reduced of length six, whereas xy has cyclic length two. They are not conjugate in the discrete free group. In the same finite S_3 quotient, xy has order three whereas its image under v is the identity, so they are not conjugate in the full profinite group either. The third-loop condition cannot therefore be inferred from the first two, or from the completion assertions just proved. □

C. Congruence quotients and arithmetic comparisons

C.r1. Exact arithmetic outer images on every congruence quotient [Proved]. *For $n \geq 2$, put $R_n = \mathbb{Z}/n$ and $K_n = \mathrm{SL}_2(R_n)$. The outer action of $G_{\mathbb{Q}}$ on K_n induced by the universal elliptic curve is the cyclotomic character followed by an injection*

$$R_n^\times/(R_n^\times)^2 \hookrightarrow \mathrm{Out}(K_n). \tag{C.r1.1}$$

Its image is exactly this square-class group. At the adelic level its kernel is $\chi^{-1}((\widehat{\mathbb{Z}}^\times)^2)$, the subgroup fixing the maximal multiquadratic extension of $\mathbb{Q}$. This proves the arithmetic statements of §39, scans22–24, folios299–300, with "outer" retained throughout.

Proof. Over $\mathcal{M}_{1,1}$ the full n-torsion local system gives a continuous representation $\pi_1(\mathcal{M}_{1,1}) \to \mathrm{GL}_2(R_n)$. The Weil pairing identifies its determinant with $\chi \bmod n$; over $\overline{\mathbb{Q}}$ the topological monodromy is $\mathrm{SL}_2(\mathbb{Z}) \twoheadrightarrow K_n$. Thus the exact arithmetic stack extension induces on K_n the outer action given by conjugation by a matrix of determinant $\chi(\gamma)$. Different lifts of γ change that matrix by an element of K_n.

A matrix commuting with the elementary matrices $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1&0\\1&1\end{smallmatrix}\right)$ is scalar: entrywise comparison forces both off-diagonal entries to vanish and the diagonal entries to coincide. Therefore the kernel of $\mathrm{GL}_2(R_n) \to \mathrm{Out}(K_n)$ is exactly $K_n \cdot (R_n^\times I)$. Taking determinants identifies the quotient with $R_n^\times/(R_n^\times)^2$ and proves injectivity in (C.r1.1). Surjectivity of the cyclotomic character modulo n follows from $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) = R_n^\times$, so the stated outer image is exact.

The same centralizer proof applies over $\widehat{\mathbb{Z}}$. Its unit group is a product of compact groups; the subgroup of squares is the product of their square subgroups and is closed. By the cyclotomic description of the abelianized absolute Galois group (Kronecker–Weber), its quotient by squares is the maximal elementary abelian two quotient. Equivalently its fixed field is the compositum of all quadratic fields. This last identification can also be read characterwise from the quadratic Dirichlet characters. This proves all the asserted kernel and image statements; it does not imply that the action on the full profinite modular group factors through this small quotient. □

n	$\lvert\mathrm{SL}_2(R_n)\rvert$	$\lvert Z(\mathrm{SL}_2(R_n))\rvert$	$\lvert\mathrm{Ad}\,\mathrm{GL}_2(R_n)\rvert$	$\lvert\mathrm{Inn}(K_n)\rvert$	arithmetic outer image
2	6	1	6	6	1
3	24	2	24	12	C_2
4	48	2	48	24	C_2
5	120	2	120	60	C_2
6	144	2	144	72	C_2
7	336	2	336	168	C_2
8	384	4	384	96	C_2^2
12	1152	4	1152	288	C_2^2
24	9216	8	9216	1152	C_2^3

All entries, every unit square, and the scalar-centralizer assertion are exhaustively checked in `finite_congruence.py`. Notice the centers of orders four and eight: replacing them by a uniform $\{\pm I\}$ would give wrong inner-automorphism counts.

C.r2. A unique multiplier cannot be recovered from an outer congruence action [Refuted]. *The claim on §39, scans25–26, folios300 verso–301, that a unique $\chi(u) \in \widehat{\mathbb{Z}}^\times$ is recovered even from an outer automorphism of $\mathrm{SL}_2(\widehat{\mathbb{Z}})$ preserving its unipotent cusp subgroup is false. The strongest conclusion justified by this datum is its square class, as in C.r1. On the full profinite modular group, the peripheral multiplier is well defined; the passage to the congruence completion destroys precisely the indicated uniqueness argument.*

Proof. For a unit a, conjugation by $D_a = \mathrm{diag}(a, a^{-1}) \in \mathrm{SL}_2$ sends $U(t) = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$ to $U(a^2 t)$. It is an inner automorphism, so represents the same outer class as the identity, while its multiplier on the displayed cusp parametrization is a^2. Modulo five, take $a = 2$; the two multipliers are 1 and 4. Adelic examples are obtained by taking $a_5 = 2$ and $a_p = 1$ at the other primes. This is an explicit counterexample to uniqueness, already in the requested finite range.

For comparison, in the full projective modular group let $T = sr$, whose square belongs to the index-six free group F. An inner conjugator normalizing $\langle T \rangle$ has reduction in the stabilizer of the corresponding cusp in S_3. That stabilizer is generated by the reduction of T, so it may be removed by a power of T. The remaining conjugator is in F. Abelianizing F then shows that a conjugacy $T^2 \mapsto (T^2)^b$ forces $b = 1$, because the cusp vector in $F^{\mathrm{ab}} \simeq \widehat{\mathbb{Z}}^2$ is primitive. Thus this ambiguity does not occur in the original profinite peripheral structure. The same argument applies to its linear central lift, whose extra central subgroup acts trivially by conjugation. □

C.r3. Genuine and outer arithmetic actions must be counted separately [Proved]. *The input $G_{\mathbb{Q}} \to \mathrm{Out}(\widehat{\mathrm{SL}_2(\mathbb{Z})})$ has a canonical outer meaning. Without choosing a section and compatible paths it does not specify a homomorphism to the genuine automorphism group. At a finite congruence level, the full arithmetic fundamental group acts with image $\mathrm{Ad}\,\mathrm{GL}_2(R_n)$, the third numeric column above; Galois acts canonically with the square-class outer image in the last column. These are different requested objects, not alternative counts for the same action.*

Manuscript question: §§39,45; fols.299–301,387–388; scans22–26,176–178.

Proof. The stack extension is an extension of $G_{\mathbb{Q}}$ by the geometric modular group. Conjugation by a chosen lift of γ is changed by an inner automorphism if that lift is changed. This gives the outer action and explains its ambiguity directly. Its torsion representation contains every determinant-one matrix by geometric monodromy and has every determinant by cyclotomic surjectivity; consequently its image is all of $\mathrm{GL}_2(R_n)$. Quotient by its scalar centralizer gives the genuine action image, and quotient further by inner conjugations gives C.r1. The distinction survives at inverse limits. The faithful action on the full profinite modular group and its centralizer are proved in C.a5–C.a7, rather than inferred from finite congruence counts. □

The congruence representation obtained from actual profinite cocycles

C.d1. Projective congruence kernels are preserved by $\widehat{\mathrm{GT}}_0$ [Known]. *Let $q : \widehat{B} \twoheadrightarrow \mathrm{SL}_2(\widehat{\mathbb{Z}})/\{\pm I\}$ be the congruence homomorphism, with the single global central subgroup $\{\pm I\}$. For every $u \in \widehat{M}^1$, there is $U(u) \in \mathrm{GL}_2(\widehat{\mathbb{Z}})/\{\pm I\}$ of determinant $\mu(u)$ such that*

$$q \circ u = \mathrm{Int}(U(u)) \circ q.$$

Consequently all projective congruence kernels are invariant under the genuine extension of $\widehat{\mathrm{GT}}_0$. This statement needs neither the pentagon nor an arithmetic-image assumption. It does not assert a canonical action on $\widehat{\mathrm{SL}_2(\mathbb{Z})}$ without the specified central pullback. Its profinite structural input is the torsion theorem for free profinite products, not an inference from a matrix cocycle at one finite level.

Manuscript question: §39; scans21–30; §44; scans150–168; §46; fols398–400,409; scans198–202,218–219.

Proof. We make the structural input explicit. In $\widehat{B} = \widehat{C}_2 \amalg \widehat{C}_3$, every finite-order element is conjugate into a factor (Herfort–Ribes, 1985, Theorem 2(i) (Herfort and Ribes, 1985), also explicitly used in Minamide–Nakamura, p.8, in the attached paper). The primary Herfort–Ribes

proof was not supplied; this imported theorem is not a proof certified by the present calculation. Applying it to ηs and ξr in B.m1 gives

$$\eta = \beta\sigma(\beta)^{-1}, \qquad \xi = \alpha\rho(\alpha)^{-1}, \qquad \beta \in \widehat{F}, \quad \alpha \in \widehat{F} \cdot \{1, J\}.$$

For the order-two factor, reduction of a conjugator in S_3 lies in the centralizer of its transposition and is removed by a power of s. For the order-three factor, the two possible conjugacy classes are represented by r, r^{-1}; their normalizer in the extended group adds sJ, which reverses r. Comparing reductions in S_3 then places the conjugator in the displayed coset union. The determinant of its matrix representative is a single sign δ. This step simultaneously excludes the spurious scalar branch at every prime.

Choose matrix representatives for α, β, f_G, with determinants $\delta, 1, 1$, and define the lifts of ξ, η by their matrix coboundaries. They are independent of the chosen global signs of α, β. With

$$\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad \eta = \begin{pmatrix} P & Q \\ Q & T \end{pmatrix},$$

conjugacy of ξR to R gives $C + D - B = 1$; conjugacy of ηS to S gives the symmetry of η. The projective loop relation lifts as

$$\xi = \varepsilon\eta \begin{pmatrix} 1 & 0 \\ \mu - 1 & 1 \end{pmatrix}, \qquad \varepsilon \in \{\pm 1\} \text{ a single global sign.}$$

Comparing entries gives $B = \varepsilon Q$, $D = \varepsilon T$, and $1 = C + D - B = \mu D$. In particular $D = \mu^{-1}$ is a unit. Put

$$V = \begin{pmatrix} \mu & \mu B \\ 0 & 1 \end{pmatrix}.$$

A direct multiplication, using $AD - BC = PT - Q^2 = 1$, proves

$$VRV^{-1} = \xi R = \alpha R \alpha^{-1}, \qquad VSV^{-1} = \varepsilon\eta S = \varepsilon\beta S\beta^{-1}.$$

The sign in the second equality is essential in the linear group and disappears in its stated projective quotient. Thus V implements $q \circ u_0$ on the two generators s, r. For $u = \mathrm{Int}(f_G^{-1})u_0$, take $U = f_G^{-1}V$. Agreement on these topological generators gives agreement of the continuous homomorphisms on all of $\widehat{B}$. Also $\det U = \mu$.

This formula has the claimed ambiguity. Gauge by ℓ_0^n replaces f_G and V by their common left translate, so leaves U unchanged; a global change of sign of f_G changes U by $-I$. For composition, the normalized representatives form the subgroup normalizing $\langle y \rangle$, and their matrices V have lower-right entry one. The product has the same property and implements the composite on R, S, so it is the normalized matrix of that composite. Indeed a matrix commuting with both R, S is scalar, as one checks entry by entry, and the lower-right entry fixes the scalar. For general representatives the cocycle formula $f''_G = f'_G u'(f_G)$ supplies precisely the corresponding multiplication law modulo the global sign. Hence $u \mapsto U(u)$ is a homomorphism.

At a finite level, an element of the projective congruence kernel is sent to the identity by the right-hand side, proving invariance. This is a genuine descent proof: the matrix U was constructed before descent was asserted. □

C.d2. The norm equations and a correction visible on complex conjugation [Proved]. *Writing* $\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ *and* $\beta = \left(\begin{smallmatrix} r & s \\ t & v \end{smallmatrix}\right)$*, the matrix loop relation implies*

$$\delta(c^2 + cd + d^2) = \varepsilon(t^2 + v^2) = \mu^{-1}, \qquad \delta(ac + bc + bd) = \varepsilon(rt + sv).$$

It follows algebraically that $\delta \equiv \mu \pmod 3$ *and* $\varepsilon \equiv \mu \pmod 4$*. These are therefore already constraints on the whole actual profinite solution group of §46. Scan198 (73) omits a factor* δ *in its first equality; the correct later formula is (88), scans202–203.*

Proof. Multiplying the matrix coboundaries gives

$$D = \delta(c^2 + cd + d^2), \quad B = \delta(ac + bc + bd), \quad T = t^2 + v^2, \quad Q = rt + sv.$$

Substitute $D = \mu^{-1}$, $D = \varepsilon T$ and $B = \varepsilon Q$ from C.d1. Modulo three the first norm is $(c-d)^2$, which is one if it is a unit. Hence $\delta \equiv \mu$ there. The second norm is an odd unit; precisely one of t, v is odd, so its residue modulo four is one, giving $\varepsilon \equiv \mu$. No class field theory is needed for these algebraic implications; identifying the signs as Galois characters is a separate arithmetic comparison.

The error in (73) is not merely possible: use complex conjugation with $\mu = -1$, $\alpha = J$, $\beta = f_G = I$. Then $\delta = \varepsilon = -1$ and both norms are one. The printed equality $c^2 + cd + d^2 = \varepsilon(t^2 + v^2)$ reads $1 = -1$, already false modulo three. The corrected equality is $c^2 + cd + d^2 = \delta\varepsilon(t^2 + v^2)$. The attached matrix check includes this counterexample. This does not affect (88). □

C.d3. The adelic solution locus is a product with global sign compatibility [Proved]. *The matrix systems of scans204–205 have the following exact coordinate description. For each prime p and fixed global signs δ, ε, let $X_p^{\delta,\varepsilon}$ consist of*

$$(\mu_p, \alpha_p, \beta_p, f_p), \quad \mu_p \in \mathbb{Z}_p^\times, \quad \det \alpha_p = \delta, \quad \det \beta_p = \det f_p = 1,$$

satisfying

$$\alpha_p R \alpha_p^{-1} R^{-1} = \varepsilon \beta_p S \beta_p^{-1} S^{-1} \begin{pmatrix} 1 & 0 \\ \mu_p - 1 & 1 \end{pmatrix}.$$

At $p = 2$ additionally require $\alpha_2, \beta_2, f_2 \equiv I \pmod 2$. The unquotiented adelic locus is

$$X = \coprod_{\delta,\varepsilon \in \{\pm 1\}} \left\{ (x_p)_p \in \prod_p X_p^{\delta,\varepsilon} : \mu_3 \equiv \delta \pmod 3,\ \mu_2 \equiv \varepsilon \pmod 4 \right\}.$$

The parameter space of triples as used in the manuscript is its quotient by the simultaneous left action $(\alpha, \beta, f) \mapsto (L_0^n \alpha, L_0^n \beta, L_0^n f)$, $n \in \widehat{\mathbb{Z}}$, and the three independent global *signs on α, β, f. Local independent signs must not silently replace those three global signs. Its matrix quotient is obtained by*

$$(\mu, \alpha, \beta, f) \longmapsto \left[f^{-1} \begin{pmatrix} \mu & \mu B \\ 0 & 1 \end{pmatrix} \right], \qquad B = (\alpha R \alpha^{-1} R^{-1})_{12}.$$

This gives a concrete product-of-local-equations description, but does not assert that every point of this locus comes from an actual profinite automorphism.

Proof. A matrix over $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ is invertible precisely when every local determinant is a unit. All the displayed equations use only addition, multiplication and inversion of unit determinants, so they hold if and only if they hold componentwise. C.d2 supplies the two residual sign compatibilities. The full sign δ must be one of the two elements of the diagonal $\{\pm 1\}$ because it comes from the single coset $\widehat{F} \cdot \{1, J\}$; the same applies to the projective equality sign ε. The simultaneous gauge is coordinatewise because $\widehat{\mathbb{Z}}$ itself is the unrestricted product. The matrix quotient is gauge invariant by the calculation in C.d1 and is unchanged by the allowed signs projectively.

These equations are the $\widehat{\mathbb{Z}}$-points of the corresponding signed affine matrix schemes, localized at the indicated determinants. Their coboundary maps are the maps $F_R(\alpha) = \alpha R \alpha^{-1} R^{-1}$ and $F_S(\beta) = \beta S \beta^{-1} S^{-1}$ studied in §48, scans256–266. Their fibers are right torsors under the centralizers T_R, T_S: if two matrices have the same coboundary, their quotient centralizes the indicated generator, and conversely. This proves the direct relationship to the group-scheme

calculation, without equating a profinite image with every adelic point. At three T_R and at two T_S have non-torus fibers, so the characteristic-zero description as maximal tori cannot be substituted integrally. □

C.d4. The adelic group is an explicit fiber product of normalizers [Proved]. *Put $K = SJ$, so that $K^2 = I$ and $KRK^{-1} = R^{-1}$. For $\mu \in \widehat{\mathbb{Z}}^\times$ write $\delta = \varepsilon_3(\mu)$, $\varepsilon = \varepsilon_2(\mu)$, and $a = (1-\delta)/2 \in \{0,1\}$. The gauge-and-sign quotient in C.d3 is canonically the following group, with componentwise multiplication:*

$$\mathcal{G}_{\mathrm{ad}} = \left\{ ([T],[W],[U]) : \begin{array}{ll} T,W,U \in \mathrm{GL}_2(\widehat{\mathbb{Z}}), & \det T = \det W = \det U = \mu, \\ TRT^{-1} = R^{\delta}, & WSW^{-1} = \varepsilon S, \\ T \bmod 2 = S^a, & W \bmod 2 = U \bmod 2 = I \end{array} \right\}.$$

Each bracket denotes quotient by its own single global sign. Thus this is a closed subgroup of a product of two specified matrix normalizers and $\mathrm{GL}_2(\widehat{\mathbb{Z}})$, modulo three central signs. Equivalently it is the subgroup of the product of its local normalizer systems with common local determinants and the two global sign characters. The isomorphism from the manuscript coordinates is

$$A = \alpha K^a, \qquad T = A^{-1}V, \quad W = \beta^{-1}V, \quad U = f^{-1}V, \qquad V = \begin{pmatrix} \mu & \mu B \\ 0 & 1 \end{pmatrix}.$$

The canonical invariants of actual $\widetilde{M}^!$ therefore define a continuous homomorphism to $\mathcal{G}_{\mathrm{ad}}$. The correction of the J-coset multiplication, omitted in scan193, is

$$\alpha'' = V'\alpha K^a (V')^{-1} \alpha' K^a, \qquad \beta'' = V'\beta (V')^{-1}\beta', \qquad f'' = f'U'f(U')^{-1}, \qquad \mu'' = \mu'\mu.$$

These are identities modulo the recorded global signs and gauge; the exponent a belongs to the right factor u in $u'' = u'u$.

Proof. First work before the sign quotient. Since $\det A = 1$, all three displayed determinants are μ. The identities in C.d1 give $AR^\delta A^{-1} = VRV^{-1}$ and $\beta S\beta^{-1} = \varepsilon VSV^{-1}$. These are exactly the two normalizer equations for T, W. Because α, β, f, V reduce to I modulo two and $K \bmod 2 = S$, the asserted parity conditions follow. A simultaneous left gauge multiplies A, β, f, V by the same L_0^n, leaving T, W, U unchanged. The three global signs act independently on the three resulting factors.

Conversely, given such (T, W, U), choose $V = \mathrm{diag}(\mu, 1)$ and set

$$\alpha = VT^{-1}K^a, \qquad \beta = VW^{-1}, \qquad f = VU^{-1}.$$

Their determinants are $\delta, 1, 1$, and all three reduce to I modulo two. The normalizer equations imply $\alpha R\alpha^{-1} = VRV^{-1}$ and $\beta S\beta^{-1} = \varepsilon VSV^{-1}$. Since $V(SR)V^{-1} = (SR)^\mu$, multiplying these equations gives exactly the loop relation of C.d3. Any other normalized upper-triangular choice is $V_b = \left(\begin{smallmatrix} \mu & b \\ 0 & 1 \end{smallmatrix}\right)$ with $b \in 2\widehat{\mathbb{Z}}$; it differs by the unique gauge $L_0^{-b/2}$. This proves bijectivity with precisely the stated quotient, including the prime two.

The target is a group intrinsically. Products of R-normalizers have exponents $\delta'\delta$; products of S-normalizers have signs $\varepsilon'\varepsilon$. Determinants multiply, and both signs are characters of the determinant. Modulo two, powers of the same S multiply as required. The defining equations are closed, so this is a profinite group. This proves associativity without assuming it for an ad hoc triple law.

To see the actual profinite compatibility, now use the lifts s, r, J in the extended free product and put $\kappa = sJ$, so that $\kappa r \kappa^{-1} = r^{-1}$. Although α may lie outside $\widehat{B}$, the element $A = \alpha\kappa^a$ lies

in $\widehat{B}$, and $u_0(r) = Ar^{\delta}A^{-1}$. Consequently $A'' = u_0'(A)A'$ is a well-typed cocycle formula on $\widehat{B}$. C.d1 permits replacing u_0' by conjugation by V' on its matrix quotient. Thus $A'' = V'A(V')^{-1}A'$; returning to α gives the formula in the statement. The formulas for β, f follow from their ordinary cocycles in $\widehat{F}$. Direct substitution then gives $T'' = T'T$, $W'' = W'W$ and $U'' = U'U$. This also proves continuity. The finite program checks the reconstruction and the corrected product for both sign cosets at primes three and five.

Finally, $N_R^{\delta} = T_R K^a$ and $N_S^{\varepsilon} = T_S J^{(1-\varepsilon)/2}$: multiplication by the indicated fixed normalizer reduces either equation to a centralizer equation. Thus the first two factors are exactly the centralizer schemes of §48 with their inversion components, with the determinant fiber product and the parity condition. Their bad fibers at three and two remain the fibers described in C.d3, not tori obtained by ignoring those primes. □

C.d5. The remaining image problem in the two centralizer factors [Open]. *C.d4 determines the adelic group and the homomorphism to it, but does not show that the image of $\widetilde{M}^!$ is all of $\mathcal{G}_{\mathrm{ad}}$. The projection to U is onto the matrices reducing to I modulo two, modulo the global sign: the geometric image supplies every determinant-one such matrix, and the arithmetic subgroup supplies every determinant through the cyclotomic character. The undecided part is simultaneous realization of the two remaining normalizer coordinates over a prescribed U.*

An exact residual instance is the determinant-one fiber over $U = I$: is every pair of classes $T \in T_R$, $W \in T_S$ with determinant one and reduction I modulo two attained by an actual loop automorphism whose congruence action is trivial? Already its reduction modulo three is not determined by the individual norm equations. C.d4 constructs every such adelic matrix solution, and the finite code verifies its reconstruction and multiplication; none of those computations asserts a lift to an automorphism of the full free profinite group. This is the remaining image statement, not a missing definition of the adelic group.

Manuscript question: §46; fol401; scans204–205.

Established projection and attempted attack. The geometric image is onto the level-two subgroup of $\mathrm{SL}_2(\widehat{\mathbb{Z}})/\{\pm I\}$ by the elementary-matrix argument in D.3. Given a target U of determinant μ, choose an arithmetic representative whose cyclotomic character is μ; its matrix has that determinant by C.d1 and the arithmetic comparison. Multiplying by a geometric matrix adjusts it to the specified U. This proves the projection claim.

For $\mu = 1$, the other two factors are the norm-one centralizers. One can freely choose their local points and use the inverse construction in C.d4, so there is no further matrix obstruction. However, a chosen matrix conjugator need not lift to a conjugator of the prescribed element in $\widehat{B}$: equality after q does not imply equality before q. The kernel of q, which Part B proves nontrivial, is precisely the missing lifting information. Establishing compatible profinite conjugators and the generation condition of B.m1 is the unresolved step. This pinpoints the smallest requested congruence level at which this image fiber is not determined here, rather than replacing it by a global surjectivity claim. □

C.r4. What the congruence proof decides, and its graded limitation [Proved]. *The projective congruence-kernel invariance in §39 and the resulting square-class action follow for all of $\widehat{\mathrm{GT}}_0$ by C.d1, with the normalizations stated there. Therefore these conditions do not distinguish Galois from $\widehat{\mathrm{GT}}$ or from $\widehat{\mathrm{GT}}_0$ in that projective setting. The signs in C.d2 are algebraic consequences, not new arithmetic restrictions. The full adelic conditions do not factor through the single rational pro-unipotent or pro-ℓ quotient used in Part B.*

Manuscript question: §39; fols.298 verso–303; scans21–30; §41; fols.315–316 verso; scans54–57; §46; fols.398–400; scans198–202.

Proof. The conjugating matrix constructed in C.d1 has determinant μ. Changing a genuine representative by an inner element changes it by the corresponding geometric matrix; changing a scalar changes its determinant by a square. Hence its projective outer class is obtained from the square class of μ. The arithmetic subgroup realizes every μ by its cyclotomic character, so the projective congruence outer images of all three groups agree. At individual projective levels this square-class map may have additional kernel, for example at level four; the linear-group table C.r1 is not used to assert otherwise.

For the graded claim, a finite quotient of a pro-ℓ group must be an ℓ-group. The congruence image modulo an odd prime generally has order divisible by several primes, so there is no factorization through that completion. A characteristic-zero unipotent group has no nontrivial finite quotient in the category of unipotent algebraic groups. Individual abelian ℓ-primary reductions can of course be read from suitable integral quotients; for instance the level-four Legendre image is a 2-group. This does not provide the missing factorization for the full adelic representation. After imposing $\lambda = 1$ and tensoring the weight-graded Lie algebra with $\mathbb{Q}_\ell$, the finite signs have zero differential. We classify the full congruence constraints as having no faithful graded form in the specified construction, and the sign differentials as vacuous. □

C.r5. The full modular action retains the group-comparison problem [Proved]. *The cusp-preserving construction of C.a15 induces an injective continuous homomorphism*

$$C = \widehat{\mathrm{GT}}_0 \hookrightarrow \mathrm{Out}(H), \qquad H = \widehat{\mathrm{SL}_2(\mathbb{Z})}.$$

Under this injection the arithmetic, pentagon, and two-cycle images are the respective embedded subgroups $G_{\mathbb{Q}} \subseteq \widehat{\mathrm{GT}} \subseteq \widehat{\mathrm{GT}}_0$. Thus their equality on every congruence outer quotient, proved in C.a16, does not decide their equality on the full modular group. This gives the precise infinite-group interpretation of C(a), beyond the finite table.

Manuscript question: §§45–46; fols.384–391; scans170–184.

Proof. Suppose a class in C gives an inner automorphism of H. Quotienting by its center gives an inner automorphism of $\widehat{B}$ with a conjugator $b \in \widehat{B}$. On $\widehat{F}$ its outer class is the class of b in $\widehat{B}/\widehat{F} = S_3$. But the original class belongs to C, the label-preserving centralizer, and the geometric S_3 acts faithfully on the three labels. Their intersection is the identity. Hence the original class is the identity in $\mathrm{Out}(\widehat{F})$ and in C, proving injectivity. Continuity follows from the continuous finite-quotient constructions and uniqueness of the central lift; a continuous injection from this compact group has closed image. The restriction statements follow from the defining inclusions and arithmetic compatibility in C.a16. This proof identifies the images as subgroups without claiming a decision on their strictness. □

Quadratic signs and the precise arithmetic meaning of Frobenius

C.a1. The two signs are already forced by the matrix norm equations [Proved]. *Let $\mu \in \widehat{\mathbb{Z}}^\times$, let $t, u, c, d \in \widehat{\mathbb{Z}}$, and let $\epsilon_2, \epsilon_3 \in \{1, -1\}$ be diagonal signs. If*

$$\mu(t^2 + u^2) = \epsilon_2, \qquad \mu(c^2 + cd + d^2) = \epsilon_3, \tag{C.a1.1}$$

then

$$\epsilon_2 = \chi_{-4}(\mu), \qquad \epsilon_3 = \chi_{-3}(\mu),$$

where $\chi_{-4}(a)$ is 1 or -1 according as $a \equiv 1$ or $-1 \pmod 4$, and $\chi_{-3}(a)$ is 1 or -1 according as $a \equiv 1$ or $-1 \pmod 3$. These are the signs of §44, folios 365 verso–367, scans 152–155, in (63), (66)–(68). In particular they do not furnish an additional arithmetic restriction on a solution for which (C.a1.1) has already been proved.

Proof. A square modulo 4 is 0 or 1. Consequently a unit represented by t^2+u^2 is 1 (mod 4). Since both sides of its equation are units, $\mu \equiv \epsilon_2 \pmod 4$. Modulo 3 we have $c^2+cd+d^2=(c-d)^2$. Its nonzero value is therefore 1, giving $\mu \equiv \epsilon_3 \pmod 3$. Each modulus distinguishes the two diagonal signs. These are exact calculations in finite rings, also exhaustively checked in `code/arithmetic_checks.py`. □

C.a2. Class field interpretation, and the sign ϵ' [Proved]. *For $\gamma \in G_{\mathbb{Q}}$ put $\mu = \chi_{\mathrm{cyc}}(\gamma)$, the cyclotomic character for the convention $\gamma(\zeta_m) = \zeta_m^{\mu \bmod m}$. Then*

$$\epsilon_2(\gamma) = \frac{\gamma(i)}{i}, \qquad \epsilon_3(\gamma) = \frac{\gamma(\sqrt{-3})}{\sqrt{-3}}, \qquad \epsilon_2(\gamma)\epsilon_3(\gamma) = \frac{\gamma(\sqrt{3})}{\sqrt{3}}. \tag{C.a2.1}$$

Here the quotients are ordinary signs, and changing any of the square roots does not change its character. If the matrix in scan 155 is written in the final form of scan 158,

$$S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad A = \mu \begin{pmatrix} u & \epsilon_2 t \\ -t & \epsilon_2 u \end{pmatrix},$$

then $ASA^{-1} = \epsilon_2 S$. Thus the ϵ' of (67) is ϵ_2. If $g\rho(g)\rho^2(g) = \epsilon_2$ in the chosen lift and $A\rho(A)^{-1} = \epsilon'' g$, then also $\epsilon'' = \epsilon_2$, as in (68 bis). For an odd prime p, its value on arithmetic Frobenius in $\mathbb{Q}(i)/\mathbb{Q}$ is $+1$ exactly when $p \equiv 1 \pmod 4$.

Manuscript question: §44; folio367; scan155; folio368 verso; scan158.

Proof. The first equality follows from $i = \zeta_4$ and the fact that the units modulo 4 are $1, -1$. Write $\sqrt{-3} = 2\zeta_3 + 1$; exchanging ζ_3 and ζ_3^{-1} negates this expression. The second equality follows, and multiplication gives the third because $i\sqrt{-3} = -\sqrt{3}$ after a choice of roots. These are the Artin characters of the indicated quadratic extensions: at an unramified prime arithmetic Frobenius sends the residue of a root of unity to its pth power. This proves the Frobenius formula without an unspecified appeal to class field theory. Geometric Frobenius has the same value because these characters have order two. Direct multiplication gives $AS = \epsilon_2 SA$; A is invertible by (C.a1.1), so the conjugation formula follows. The product of the three ρ-translates of a coboundary $A\rho(A)^{-1}$ is 1; the same product for $\epsilon'' g$ is $\epsilon''\epsilon_2$, giving the asserted second sign.

Alternatively, the endomorphism $[i]$ of $y^2 = x^3 - x$ is $(x, y) \mapsto (-x, iy)$; Galois conjugation takes it to $[i]$ or $[-i]$ according to its action on i. This realizes the same sign intrinsically on the Tate modules, whenever the matrix represents this complex multiplication symmetry. The analogous order-three symmetry of $y^2 = x^3 - 1$ is defined over $\mathbb{Q}(\zeta_3)$. □

The table, verified by the program, is

$\mu \bmod 12$	1	5	7	11
ϵ_2	$+1$	$+1$	-1	-1
ϵ_3	$+1$	-1	$+1$	-1
$\epsilon_2\epsilon_3$	$+1$	-1	-1	$+1$

The prime 2 is ramified in $\mathbb{Q}(i)$, and the prime 3 in $\mathbb{Q}(\sqrt{-3})$; there is no unramified Frobenius assertion there. Nor is a global Frobenius multiplier literally the integer p in $\widehat{\mathbb{Z}}^\times$: p is not a unit at p. The statement uses its residue modulo the conductor, away from p.

C.a3. The 2-adic square criterion in scan 152 [Refuted]. *The sentence in §44, folio 365 verso, scan 152 identifying $\epsilon_p(\mu) = 1$ with $\mu_p \in (\mathbb{Z}_p^\times)^2$ cannot apply to the sign ϵ_2 defined there. The strongest valid uniform replacement is the explicit pair of conductor-4 and conductor-3 characters in C.a1; for $p = 2$ the square test requires modulus 8.*

Proof. For example $5 \equiv 1 \pmod 4$, so $\chi_{-4}(5) = 1$, whereas the square of every odd integer is $1 \pmod 8$; therefore $5 \notin (\mathbb{Z}_2^\times)^2$. If an actual profinite unit is wanted, choose its 2-component equal to 5 and every odd component equal to 1. The finite counterexample is printed by the program. Conversely an odd 2-adic unit is a square exactly when it is $1 \pmod 8$: necessity is as above; for sufficiency, starting with a square root modulo 8, replace an odd root r modulo 2^k by $r + 2^{k-1}$ to change its square by $2^k r$ modulo 2^{k+1}, and choose the sign of the correction to lift one step. Compactness gives a 2-adic root. For an odd prime the usual lifting argument has invertible derivative $2r$ and proves the square criterion modulo p; at $p = 3$ it agrees with C.a1. □

No handwriting is silently changed by this correction. The scan-155 image supports the displayed modulus 4; the mathematical error is the overly uniform square interpretation of a different passage, not that modulus. The later formula $\mu \equiv \epsilon_2 \pmod 2$ in scan 165 determines no sign and must be strengthened to modulus 4 if it is to do the work claimed for it.

Faithfulness on the modular group and its centralizer

Throughout this subsection $U = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$, $F = \pi_1(U_{\overline{\mathbb{Q}}}) \simeq \widehat{F}_2$, $H = \widehat{\mathrm{SL}_2(\mathbb{Z})}$, and $P = H/\langle -I \rangle \simeq \widehat{\mathrm{PSL}_2(\mathbb{Z})}$. Every automorphism is continuous and every outer automorphism is taken modulo conjugation by an element of its target. Compatible geometric base points identify F with $\ker(P \to \mathfrak{S}_3)$, up to P-conjugacy. We use the arithmetic actions arising from the stacks $\mathcal{M}_{1,1}$ and its rigidification by $\{\pm 1\}$.

C.a4. Exactly what faithfulness modulo geometric monodromy means [Known]. *Let $\Delta = \pi_1(E_{\overline{\mathbb{Q}}} \setminus \{0\})$ for an elliptic curve $E/\mathbb{Q}$. Let*

$$\Psi : \pi_1(\mathcal{M}_{1,1}) \longrightarrow \mathrm{Out}(\Delta), \quad N_{\mathrm{ar}} = \operatorname{im} \Psi, \quad \Sigma = \Psi(\pi_1(\mathcal{M}_{1,1,\overline{\mathbb{Q}}})).$$

Then the canonical map $G_{\mathbb{Q}} \to N_{\mathrm{ar}}/\Sigma$ is an isomorphism of profinite groups. This is faithfulness modulo the geometric orientation-preserving monodromy image, as asked in §35–36, folios 254–255 verso, scans 502–505. It is not by itself the assertion $G_{\mathbb{Q}} \hookrightarrow \mathrm{Out}(H)$.

Proof. The chosen elliptic curve gives a section over $G_{\mathbb{Q}}$ of the arithmetic stack extension, and hence identifies its action on Δ with the usual elliptic-curve outer action. Apply Hoshi and Mochizuki (2011, Corollary 6.4) with $k = \mathbb{Q}$ and $(g, r) = (1, 1)$. Its precise conclusion is $\rho_{E\setminus 0}^{-1}(\Sigma) \subseteq \ker(G_{\mathbb{Q}} \to G_{\mathbb{Q}}) = 1$. Surjectivity of $G_{\mathbb{Q}} \to N_{\mathrm{ar}}/\Sigma$ follows from surjectivity of the stack extension onto $G_{\mathbb{Q}}$. A continuous bijection between compact Hausdorff groups is a topological isomorphism. The theorem concerns Σ, whether or not it has already been identified with H; Hoshi and Mochizuki (2011, Corollary 6.5) explicitly separates injectivity of total monodromy from that geometric congruence question. It also applies with any number field in place of $\mathbb{Q}$. □

C.a5. Faithfulness on H and already on P [Proved]. *The natural continuous outer actions*

$$G_{\mathbb{Q}} \longrightarrow \mathrm{Out}(P), \qquad G_{\mathbb{Q}} \longrightarrow \mathrm{Out}(H)$$

are injective. This proves the stronger statement eventually written in §36, folio 270 verso, scan 530, as well as answering the earlier doubt.

Proof. The ordered level-two covering of the rigidified modular stack is defined over $\mathbb{Q}$. It gives the Galois-compatible exact sequence $1 \to F \to P \to \mathfrak{S}_3 \to 1$ of scans 521–523. If γ acts on P by conjugation by p, its action on F, modulo F-conjugacy, equals the anharmonic permutation represented by $pF \in \mathfrak{S}_3$. On $F^{\mathrm{ab}} \simeq \widehat{\mathbb{Z}}^2$, Galois acts by the scalar $\chi_{\mathrm{cyc}}(\gamma)$, whereas $\mathfrak{S}_3$ acts by its

standard action on the three vectors e_0, e_1, e_∞ with $e_0 + e_1 + e_\infty = 0$. A scalar permuting these three distinct vectors is the identity: it cannot send e_0 to e_1 or $-e_0 - e_1$, so fixes e_0, and then fixes all three. Thus $p \in F$. Consequently γ acts trivially in $\mathrm{Out}(F)$. Belyi's faithfulness theorem (Belyi, 1980, Theorem 4 and its Corollary), or the directly applicable Hoshi and Mochizuki (2011, Corollary 6.3(i)), gives $\gamma = 1$. An inner action on H induces an inner action on P, so the second assertion follows from the first. This argument needs no assertion that $\mathrm{Out}(H)$ is $\widehat{\mathrm{GT}}$. □

C.a6. The tripod centralizer used below [Known]. *For its standard arithmetic action,*

$$C_{\mathrm{Out}(F)}(G_{\mathbb{Q}}) = \mathfrak{S}_3.$$

Manuscript question: §36; folios266–270 verso; scans521–530 (cote140-2).

Proof. We give the reduction to the primary theorem actually opened, rather than importing the assertion from LM3. In Hoshi and Mochizuki (2011, Corollary 6.3(iii),(iv)), take $X = U$, $k = k_0 = \mathbb{Q}$, and $n = 1$. The tripod is of quasi-Belyi type, and the fibration-preserving qualification Out^F is empty for one point. The theorem says that $A = \mathrm{Aut}(U_{\overline{\mathbb{Q}}}/\mathbb{Q}) = G_{\mathbb{Q}} \times \mathfrak{S}_3$ embeds and is commensurably terminal in $\mathrm{Out}(F)$: its commensurator equals A. If a centralizes $G_{\mathbb{Q}}$, the intersection $A \cap aAa^{-1}$ contains $G_{\mathbb{Q}}$, of index six in each group. Hence $a \in A$. It remains to see that $Z(G_{\mathbb{Q}}) = 1$. Choose the real algebraic closure $R \subset \overline{\mathbb{Q}} \subset \mathbb{C}$ and its complex conjugation c. An element commuting with c preserves R. Every automorphism of R preserves its order because positive elements are squares, and fixes its dense rational subfield, hence fixes R pointwise. Thus $C_{G_{\mathbb{Q}}}(c) = \{1, c\}$. The image of c in the splitting field of $X^3 - 2$ is a noncentral transposition of $\mathfrak{S}_3$, so c is not central in $G_{\mathbb{Q}}$. Therefore $Z(G_{\mathbb{Q}}) = 1$, and the centralizer inside A is exactly its second factor, as asserted. □

C.a7. The centralizer on the modular group is a central transvection [Proved]. *With the arithmetic actions of C.a5,*

$$C_{\mathrm{Out}(P)}(G_{\mathbb{Q}}) = 1, \qquad C_{\mathrm{Out}(H)}(G_{\mathbb{Q}}) \simeq C_2. \tag{C.a7.1}$$

The second group is generated by the outer class of

$$\theta(h) = (-I)^{\eta(h)} h, \qquad \eta : H \longrightarrow C_2, \tag{C.a7.2}$$

where η is the unique nonzero character, equivalently the sign of reduction modulo 2 in $\mathfrak{S}_3$.

Manuscript question: §36; folios266–270 verso; scans521–530 (cote140-2).

Proof. We spell out the group-theoretic comparison. Since $P = C_2 \widehat{*} C_3$, every surjection $P \to \mathfrak{S}_3$ sends the order-two generator to a transposition and the order-three generator to a three-cycle. The six choices are conjugate, so their common kernel F is characteristic. The induced action of P/F on F^{ab} is faithful, and the free profinite group F is centre-free; thus $C_P(F) = 1$. The centre-free property is the Corollary to Lemma 1 of Belyi (1980, §1, pp. 248–249); its proof uses the universal lifting property of $\widehat{F}_2$ and noncommuting central extensions of elementary abelian finite quotients. An automorphism of P restricting to the identity on F is the identity: for $p \in P$, its discrepancy $\alpha(p)p^{-1}$ centralizes F. It follows that restriction gives an injection

$$\mathrm{Out}(P) \hookrightarrow N_{\mathrm{Out}(F)}(\mathfrak{S}_3)/\mathfrak{S}_3. \tag{C.a7.3}$$

If $a \in \mathrm{Out}(P)$ centralizes Galois, choose a restriction $\widetilde{a} \in N_{\mathrm{Out}(F)}(\mathfrak{S}_3)$. For every $\gamma \in G_{\mathbb{Q}}$ the commutator $[\widetilde{a}, \rho_F(\gamma)]$ belongs to $\mathfrak{S}_3$. Both $\rho_F(\gamma)$ and its $\widetilde{a}$-conjugate centralize $\mathfrak{S}_3$, since the

anharmonic maps are defined over $\mathbb{Q}$. This commutator consequently lies in $Z(\mathfrak{S}_3) = 1$. C.a6 yields $\widetilde{a} \in \mathfrak{S}_3$, and (C.a7.3) gives $a = 1$.

Now $Z(P) = 1$ by $C_P(F) = 1$, so $Z(H) = \langle -I \rangle$ is characteristic. If an automorphism of H is inner after passage to P, remove the inner automorphism by lifting its conjugator to H. The result has the form $h \mapsto (-I)^{b(h)}h$ for a continuous character $b : H \to C_2$ vanishing on $-I$. Abelianizing the presentation $S^4 = 1$, $S^2 = (ST)^3$ gives $H^{\mathrm{ab}} = \mathbb{Z}/12$, with T a generator and $-I$ corresponding to 6. Therefore exactly two such characters occur, zero and η. Their automorphisms are invertible; $\theta^2 = 1$ because $\eta(-I) = 0$. The nonzero one is not inner, since it acts on H^{ab} by multiplication by 7, rather than 1. It commutes with every automorphism of H, because both $-I$ and η are characteristic. The kernel of $\mathrm{Out}(H) \to \mathrm{Out}(P)$ is therefore this central C_2. Combining it with the already proved centralizer on P gives (C.a7.1). □

The program checks (C.a7.2) on every element and a generating pair of $\mathrm{SL}_2(\mathbb{Z}/n)$ for $n = 2, 4, 6, 8, 12, 24$, and searches every inner conjugator. It is trivial at $n = 2$ and noninner at each of the other five levels. This finite check supports the explicit formula; the proof above is the assertion about the full profinite group. The theorem of Minamide and Nakamura (2019, Theorem B, Theorem 4.3 and Corollary C) concerns four or more strands and $\widehat{\Gamma}_{1,2}$. It must not be specialized to three strands or used to identify $\mathrm{Out}(H)$ with $\widehat{\mathrm{GT}}$.

The lifts through $\{\pm 1\}$ and the withdrawn nonexistence doubt

C.a8. The obstruction and the torsor of lifts [Proved]. *Let $1 \to A \to E \xrightarrow{q} B \to 1$ be a central extension of profinite groups with $A = C_2$, and let $j : D \to B$ be continuous. The obstruction to a continuous lift $\widetilde{j} : D \to E$ is the pullback class $j^*[E] \in H^2_{\mathrm{cont}}(D, A)$. When a lift exists, its set is a torsor under $H^1_{\mathrm{cont}}(D, A) = \mathrm{Hom}_{\mathrm{cont}}(D, A)$. For $D = \widehat{F}_2$ this obstruction vanishes and there are exactly four lifts of a fixed j. For $D = F_2$ the same statements hold in the discrete category.*

Manuscript question: §36; folios267–268; scans523–525 (cote140-2).

Proof. A finite-kernel surjection of profinite groups admits a continuous set-theoretic section: take an open subgroup meeting A trivially, use its inverse section over its open image, and choose representatives on finitely many clopen cosets. Such a section gives a continuous factor set $c \in Z^2(B, A)$. A lift is a function $b : D \to A$ with $\delta b = j^*c$ (up to the immaterial inverse convention for c). This proves the obstruction criterion. Given two lifts, their pointwise ratio is a continuous homomorphism $D \to A$ because A is central; every such homomorphism changes a lift to another, freely and transitively. A lift for $\widehat{F}_2$ is obtained by choosing arbitrary lifts of its two free generators and using the universal property. The choices in each fiber are exactly two. The same argument works for the discrete free group. □

For specified genuine actions of a group Γ on this diagram, the set of equivariant lifts, if nonempty, is the fixed torsor under $H^1(D, A)^\Gamma$. A fixed point is an extra condition on an affine Γ-action, not an automatic consequence of $H^2(D, A) = 0$. For outer arithmetic actions one must use the full arithmetic extensions; scan 524 makes precisely this correction. The next result solves that extra condition in the case actually considered there.

C.a9. The family doubted on scan 524 exists over $\mathbb{Q}$ [Refuted]. *The nonexistence suggested in §36, folio 267 verso, scan 524 is false. Indeed it is explicitly withdrawn in the checked continuation at scans 527–528, folio 269. Over $U = \mathrm{Spec}\,\mathbb{Q}[\lambda, 1/\lambda, 1/(\lambda - 1)]$,*

$$E_\lambda : \quad Y^2Z = X(X - Z)(X - \lambda Z) \tag{C.a9.1}$$

with origin $[0:1:0]$ *and ordered nonzero two-torsion points* $(0,0),(1,0),(\lambda,0)$ *is the required family. It gives a morphism* $U \to \mathcal{M}_{1,1}$ *lifting the tautological map to its* $\{\pm1\}$*-rigidification, and therefore a morphism of arithmetic fundamental-group extensions over* $G_{\mathbb{Q}}$*.*

Proof. The discriminant is $16\lambda^2(\lambda-1)^2$, a unit on U; the projective cubic is smooth, with the stated origin and three distinct nonzero points of order two. Its quotient by negation is the projective X/Z-line branched at $0,1,\lambda,\infty$, so its ordered level-two invariant is exactly λ. Functoriality of étale π_1 gives the asserted homomorphism, continuous and well-defined up to conjugation by the geometric target group. This is the homomorphism (40) on scan 528. The polynomial discriminant is also checked symbolically by the program. The construction works over $\mathbb{Z}[1/2]$ with λ and $\lambda-1$ inverted, including characteristic 3; it makes no assertion about ordinary full level two in characteristic 2. □

C.a10. All four geometric lifts are arithmetic and geometric [Proved]. *Fix the quotient map* $U \to \mathcal{M}_{1,1}^{\mathrm{rig}}$ *just used and fix the Legendre lift as origin. The isomorphism classes of its lifts through the* C_2*-gerbe* $\mathcal{M}_{1,1} \to \mathcal{M}_{1,1}^{\mathrm{rig}}$ *are in bijection with*

$$H^1_{\mathrm{et}}(U,\mu_2) = \mathbb{Q}^\times/\mathbb{Q}^{\times 2} \ \oplus\ (\mathbb{Z}/2)^2. \tag{C.a10.1}$$

They are realized by

$$a\lambda^e(\lambda-1)^f Y^2Z = X(X-Z)(X-\lambda Z), \quad a \in \mathbb{Q}^\times/\mathbb{Q}^{\times 2}, \quad e,f \in \{0,1\}. \tag{C.a10.2}$$

The restriction to geometric fundamental groups gives exactly the four lifts $F \to H$ *of* $F \hookrightarrow P$ *in C.a8. All four are fixed under the outer* $G_{\mathbb{Q}}$*-action, and all arise from (C.a10.2) already over* $\mathbb{Q}$*.*

Manuscript question: §36; folios267–269 verso; scans523–528 (cote140-2).

Proof. A choice of object in a gerbe banded by μ_2 identifies its other objects with μ_2-torsors by their sheaves of isomorphisms. Here the Legendre family is that object, and the band is μ_2: after rigidifying by all labels, the only automorphisms are ±1. Let $R = \mathbb{Q}[\lambda,\lambda^{-1},(\lambda-1)^{-1}]$. This is a localization of a principal ideal domain, so $\mathrm{Pic}(R)=0$; its units are exactly $a\lambda^m(\lambda-1)^n$, with $a \in \mathbb{Q}^\times$ and $m,n \in \mathbb{Z}$. The Kummer sequence therefore gives (C.a10.1), and descent along $w^2 = a\lambda^e(\lambda-1)^f$ gives precisely the quadratic twists (C.a10.2). Over $\overline{\mathbb{Q}}$ the constant a becomes a square, whereas λ and $\lambda-1$ have independent odd valuations at 0 and 1; they give the two independent characters of F into C_2. Their four combinations exhaust C.a8. They are defined over $\mathbb{Q}$, so all four have arithmetic representatives and are Galois-fixed. They remain distinct modulo conjugation: a conjugator identifying two lifts while fixing their quotient inclusion would project to an element of $C_P(F)=1$, and hence would be central in H and act trivially. □

C.a11. The precise anabelian converse here needs no section conjecture [Proved]. *For the fixed tautological quotient map in scans 523–526, every lift of the arithmetic fundamental-group homomorphism through* $\{\pm1\}$ *is induced by a geometric lift, up to its stated geometric inner ambiguity; the same is true of every outer-Galois-compatible geometric lift* $F \to H$*. No assertion that every section of* $\pi_1(\mathcal{M}_{1,1}) \to G_{\mathbb{Q}}$ *is geometric is used or implied.*

Manuscript question: §36; folios267–269; scans523–528 (cote140-2).

Proof. For arithmetic fundamental groups, fix the lift furnished by C.a9. The ratio of any other lift is a continuous C_2-character of $\pi_1(U)$, by C.a8. Finite étale torsors are classified by the fundamental group, so

$$H^1_{\mathrm{cont}}(\pi_1(U),C_2) = H^1_{\mathrm{et}}(U,\mu_2).$$

The corresponding twist in C.a10 induces that character and hence that lift. For geometric groups, C.a10 realizes all four possible lifts and therefore all Galois-compatible ones. The weaker datum of a geometric lift does not determine the arithmetic lift uniquely: the kernel of restriction in (C.a10.1) is $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$. This explains exactly the ambiguity that the manuscript starts to detect on scan 524. The H^2 comparison suggested on scan 526 is unnecessary after the explicit neutralization by the Legendre family; if one uses it, both obstruction classes are already zero. □

Thus the particular request C(h) is settled more strongly than a reduction to a section conjecture. A classification of arbitrary sections of a modular curve is a separate assertion with different input data. It would be a mistake to transfer an unverified statement about such sections to this finite-band gerbe problem, or conversely to deduce a section theorem from the elementary lifting result proved here.

A characterization of the arithmetic image with specified additional data

C.a12. Pop's theorem gives an exact extendability criterion [Known]. *Let $X = \mathbb{A}^2_{\overline{\mathbb{Q}}}$ with coordinates u, v, and let $\mathcal{V}$ be the category with objects U and a basis of sufficiently small nonempty principal open subsets of X defined over $\mathbb{Q}$. Include all inclusions between these opens and the maps to U defined by the three rational functions $u, v, v-u$, whenever regular with values in U. Put $\pi_{\mathcal{V}}(Y) = \pi_1(Y_{\overline{\mathbb{Q}}})$ in the category of profinite groups and outer continuous homomorphisms. Then evaluation at U identifies*

$$\mathrm{Aut}(\pi_{\mathcal{V}}) \xrightarrow{\sim} \rho_U(G_{\mathbb{Q}}) \subset \mathrm{Out}(F). \tag{C.a12.1}$$

In particular an element of any of Grothendieck's candidate groups acting on F is arithmetic exactly when its outer action extends to such a compatible family. This answers the characterization question of §45, scans 170,178,185, by specifying data beyond the two types.

Manuscript question: §§45–46; fols.384,388,391 verso; cote140-3 scans170,178,185.

Proof. We verify Hypothesis (H) of Pop (2018, Definition/Remark 2.2). The function field $\overline{\mathbb{Q}}(u, v)$ has transcendence degree 2 and is generated by the indicated functions. A compatible birational automorphism has one common automorphism of the tripod on all three maps. Its restriction to constants is some $\sigma \in G_{\mathbb{Q}}$; after removing σ, the tripod automorphism is one of the six anharmonic fractional linear functions r. Compatibility forces

$$r(v) - r(u) = r(v - u).$$

Only $r(t) = t$ satisfies this rational identity. For $r = 1 - t$ its discrepancy is -1; for the four remaining nonidentity functions it is a nonzero rational function, listed in the exact symbolic output of the program. Thus the birational automorphism fixes u, v and is exactly σ on constants. Conversely every σ fixes every rationally defined open and the three maps. This proves rigidity and hence Hypothesis (H). Apply Pop (2018, Theorem 2.7(1)); characteristic zero makes its valuation-tame fundamental groups equal to the full groups under consideration. It gives $G_{\mathbb{Q}} \simeq \mathrm{Aut}(\pi_{\mathcal{V}})$. Evaluation is the usual action on F, faithful by C.a5's tripod input, proving (C.a12.1). □

The extra datum is not a finite list of finite quotients: even though there are only three designated tripod functions, the neighborhood category includes complements of arbitrarily many divisors in a surface. The theorem does not say that checking the two original modular types or any fixed level in the requested computations suffices.

C.a13. The combinatorial characterization determines a conjugacy class [Known]. *Let $\mathcal{A}$ be the collection of closed subgroups of $\widehat{\mathrm{GT}}$ satisfying the AA-property of Hoshi et al. (2025, Definition 5.12), including its COF, RGC and ArBC prerequisites. The maximal elements of $\mathcal{A}$ are exactly the $\widehat{\mathrm{GT}}$-conjugates of $\rho_U(G_{\mathbb{Q}})$. This is a characterization of a conjugacy class of subgroups, not a canonical distinguished embedding or an elementwise equation on (λ, f).*

Manuscript question: §§45–46; fols.384,388,391 verso; cote140-3 scans170,178,185.

Proof. The precise theorem is Hoshi et al. (2025, Theorem 5.17(ii), also Theorem C(ii)). We record its reduction to prevent a stronger reading. The arithmetic subgroup itself satisfies AA (their Remark 5.12.1). Their Theorem 5.15(iv) puts any AA subgroup in a conjugate of the arithmetic subgroup. Their Lemma 5.16 rules out a proper containment between the relevant conjugates of the latter. Thus maximal AA subgroups and exactly those conjugates coincide. Transport the statement along the chosen identification $F \simeq \widehat{\pi}_{0,3}$ and, when appropriate, along the quotient from Grothendieck's extension group to its outer loop action. Transport changes the embedded subgroup by the declared conjugacy ambiguity, and changes nothing about maximality or closure. □

The input from which the theorem reconstructs this collection is the abstract topological group of the second configuration space of the tripod, $\pi_1(M_{0,5,\overline{\mathbb{Q}}})$, together with its intrinsically reconstructed structure. Definition 5.12 quantifies over all normal open subgroups and cusp sets and requires nondegenerate Kummer data and compatible fields of functions. These conditions are substantially more than the order-two and order-three equations on one copy of F. We opened the stated theorem, the reduction just cited, and Definition 5.12; we do not claim an independent verification of all 125 pages of the argument.

C.a14. The arithmetic conclusions have the following graded limits [Proved]. *On the subgroup $\lambda = 1$, the characters ϵ_2 and ϵ_3 of C.a1–C.a2 are identically $+1$. Their infinitesimal conditions are therefore vacuous. The order-two transvection in C.a7, the finite-band lifting choices in C.a8–C.a11, and the inverse-limit extendability condition of C.a12 are not recovered by merely taking rational weight-graded Lie algebras of F.*

Manuscript question: §44; fols.365 verso–368 verso; cote140-3 scans152–158; §36; fols.267–269 verso; cote140-2 scans523–528.

Proof. The first assertion is evaluation of the two characters at 1. Any map from a uniquely divisible unipotent group over a characteristic-zero field to a group of exponent two is trivial; equivalently the tangent space of the constant finite group C_2 is zero. Consequently neither four distinct C_2-twists nor the nonzero transvection can be encoded by that tangent space. The criterion C.a12 includes the full geometric fundamental groups of every object and all comparison maps, whereas the stated graded construction only retains the lower-central successive quotients of F. No implication from the latter data to the former is part of Pop's theorem. This last observation identifies the missing comparison map; it is not an assertion that no other Lie-theoretic reconstruction could ever exist. □

Removing the central ambiguity: the full linear modular action

C.a15. The unique cusp-preserving lift from the projective modular group [Proved]. *Write*

$$H = \widehat{\mathrm{SL}_2(\mathbb{Z})} = \langle S, R \mid S^4 = 1,\ S^2 = R^3 \rangle_{\mathrm{prof}}, \quad z = S^2 = R^3 = -I,$$

$P = H/\langle z\rangle = \langle s, r \mid s^2 = r^3 = 1\rangle_{\mathrm{prof}}$, and $T = SR$, $e = sr$. Let $u \in \mathrm{Aut}(P)$ satisfy

$$u(e) = f^{-1} e^{\mu} f, \qquad \mu \in \widehat{\mathbb{Z}}^{\times}, \quad f \in P. \tag{C.a15.1}$$

For the linear congruence coordinates we choose

$$q_H(S) = -S_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad q_H(R) = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad q_H(T) = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = E,$$

where S_0 is the matrix used for the projective involution in C.d1. Changing its lift from S_0 to $-S_0$ leaves every projective cocycle and conjugation formula of C.d1 unchanged and makes the chosen full cusp generator map to the unipotent E. Then precisely one of the two lifts of u to $\mathrm{Aut}(H)$ satisfies

$$\widetilde{u}(T) = \widetilde{f}^{-1} T^{\mu} \widetilde{f}, \tag{C.a15.2}$$

where $\widetilde{f}$ is any lift of f. It sends S to a conjugate of $S^{\mu \bmod 4}$ and R to a conjugate of $R^{\mu \bmod 6}$. The two lifts differ by the central transvection of C.a7. The lift (C.a15.2) respects composition and inner automorphisms, and hence induces an action of $\widehat{\mathrm{GT}}_0$ on H by outer automorphisms.

Manuscript question: §36; fols.266–270 verso; cote140-2 scans521–530; §§45–46; fols.387–391; cote140-3 scans176–184.

Proof. Use the same finite-torsion conjugacy theorem as in C.d1. The element $u(s)$ is conjugate to s, and $u(r)$ to either r or r^{-1}. Abelianizing (C.a15.1) in $P^{\mathrm{ab}} = C_2 \times C_3$ shows which: $u(r)$ is conjugate to $r^{\mu \bmod 3}$. Thus choose $a, b \in P$ with $u(s) = asa^{-1}$ and $u(r) = br^{\mu}b^{-1}$, and choose lifts $\widetilde{a}, \widetilde{b} \in H$. The assignments

$$S \longmapsto \widetilde{a} S^{\mu} \widetilde{a}^{-1}, \qquad R \longmapsto \widetilde{b} R^{\mu} \widetilde{b}^{-1} \tag{C.a15.3}$$

respect the displayed presentation of H: the square of the first image and cube of the second are both z, since μ is odd. They therefore define a continuous endomorphism $\widetilde{u}$ with quotient u and with z fixed. It is an automorphism: its kernel maps trivially to P and meets $\langle z\rangle$ trivially; for surjectivity lift a preimage in P and correct by z in the center. Compactness makes the resulting bijection a topological isomorphism.

The quotient equation implies $\widetilde{u}(T) = z^{\epsilon} \widetilde{f}^{-1} T^{\mu} \widetilde{f}$ for some $\epsilon \in \{0, 1\}$. In the cyclic group $H^{\mathrm{ab}} = \mathbb{Z}/12$, the class t of T generates, the class of S is $3t$, that of R is $-2t$, and that of z is $6t$. These identities follow by abelianizing $2S = 3R$ and $4S = 0$; their converse defines a surjection onto $\mathbb{Z}/12$, so no extra relation has been imposed. Formula (C.a15.3) therefore sends t to $3\mu t - 2\mu t = \mu t$. Comparison with the previous display gives $6\epsilon = 0 \pmod{12}$, so $\epsilon = 0$. In particular the result is independent of the choices of a, b and their lifts.

Any two lifts of u differ by a character $H \to C_2$ fixing z, as in C.a7. There are exactly two; the nontrivial one is η and $\eta(T) = 1$. It changes the right side of (C.a15.2) by z, so only one lift satisfies that equation. The lift of a composite satisfies the same condition with multiplier the product of the two multipliers; uniqueness proves compatibility with composition. A lift of an inner automorphism of P is the inner automorphism defined by a lift of its conjugator, and satisfies the cusp condition; therefore the construction descends to outer classes. Apply it to the automorphisms of P in B.m1–B.m2. □

The uniqueness in this assertion is a uniqueness of a central lift with its full cusp generator specified. It does not identify or split the different, noncentral extension asked about in Part D. On outer classes one can equivalently require that T be sent to a conjugate of T^{μ}: T^{μ} and zT^{μ} have different abelianizations, so they cannot be conjugate.

C.a16. The congruence descent holds in the full linear group [Proved]. *Let $q_H : H \twoheadrightarrow \mathrm{SL}_2(\widehat{\mathbb{Z}})$ be the full congruence homomorphism, and take the matrices $U(u)$ of C.d1. For the cusp-preserving lift of C.a15,*

$$q_H \circ \widetilde{u} = \mathrm{Int}(U(u)) \circ q_H. \tag{C.a16.1}$$

Thus the full, not only projective, congruence kernels are preserved by this action of $\widehat{\mathrm{GT}}_0$. Its outer congruence image, that of $\widehat{\mathrm{GT}}$, and the arithmetic image agree at every finite level: they are the conjugations by $\mathrm{GL}_2(\mathbb{Z}/n)$ modulo the inner conjugations by $\mathrm{SL}_2(\mathbb{Z}/n)$, with determinant parameter $(\mathbb{Z}/n)^\times/((\mathbb{Z}/n)^\times)^2$.

Manuscript question: §39; fols.298 verso–301; cote140-3 scans21–26; §41; fols.315–316 verso; scans54–57.

Proof. First normalize u_0 so that $u_0(e) = e^\mu$. In C.d1 its matrix is

$$V = \begin{pmatrix} \mu & \mu B \\ 0 & 1 \end{pmatrix}, \qquad q_H(T) = E = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

The two continuous homomorphisms $q_H \circ \widetilde{u}_0$ and $\mathrm{Int}(V) \circ q_H$ have equal images after passage to the quotient by the single global $\{\pm I\}$, by C.d1. Their pointwise ratio is therefore a continuous character $\kappa : H \to \{\pm I\}$: centrality makes this ratio multiplicative. But C.a15 gives $q_H(\widetilde{u}_0(T)) = E^\mu$, while direct matrix multiplication gives $VEV^{-1} = E^\mu$. Hence $\kappa(T) = 1$. Every character to C_2 factors through $H^{\mathrm{ab}} = C_{12}$, generated by T, and consequently $\kappa = 1$. This proves (C.a16.1) for u_0. Removing the normalization by the same inner conjugator as in C.d1 proves it for u. The alternative lift differs by θ and does not satisfy this equality: it sends T to zT^μ.

At level n, equality (C.a16.1) proves descent before an action on the quotient is asserted. A matrix commuting with all of $\mathrm{SL}_2(\mathbb{Z}/n)$ commutes in particular with both elementary unipotents; comparing entries shows that it is a scalar matrix. Thus two GL_2 conjugations differ by an inner SL_2 conjugation exactly when their determinants differ by the square of a unit. Reduction $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/n)$ is surjective, as follows from elementary row operations over each prime-power local factor and the Chinese remainder theorem; hence these are exactly the inner conjugations coming from H as well. C.d1 gives $\det U(u) = \mu(u)$. The cyclotomic character $G_{\mathbb{Q}} \to \widehat{\mathbb{Z}}^\times$ is surjective, so its finite residue characters realize every determinant square class. The inclusions of the arithmetic image into $\widehat{\mathrm{GT}} \subseteq \widehat{\mathrm{GT}}_0$ then force equality of all three finite outer images. The arithmetic lift is the cusp-preserving one because geometric tame inertia at the modular cusp is acted on by the cyclotomic character; the Tate family fixes this full inertia subgroup over $\mathbb{Q}$. Equivalently its homological congruence action sends a unipotent to a conjugate unipotent, which excludes the alternative zT^μ already over characteristic zero. □

D. Splittings, their ambiguity, and the sign of the norm

D.1. The extension splits, with a unique Belyi normalization [Known]. *The §46 extension in question is*

$$1 \longrightarrow \widehat{F} \longrightarrow \widetilde{M}^! \longrightarrow \widehat{\mathrm{GT}}_0 \longrightarrow 1,$$

not an extension denoted $\widetilde{N}$ by $\widehat{F}$. It has the section whose representative is $x \mapsto x^\mu$, $y \mapsto f^{-1}y^\mu f$, $f \in \widehat{F}'$. That section is unique among sections all of whose representatives have this normalization. It is Harbater–Schneps (2000), §1.2, Proposition 2 and Theorem 1, applied through A.3. The statements of uniqueness on scans208–209 concern the further matrix normalizer or torus constraint and are accompanied by explicit withdrawal margins.

Manuscript question: §46; fols403–403 verso; scans208–209.

Proof. We give the normalization argument, since it identifies the ambiguity. Start with a loop automorphism and conjugate it so that it sends x to x^{μ}. Write its image of y as $g^{-1}y^{\mu}g$. Multiplying g on the left by a power of y does not change this image, and composing the automorphism with an inner power of x changes its right x exponent. These two changes kill both abelianization coordinates of g. If two normalized representatives differ by $\mathrm{Int}(h)$, comparison on x gives $h = x^{c}$; comparison on y then gives $gx^{c}f^{-1} = y^{d}$. Abelianization and $f, g \in \widehat{F}'$ give $c = d = 0$, so both representatives and both words coincide. The composite of two normalized representatives is again normalized: its conjugator is a product of commutator-subgroup elements, since that subgroup is characteristic. Thus the unique choice is a group homomorphism, and the closed conditions defining its graph make it continuous. This is also the proof of the cited theorem in the present coordinates.

The scan209 ink names $\widetilde{M}^{!}$ and says "Je suspecte que c'est le seul scindage"; the margin reads "canulé, pour mêmes raisons que tout à l'heure". The preceding scan208 already retracts its analogous claim with an isogeny objection. We therefore distinguish these withdrawn conjectures from the theorem of unique normalized representatives. □

D.2. Unqualified strict uniqueness of a splitting [Refuted]. *There is more than one strict continuous section of the displayed extension. The strongest replacement furnished by the equations is uniqueness after the Belyi normalization of D.1, or a classification by a nonabelian continuous H^1 after quotienting by conjugation. In particular, "unique" without naming the normalization or the conjugation ambiguity is false.*

Manuscript question: §46; fols403–403 verso; scans208–209.

Proof. Let s be the normalized section and $c = (-1, 1)$. For any $k \in \widehat{F}$ put $s_k(g) = ks(g)k^{-1}$. This is a section because k lies in the geometric kernel. For $k = x$, since $s(c)(x) = x^{-1}$,

$$s_x(c) = \mathrm{Int}(x^2)\, s(c) \neq s(c).$$

The inequality follows from $x^2 \neq 1$ and the trivial center of $\widehat{F}$. Thus the sections differ; they are nevertheless conjugate by construction. For a general section t, the quotient $a(g) = t(g)s(g)^{-1}$ belongs to $\widehat{F}$ and satisfies

$$a(gh) = a(g)\, s(g)(a(h)).$$

Conversely this cocycle equation makes $a(g)s(g)$ a section. Conjugating by k changes $a(g)$ to $ka(g)s(g)(k^{-1})$. These are exactly the defining equivalence relations of $H^1_{\mathrm{cts}}(\widehat{\mathrm{GT}}_0, \widehat{F})$ for the action determined by s. □

D.3. The strict normalizer uniqueness already fails modulo seven [Refuted]. *The matrix normalizer condition on scan208 does not restore strict uniqueness. Let $K_{1/2}$ be the section centralizing σ_{∞}. There is a distinct conjugate section whose image still maps into the normalizer of $\widehat{\mathbb{Z}}[S]^{\times}$ in the projective matrix group. Distinctness is detected by the two involutions*

$$J = \begin{pmatrix} 1 & 0 \\ 0 & 6 \end{pmatrix}, \qquad J' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \textit{in } \mathrm{PGL}_2(\mathbb{F}_7).$$

This is a counterexample to strict uniqueness; it does not refute uniqueness modulo geometric conjugation or the distinct order-three torus question of scan209.

Manuscript question: §46; fol403; scan208.

Proof. The cyclic finite-subgroup theorem for free profinite products says that an order-two element of $\widehat{B} = \widehat{C}_2 \amalg \widehat{C}_3$ is conjugate to s, with centralizer $\langle s \rangle$. Applied to the image of s under a pure automorphism, its conjugator may be taken in $\widehat{F}$: its reduction in S_3 centralizes the transposition and is removed by a power of s. The intersection of this centralizer with $\widehat{F}$ is trivial. Consequently each class has a unique representative centralizing s, so these representatives form the continuous section $K_{1/2}$. This is the precise profinite input behind the scan206–208 construction. The torsion part is Herfort–Ribes (1985), Theorem 2(i), explicitly cited in the opened Minamide–Nakamura paper, p.8; the primary paper was not supplied and its proof has not been independently checked here. The normal-form centralizer assertion is the corresponding free-product centralizer theorem. We isolate this standard structural input instead of inferring it from a finite matrix test.

Here is the counterexample once that section is fixed. At 7 set

$$k_7 = 2I + 2S = \begin{pmatrix} 2 & 5 \\ 2 & 2 \end{pmatrix}.$$

It has determinant one, lies in the S-torus, and $k_7 J k_7^{-1} = J'$. Both J, J' square to I and conjugate S to $-S$. For a global lift, take $t_7 = 2$ and the unique Hensel lift $u_7 \in \mathbb{Z}_7$ of 2 with $u_7^2 = -3$; its derivative $2u_7$ is a unit, so the lift exists. Set $k_7^{\mathrm{ad}} = u_7 I + t_7 S$ and $k_p^{\mathrm{ad}} = I$ for all $p \neq 7$. Then $k^{\mathrm{ad}} \in \mathrm{SL}_2(\widehat{\mathbb{Z}})$, is the identity modulo two, and commutes with S.

Reduction $\widehat{F} \to \ker((\mathrm{SL}_2(\widehat{\mathbb{Z}})/\{\pm I\}) \to \mathrm{SL}_2(\mathbb{F}_2))$ is onto. Indeed $\mathrm{SL}_2(\mathbb{Z})$ surjects onto $\mathrm{SL}_2(\mathbb{Z}/n)$ by elementary row operations and the Chinese remainder theorem; restricting to the inverse image of the identity mod two and then taking compact inverse limits proves the assertion. Hence choose $k \in \widehat{F}$ over k^{ad}. Conjugate $K_{1/2}$ by k. Its matrix image remains in the normalizer, since conjugation by an element of the torus preserves its normalizer. On complex conjugation the reductions of the two sections are J, J', which are not equal up to sign. This proves distinctness. The attached program verifies every finite identity; at level seven one representative word for k_7 is $L_0^{-2} L_1 L_0^{-2}$. □

D.4. Exact finite restricted splitting counts and a graded obstruction calculation [Proved]. *The following numbers count sections of the* restriction to $C_2 = \langle c \rangle$ *of the projective matrix extension $H_n \rtimes \langle c \rangle$, with H_n the image of F generated by L_0, L_1 and c acting by conjugation by J. They do not count sections over an unspecified "finite GT group".*

n	2	3	4	5	6	7	8	12	24
$\lvert H_n \rvert$ *projectively*	1	12	4	60	12	168	32	96	768
Sections over C_2	1	6	4	16	6	28	24	48	288
Normalizing the S*-algebra*	1	2	4	4	2	4	8	16	32

For a lower-central graded piece $V_n = \mathrm{gr}_n(F^{(\ell)}) \simeq L_n(\mathbb{Z}_\ell^2)$, complex conjugation acts by $(-1)^n$. Its continuous cohomology is

$$H^1(C_2, V_n) = \begin{cases} V_n/2V_n & n \text{ odd}, \\ 0 & n \text{ even}, \end{cases} \qquad H^2(C_2, V_n) = \begin{cases} 0 & n \text{ odd}, \\ V_n/2V_n & n \text{ even}. \end{cases}$$

Thus both vanish for odd ℓ and over every $\mathbb{Q}_\ell$. At $\ell = 2$ the first possible conjugacy ambiguity is in degree one, of dimension two over $\mathbb{F}_2$; the first possible lifting obstruction is in degree two, of dimension one. These are cohomological obstruction groups, not assertions that every class is realized by the full extension.

Manuscript question: §46; fols403–403 verso; scans208–209.

Proof. A section over C_2 sends its generator to (h, c), so its necessary and sufficient condition is $h\,c(h) = 1$ in H_n. Equivalently $(hJ)^2$ is scalar $\pm I$. The program enumerates the finite subgroup generated by L_0, L_1, identifies only global $\pm I$, and tests this equality for each element. It then tests whether conjugation sends S into the two-dimensional algebra spanned by I, S. Since that conjugation is invertible and the image of S still generates the same algebra in the tested cases, this is the stated normalizer condition. Each displayed number is printed in `dictionary_matrices.out`; no inference from these restricted counts to full profinite sections is made.

The two-periodic resolution of C_2 has alternating maps $c - 1$ and $1 + c$. Hence $H^1 = \ker(1 + c)/\operatorname{im}(c - 1)$ and $H^2 = \ker(c - 1)/\operatorname{im}(1 + c)$. Substitute the scalar $(-1)^n$ and use that V_n is torsion-free. The action on the degree-one generators is minus identity; functoriality of a bracket of length n gives the asserted scalar in degree n. The Lyndon-word basis proves that these modules are free with Witt ranks, which are computed in the attached script. □

D.5. The plus sign is correct in the manuscript's matrix coordinates [Proved]. *With* $R = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 1 \end{smallmatrix}\right)$*, the centralizer in* $\mathrm{GL}_2(A)$*, for every commutative ring* A*, is*

$$T_R(A) = \left\{ cR + dI : \ c, d \in A, \quad c^2 + cd + d^2 \in A^\times \right\}.$$

Thus the plus sign in scan209 (109) agrees with the explicit matrix convention of scans94,154,199 and258. The minus sign is the norm in the basis $(1, \zeta_3)$*; the plus sign is the norm in* $(1, \zeta_6)$*. The manuscript uses* R *corresponding to* ζ_6*, not to* ζ_3*. The parenthetical in the supplied second-pass corrections suggesting that (147)–(150) require a minus sign is mathematically incorrect with this fixed* R*.*

Manuscript question: §42; fol336; scan94; §46; fol403 verso; scan209(109); §48; scans258–259.

Proof. Writing $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ and solving $MR = RM$ entry by entry gives $c = -b$, $d = a + b$. Hence $M = aI + bR$, and $\det(aI + bR) = a^2 + ab + b^2$. There was no division, so the formula holds also in characteristics two and three and over nonreduced rings. The characteristic polynomial is $T^2 - T + 1$, so its roots are ζ_6, ζ_6^{-1} and the displayed determinant is their norm. For $\zeta_3 + \zeta_3^{-1} = -1$, the norm of $d + c\zeta_3$ is $d^2 - cd + c^2$. Replacing R by $-R$ exchanges the conventions, but that is a change of matrix, not an alternative sign for the same matrix.

The program enumerates all n^4 matrices for $n = 2, 3, 4$ and verifies that their invertible centralizer is exactly this norm-unit set; its sizes are $3, 6, 12$. At $n = 3$, $c = d = 1$ gives a singular matrix: the correct norm is 0, whereas the minus-sign expression is 1. At $n = 2$ the two expressions coincide, so level two alone cannot decide the reading. At $n = 4$ their values can differ although their unit/nonunit parity is the same. At the prime three this smooth centralizer group is not a torus: the algebra $\mathbb{F}_3[R]$ is the dual-number algebra because $T^2 - T + 1 = (T + 1)^2$; its units have a nontrivial additive subgroup. This is exactly the caveat Grothendieck records in §48, scan259. □

D.6. The remaining order-three and global splitting problem [Open]. *The work above does not classify sections of* $\widetilde{M}^!$ *over all of* $\widehat{\mathrm{GT}}_0$ *up to* $\widehat{F}$*-conjugacy, nor does it classify sections over* $C(3) = \ker(\mu \bmod 3)$ *whose matrix image lies in* T_R*. The exact remaining question is whether the subset of* $H^1_{\mathrm{cts}}(C(3), \widehat{F})$ *satisfying that torus constraint, for the action of the centralizing section, consists of one class. For a compatible lower-central or finite characteristic quotient system, a lift at the next abelian kernel* V *is obstructed by its explicit extension class in* $H^2_{\mathrm{cts}}(C(3), V)$*; if zero, choices form an* H^1*-torsor. Already* $V = \widehat{F}^{\mathrm{ab}} = \widehat{\mathbb{Z}}(1)^2$ *is an undecided global instance in this return. D.4 computes the full analogous obstruction groups after restriction to the involution subgroup; that subgroup is not contained in* $C(3)$ *and is not a substitute for it.*

Manuscript question: §46; fol403 verso; scan209.

Established reduction and attempted attack. D.1 proves existence for the whole extension without a torus constraint, and the C_3-centralizer construction restricts its domain as in B.m4. For any successive extension $1 \to V \to E' \to E \to 1$, pull it back along the proposed section of E. Choose a continuous local set-theoretic lift b when one exists and form $\kappa(g,h) = b(g)b(h)b(gh)^{-1}$. Associativity gives the two-cocycle equation. Changing the lift changes κ by a coboundary, so a homomorphic lift exists precisely when its class vanishes; differences of lifts satisfy the one-cocycle equation. This is the stated obstruction, and the finite restricted calculation tests its first involution instances explicitly. The failed further step is to compute this class and its constrained H^1 for the full profinite domain $C(3)$, not to multiply a longer list of 2×2 matrices. A finite quotient of the target alone supplies neither that domain nor compatible transition sections. We leave the exact global instance visible rather than infer uniqueness from the successful rational averaging argument. □

D.7. Strict uniqueness with the order-three torus constraint also fails [Refuted]. *Let $C(3) = \ker(\mu \bmod 3)$ and let $s_R : C(3) \to \widetilde{M}^!$ be the section centralizing r. There is a distinct conjugate section whose matrix image still lies in T_R; indeed the two sections have identical images at every congruence level. Their difference is detected in the characteristic abelian quotient $\widehat{F}^{\mathrm{ab}}/9$. Thus the strict uniqueness suggested and withdrawn on §46, fol.403′, scan209 is false even with its stated order-three torus constraint. This does not decide uniqueness modulo geometric conjugation.*

Proof. For $\mu \equiv 1 \pmod 3$, the conjugator of $u(r)$ may be chosen in $\widehat{F}$, by the factor-torsion argument in C.d1. Its ambiguity is $C_{\widehat{B}}(r) \cap \widehat{F} = \langle r \rangle \cap \widehat{F} = 1$. Adjusting by that unique conjugator gives a unique representative centralizing r; these representatives form the continuous section s_R.

Take the determinant-one adelic matrix k^{ad} with component R^2 at three and I at every other prime. It reduces to I modulo two and centralizes R, so the geometric surjectivity proved in D.3 gives $k \in \widehat{F}$ with that matrix image. In $\mathrm{SL}_2(\mathbb{F}_3)/\{\pm I\} \simeq A_4$, the image of R^2 has order three and is nontrivial in $A_4^{\mathrm{ab}} \simeq C_3$. Therefore $[k] \bmod 3 \neq 0$ in $\widehat{F}^{\mathrm{ab}}/3$. The program independently computes this derived subgroup and tests membership.

Choose the unit $\mu \in \widehat{\mathbb{Z}}^\times$ with $\mu_3 = 4$ and $\mu_p = 1$ for $p \neq 3$. Surjectivity of the cyclotomic character gives an arithmetic element and hence $g \in C(3)$ with this multiplier. Since every pure loop automorphism acts on $\widehat{F}^{\mathrm{ab}}$ by its multiplier, the cocycle comparing s_R and $k s_R k^{-1}$ at g has abelianization

$$[k] - \mu[k] = (1 - \mu)[k],$$

whose three-adic component is $-3[k]$ and is nonzero modulo nine. Thus the two sections are strictly distinct. The finite calculation checks this for every vector modulo nine whose reduction modulo three is nonzero.

On the other hand, $U(s_R(h))$ centralizes R for every $h \in C(3)$. A possible global negative sign in the projective equality is ruled out by $\operatorname{tr} R = 1$. The centralizer algebra $\widehat{\mathbb{Z}}[R]$ is commutative, so k^{ad} commutes with all these matrices. Consequently conjugating the section by k changes none of its congruence matrices. Both sections satisfy the exact torus constraint. This explains why a search solely in the congruence quotients would miss the counterexample, although the abelian characteristic quotient detects it. □

E. The remaining conditions and what the graded comparison sees

E.1. The extra orbifold compatibility is arithmetic [Proved]. *The outer action of $G_{\mathbb{Q}}$ satisfies the compatibility of B.m3, understood first modulo inner automorphisms of the target $\widehat{B}$. After its allowed cusp normalization it satisfies the equations (7),(10),(15),(47),(51),(54) of §47, with the sign and conjugator corrections specified in B.m5–B.m6. These are formulations and consequences of one compatibility, not six independent conjectural constraints.*

Manuscript question: §41; fols.315–316 verso; scans54–57; §47; fols.415–425; scans229–250 (cote140-3).

Proof. Let $\mathcal{X}$ be the orbifold whose coarse curve is $\mathbb{P}^1_{\mathbb{Q}} \setminus \{0\}$, with stabilizer orders two at 1 and three at ∞. The coordinate $T = 1728/j$ identifies it with the rigidification of $\mathcal{M}_{1,1}$ by its generic elliptic involution; this can equally be seen from the weighted projective presentation with weights two and three. Its geometric fundamental group is $\widehat{B} = C_2 \widehat{*} C_3$.

There are two maps over $\mathbb{Q}$ relevant here. The level-two cover from the Legendre coordinate is

$$U \longrightarrow \mathcal{X}, \qquad T = \frac{27\lambda^2(1-\lambda)^2}{4(1-\lambda+\lambda^2)^3};$$

it induces the inclusion $F \hookrightarrow B$ and the action used in the dictionary. The open immersion $\mathbb{P}^1_{\mathbb{Q}} \setminus \{0,1,\infty\} \hookrightarrow \mathcal{X}$ induces the surjection φ, because a meridian about 1 acquires order two and a meridian about ∞ acquires order three. Identify the two tripod coordinates over $\mathbb{Q}$; this is the identification used in the manuscript. Functoriality of the arithmetic fundamental group for the immersion gives a homomorphism of arithmetic extensions. Conjugating by a lift of a Galois element proves equivariance up to geometric conjugation in $\widehat{B}$. This proves the outer assertion without assuming a description of $G_{\mathbb{Q}}$ by equations.

For the explicit normalized equations, write the images of the two free generators using B.m1. The first equation determines the possible conjugator up to the procyclic centralizer of $e = sr$; write that conjugator as e^t. Replacing u_0 by $\mathrm{Int}(y^n)u_0$ changes it to $\varphi(y^n)e^t y^{-n} = e^{t-n}$, so the allowed choice $n = t$ removes it. The centralizer of e is procyclic: reduce in S_3, remove a power of e, and use the procyclic centralizer of $e^2 = y$ in $\widehat{F}$. The second is then exactly B.m3. Checking instead the other free basis gives the order-three formula (47); reducing these equalities modulo F gives (51), and their product gives (54). This is the generator-by-generator reduction of B.m6. No assertion that (I),(II),(III) imply this compatibility is used in the argument. □

E.2. The two-evaluation finite test through levels two and three [Proved]. *There are twelve generating cycle residues in the characteristic $S_3 = \mathrm{PSL}_2(\mathbb{F}_2)$ envelope and twenty-four in the characteristic $A_4 = \mathrm{PSL}_2(\mathbb{F}_3)$ envelope. Every one admits a gauge satisfying the normalized additional equation B.m3 after evaluation in the respective projective congruence group. Thus the extra equation removes none of these enumerated residues at these two levels. In particular every actual profinite $\widehat{\mathrm{GT}}_0$ element admits an allowed gauge with $\varphi^\bullet(\xi) \in \{1, s\}$ in S_3; this finite consequence of scan233 is therefore derived from (I),(II) and the automorphism condition.*

Manuscript question: §47; fols.417–420; scans233–240.

Proof. This calculation uses two homomorphisms from the *same* characteristic envelope to Q_n, not an ill-defined application of φ to an element of H_n. They are

$$j : (x,y) \mapsto (L_1, L_0), \qquad v : (x,y) \mapsto (S, SR).$$

For $n = 3$ the characteristic envelope has order 2304, exponent six, and derived group of order 256. The analogous data at $n = 2$ are in B.f3; the map j is trivial there and is still a well-defined map of the envelope. The program constructs both maps by Cayley-graph propagation and checks every relation. It enumerates (I),(II) and generation exactly, with λ modulo twelve and f in the computed derived group. For each solution put $\nu = (\lambda - 1)/2$, $\xi = y^{\nu} f$ and, for every a modulo six, $\xi_a = y^a \xi x^{-a}$. It then tests

$$j(\xi_a) = v(\xi_a) S v(\xi_a)^{-1} S^{-1} L_1^{\nu}. \tag{E.2.1}$$

The period six includes every possible profinite gauge residue. The recorded sets of successful gauges are nonempty for every one of the twelve and twenty-four solutions, respectively. All coefficients, all solutions, and all six defects per solution are retained in `finite_phi.json`; `finite_phi.py` is the complete enumeration. The assertion is exactly this necessary finite test; it does not state that every such residue has an infinite lift.

For the last assertion, every actual profinite pair projects into this list: the envelope is characteristic, the induced map is an automorphism, and f projects into its derived group. Its multiplier modulo twelve determines ν modulo six. At level two j and L_1 are trivial, so (E.2.1) says that $v(\xi_a)$ centralizes S. That centralizer in S_3 is $\{1, S\}$. A successful residue a mod 6 lifts to an allowed $a \in \widehat{\mathbb{Z}}$. This is only a finite gauge condition; it need not solve all the higher or full profinite equations. □

E.3. Does the pentagon imply the additional orbifold compatibility? [Open]. *With the precise groups and normalizations of B.m3 and B.m5, the remaining statement is*

$$\widehat{\mathrm{GT}} \;\subseteq\; C^{\varphi} \;\subseteq\; \widehat{\mathrm{GT}}_0. \tag{E.3.1}$$

The second inclusion is definitional and the arithmetic image lies in C^{φ} by E.1. In this return we prove neither the first inclusion nor its negation. The direct finite attack E.2 finds no violation at target levels two and three. The next simultaneous-evaluation target not enumerated for this equation is $Q_4 \simeq S_4$, whose characteristic envelope has the explicitly constructed order 442368. Its outer shadow is arithmetic by B.s4.1, but that outer shadow alone does not retain the normalizing multiplier and conjugators needed in (E.2.1).

Manuscript question: §41; fols.315–316 verso; scans54–57; §47; fols.415–425; scans229–250 (cote140-3).

Attack and precise remaining step. The proof of B.m2 derives every §46 loop and symmetry equation from (I),(II), with an inverse change of variables. It stops before applying φ to an unknown profinite word. Applying the congruence homomorphism also does not supply that missing value: j and v are different maps. E.2 repairs this problem at two explicit finite targets by retaining both evaluations. At level four the next required calculation is to range over the normalized pairs in the characteristic S_4 envelope, retaining their residue multiplier, and test (E.2.1) for every gauge; the normalizer calculation of B.s4.1 does not by itself perform that test. Even a violating finite pair would still require a compatible pentagon lift before it could refute the first inclusion in (E.3.1). Conversely finitely many passing targets would not prove the inverse-limit inclusion.

We also compared the actual presentation with Nakamura and Schneps (2000, Theorems 1.1–1.4 and relations (III′),(IV)). That source constructs an arithmetic-containing subgroup of $\widehat{\mathrm{GT}}$ using additional genus-one relations, including a Kummer cocycle for the roots of two. It does not give an identified substitution proving (E.3.1). We therefore do not rename its subgroup as C^{φ} or claim an equivalence of the additional relations without the missing comparison. This is the exact literature-comparison obstruction, in addition to the explicit finite calculation still required. □

E.4. Why the additional compatibility has no faithful graded form here [Proved]. *The full condition E.1 cannot be faithfully linearized by replacing both occurrences of the group by the single completed free Lie algebra of Part B. The characteristic-zero pro-unipotent completion of $B = C_2 * C_3$ is trivial. Its pro-ℓ completion is C_2 for $\ell = 2$, C_3 for $\ell = 3$, and trivial for every other prime. Consequently the rational Lie algebra obtained from that target is zero in every case. A relative completion retaining the S_3 quotient would be additional data, not this absolute completion.*

Manuscript question: §47; fols.415–420; scans229–240.

Proof. An element of finite order in a characteristic-zero unipotent group is the identity: taking its logarithm makes its order times that logarithm zero. Any homomorphism from $C_2 * C_3$ to such a group therefore kills both generators. For a pro-ℓ target a generator of order prime to ℓ is killed; the one surviving cyclic factor at $\ell = 2$ or 3 gives precisely the claimed completion by its universal property. These finite groups have zero rational Malcev Lie algebra. Meanwhile $v : F \to S_3$ in E.2 is nontrivial, so it cannot factor through a connected unipotent algebraic group into the component group of a relative completion. One can restrict to $\ker(v)$ and retain its larger free Lie algebra and comparison maps, but this changes the input and the filtration. No such replacement is silently made in W^{lac}. □

E.5. The sorted inventory has one unresolved compatibility family [Proved]. *Within the explicitly stated equations of §§38–47, the loop and symmetry equations are (I),(II); the congruence-kernel and square-class conditions follow for the whole actual profinite $\widehat{\mathrm{GT}}_0$ by C.d1 and C.a15–C.a16; the quadratic signs follow algebraically by C.d2; and the literal uniqueness and multiplier claims have the corrections proved in C.r2 and Part D. The additional equations still not derived here from $\widehat{\mathrm{GT}}$ form the orbifold-compatibility family in the following table. They are recorded separately in the condition CSV because they retain different data, and their dependence is made explicit rather than presented as independent discoveries.*

Manuscript question: §46; fols.390–413; scans182–227; §47; fols.415–427; scans229–253; inventory of §§38–47.

Entry	Condition and location	Graded class	Decision and attack
E01	Outer φ compatibility; §41 scans55–57 and §47 scans229–231, (5),(7).	(c)	Arithmetic by E.1; $\widehat{\mathrm{GT}}$, $\widehat{\mathrm{GT}}_0$ undecided as in E.3.
E02	Normalized equation $\xi = [\varphi(\xi), s]x^{\nu}$; scans234,240, (15),(30).	(c)	Equivalent normalized version; complete simultaneous tests at levels2,3 in E.2.
E04	Cyclic order-three formula and its S_3 reduction; scans244–248, (47),(51).	(c)	Equivalent free-basis formulation after normalization; same unresolved global comparison.
E05	The pair of equations (57),(58) when $\mu \equiv 1 \pmod 3$; scan250.	(c)	Their conjunction follows from E01; no claim that either single equation suffices.

Here (a) would mean a proved derivation on the rational graded pentagon Lie algebra, (b) would mean only a conditional use of the 2026 dimension equality, and (c) means there is no

faithful graded form in the specified absolute completion. All four remaining entries are of class (c), by E.4. No entry is assigned to (b) merely to conceal an unavailable completion map. The special-derivation identity itself belongs to (a), by the direct pentagon derivation B.g8. The conditional equality and its exact hypotheses are isolated in B.g9. The finite signs restrict to $+1$ on $\lambda = 1$ and have vacuous differentials, as proved in C.a14.

The order-two finite consequence of scan233, formerly designated E03, is excluded from this unresolved list: E.2 derives its existential-gauge form for the whole $\widehat{\mathrm{GT}}_0$. It remains in the complete condition inventory. This distinction prevents a successful finite test from being reported as an unresolved condition, or from being inflated into full compatibility.

For the auxiliary labels on scans247–248, B.m6 proves $i \equiv \nu \pmod 2$, equivalently $i = (1 - \epsilon_2(\mu))/2$, by applying the sign character to $\varphi(\xi) = \beta s^i$. Thus this order-two label is already determined algebraically once compatibility holds. The remaining question whether $j' \in C_3$ is a function of the multiplier is not an additional equation already proved on Galois. The explicit complex-conjugation case gives $(i, a_3, j') = (1, 1, -1)$, while the identity gives $(0, 0, 0)$. The dependence of j' is retained in the finite-normalization problem E.3, without a guessed character.

E.6. No inference from the graded equality to profinite equality is used [Proved]. *Even if the claimed 2026 upper bound is correct, it cannot on its own decide (E.3.1), remove the congruence kernel of B.f6, distinguish the four central lifts of C.a10, or classify full profinite torsion. It also does not turn the finite Lazard counterexample of B.g15 into an element of* $\widehat{\mathrm{GT}}_0 \setminus \widehat{\mathrm{GT}}$.

Manuscript question: §46; fols.390–413; scans182–227; §47; fols.415–427; scans229–253; inventory of §§38–47.

Proof. Each proposed inference needs a comparison that is absent. E.4 shows that the orbifold target disappears in the absolute Lie construction. B.f6 supplies a noncongruence finite quotient missing from the matrix completion. The four lifts differ by finite characters with zero rational tangent space. A finite group quotient or a weight-truncated Lie algebra does not produce a compatible inverse system of lifts. Finally B.f8 gives an explicit endomorphism which is an automorphism in every nilpotent and pro-ℓ realization but not in the full profinite one. These are concrete obstructions to the respective inferences, independent of the truth of the rational dimension theorem. □

F. Boundary cases, the explicit-element lead, and the audit boundary

F.1. The exceptional types and centers do not permit silent transfer [Proved]. *Over an algebraically closed field of characteristic zero, the geometric fundamental groups of types* $(0,0), (0,1), (0,2), (1,0)$ *are, respectively,* $1, 1, \widehat{\mathbb{Z}}, \widehat{\mathbb{Z}}^2$. *Consequently none is a center-free hyperbolic fiber to which C.a4 applies. For type* $(0,2)$ *over* $\mathbb{Q}$ *the outer action is the cyclotomic character, and is not faithful. For an elliptic curve of type* $(1,0)$*, its action on the last group is the Tate-module action of that particular curve; it is not the action on the universal modular group* H.

Proof. The complex sphere and plane are simply connected, the punctured plane retracts onto a circle, and a complex elliptic curve is a quotient of $\mathbb{C}$ by a rank-two lattice. Finite covering theory and the characteristic-zero comparison give the four completions. The covers $t \mapsto t^n$ of $\mathbb{G}_m$ identify its arithmetic action with the action on all roots of unity. This factors through an abelian group. Its kernel is nontrivial: the splitting field of $t^3 - 2$ has Galois group S_3 (irreducibility, and

its nonsquare discriminant -108, give this elementary cubic criterion), whereas every quotient of an abelian group is abelian. Thus a faithful cyclotomic action would contradict this quotient. For an elliptic curve the finite coverings are its isogenies; the inverse system of multiplication-by-n kernels is its Tate module. This identifies the representation without importing any assertion about the universal stack. The nontrivial centers of the latter two groups prevent the center-free reductions used earlier.

Both $(0,3)$ and $(1,1)$ satisfy $2g-2+\nu>0$ and have free rank-two geometric fiber groups. Their moduli and mapping-class groups are different objects: the central μ_2 in the elliptic moduli stack and the center of H were explicitly retained in A.1 and C.a7–C.a16. At finite congruence level the center can be larger still, as C.r1 computes. The proof of D.5 uses no division, includes characteristics two and three, and identifies the non-toric fiber at three. The Legendre gerbe is asserted only over $\mathbb{Z}[1/2]$; it includes characteristic three, but not the invalid four-etale-branch-point construction in characteristic two.

A smooth complex curve is canonically oriented, so a nonorientable surface is not another complex type (g,ν) to which these theorems apply. Orientation-reversing symmetries are instead represented by $B^{\pm}$ and J in A.2. Over $\mathbb{R}$, the arithmetic quotient is $G_{\mathbb{R}}=C_2$ and for the tripod in our normalization its outer action is represented by the same $c=(-1,1)$; a chosen real or tangential base point supplies a genuine lift. This covers the real involution in the manuscript without identifying the fundamental group of a real locus with that of the complex curve. The latter identification is generally false, since the real locus is a union of one-dimensional manifolds. □

F.2. The explicit complex-conjugation part of Lead 3 [Proved]. *With a real tangential base point and the standard real access paths to* $0,1$*, complex conjugation is the normalized pair* $(-1,1)$*. The identity and this element are distinct full profinite solutions; the latter has order two. These statements answer the literal examples of §42, fol.349′, scan120, and §47, fols.426–427, scans252–253.*

Proof. Reflection in the real axis reverses the orientation of each small peripheral circle while fixing the chosen real access paths. Thus its based action is $x\mapsto x^{-1}, y\mapsto y^{-1}$. The normalized commutator conjugator is 1, by the uniqueness proved in D.1. This action squares to the identity and acts by minus identity on $\widehat{\mathbb{Z}}^2$, so is nontrivial. Its cycle and pentagon equations hold because it is an arithmetic action; alternatively, the pentagon is the identity when $f=1$, and the two cycle equations follow by direct cancellation with $\nu=-1$. The self-centralizer calculation in B.m7 gives further information about this full profinite element, independently of a classification of all torsion. □

F.3. The exact limit of the explicit-element and full-image comparison [Open]. *We have not constructed an element of the full profinite* $\widehat{\mathrm{GT}}_0\setminus\widehat{\mathrm{GT}}$*, nor proved that this difference is empty. The finite non-arithmetic outer solution of B.g15 is a solution of the specified finite two-cycle system; it is not asserted to lift to* $\widehat{\mathrm{GT}}_0$*. The strongest explicit lifting result here is the weight-seven rational solution through class twelve in B.g11. Its first uncomputed rational extension is degree thirteen, with the group equations and the preceding coefficients retained. The first undecided integral extension at the prime thirteen also leaves the range of the class-less-than-prime BCH construction.*

Manuscript question: §42; fol.349 verso; scan120.

Attack and unresolved comparison. The raw weight-four candidate fails at degree five by the exact rank obstruction in B.g4. The special weight-seven candidate has a nonzero pentagon defect,

and solving all the retained higher coefficient equations simultaneously gives the class-twelve jet; the failed greedy attempt is retained in the code. Reduction modulo thirteen gives a genuine finite group, and the actual pure-braid action in B.g15 proves its arithmetic exclusion. To pass from this object to the full pro-13 group would require a compatible solution at every further characteristic quotient; no inverse-limit existence theorem follows from the twelve solved degrees. Passing from one pro-13 group to the full profinite group requires further simultaneous compatibilities. B.f8 shows concretely why automorphism tests in every nilpotent completion cannot replace those requirements.

The sentence "no other element is known explicitly" is a statement about available descriptions in a literature, not a mathematical nonexistence assertion. We do not turn the failed construction into such a theorem. The named comparison is Lochak and Schneps (1997), *A cohomological interpretation of the Grothendieck–Teichmüller group*, Proposition 4(i),(ii), whose proof uses only (II) after the fixed-subgroup input; see B.m7 and Lochak and Schneps (1997). It proves a centralizer statement and does not by itself classify all finite-order elements. Pro-ℓ torsion, finite-shadow orders, and full profinite torsion remain distinct questions, as B.g10 explains. □

Scope of the source and computational audit

The accompanying report records the primary sources opened, the specific imported theorem numbers, the original scans inspected, and the early failed live-source requests. The checked transcriptions, their research notes, the LM3 return and review, and the BISITE return and review were treated as supplied leads. Our calculations were rerun from the definitions; no assertion receives its status merely because it had that status in an earlier return.

The bibliography marks *verified* for a source opened, not for an entire proof independently certified. The exact partial audit of the attached 2026 preprints is B.g9. The comparator file ending in `sorry` is not a supplied proof certificate; this statement does not assert that a separate proof module is absent elsewhere. The live primary arXiv record confirms the attached Willwacher version-four date of 28 October 2026. A live request to the OpenAI repository failed; no external formalization result is imported on the basis of an inaccessible primary file or an unreviewed secondary summary.

The program directory contains all mathematical programs used for the reported calculations, their text outputs, and machine-readable certificates. Exact integer or rational arithmetic is used; finite field and permutation operations are checked by exhaustive enumeration where the claims say exhaustive. Linear dimensions come from rational rank computations and independent character formulas. A test at finitely many levels is used as a proof of those finite cases only. The failed and incomplete jet-solving attempts are retained and identified as such; their failure is never used as an impossibility proof. A deterministic build and inventory audit checks that every numbered claim has exactly one row in the results register.

The unresolved problems are not interchangeable: E.3 concerns the additional orbifold map, C.d5 concerns simultaneous profinite realization of adelic normalizer coordinates, D.6 concerns conjugacy classes of sections over a specified domain, B.f7 concerns an arithmetic finite-shadow image, and B.g11/F.3 concern an infinite lifting problem. Their finite attacks and smallest uncomputed instances are stated separately. We make no claim of priority.

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